

This document for internal discussion purposes only



2021 Hyundai Sonata Chassis Dynamometer Dynamic Response Tests



Argonne National Laboratory
Center for Transportation Research

April 8, 2022

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata - Enabling acceleration override on CAN



- The 2021 Sonata is equipped with a **Smart Cruise Control (SCC)** functionality^(*)
 - *Cruise Speed control* : vehicle maintains target speed when no vehicles ahead are detected
 - *Retardation control* : vehicle decelerates if vehicle ahead is detected
 - *Following Distance control* : vehicle maintains selected following distance
 - *Acceleration control* : vehicle accelerates to selected speed if vehicle ahead is not detected



ESCM : Electronic Stability Control Module

ECM : Engine Control Module

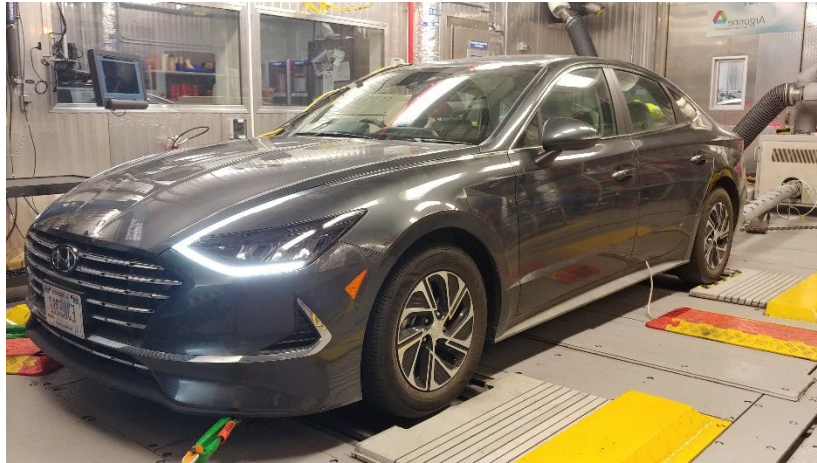
TCM : Transmission Control Module

(*) Source :

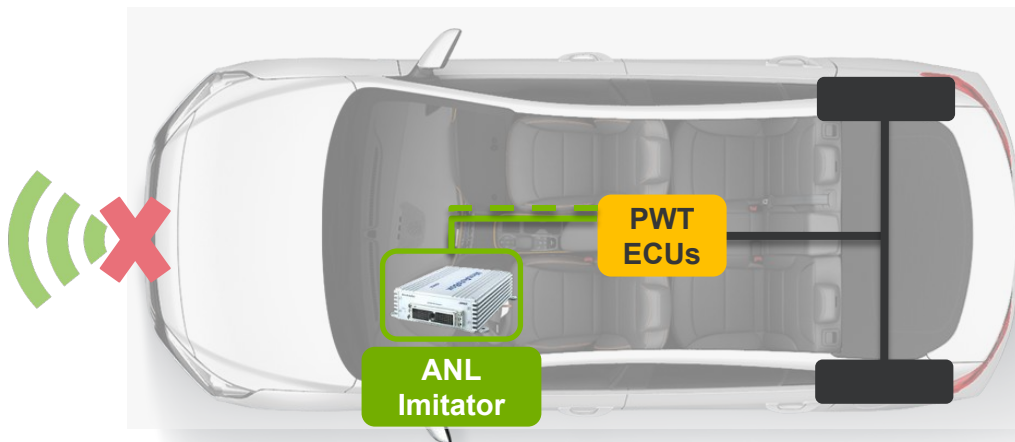
https://www.hsonata1f.org/description_and_operation-1155.html

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata - Enabling acceleration override on CAN



- For this specific vehicle CAN configuration it was possible to replace the stock vehicle radar with a “Radar Imitator”
- ANL Radar Imitator emulates all the signals originally transmitted from the radar to the rest of the powertrain modules
- Emulated signals include:
 - **Adaptive Cruise Control (ACC) status**
 - **Acceleration request**
 - Jerk Upper/Lower limits
 - Object status and properties
 - Autonomous Emergency Braking (AEB)
 - Forward Collision Assistance system

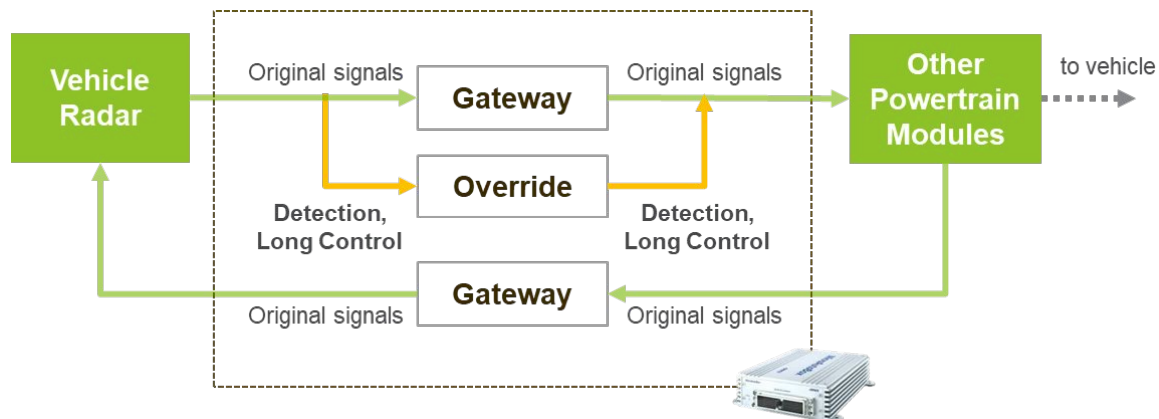


Task 1- Expanding the XIL Research Fleet

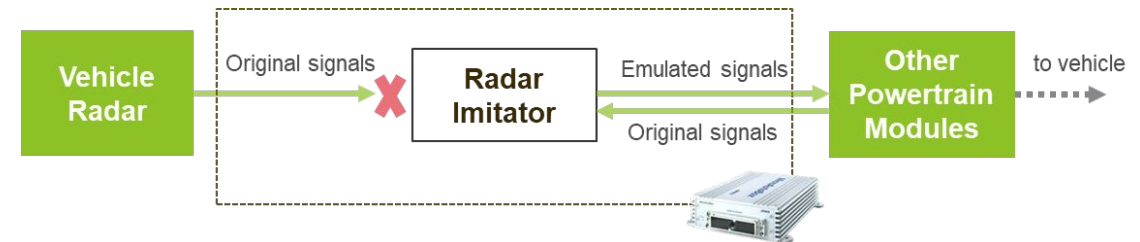
2021 Hyundai Sonata - Enabling acceleration override on CAN

- No need to create a “Man-in-the-Middle” configuration used in previous VIL vehicles
 - No invasive in-vehicle modifications required
 - Significantly lower amount of signals to be transmitted/received
 - Faster integration, ideal for rapid prototyping efforts

“Man-in-the-Middle” Configuration

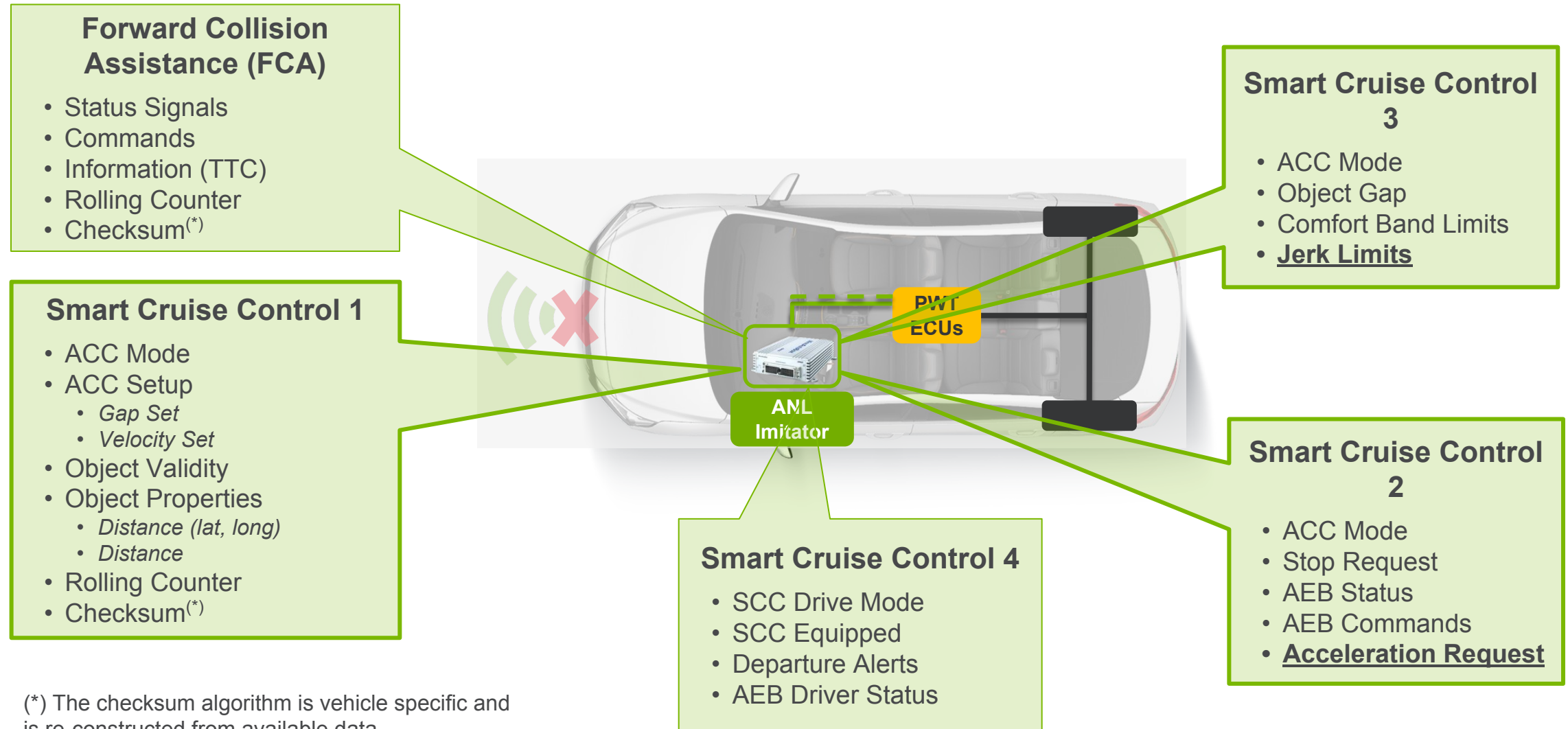


“Radar Imitator” Configuration



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata - Enabling acceleration override on CAN



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

■ Test Objectives

- Evaluate the dynamic behavior of the vehicle when longitudinal control is performed using CAN override method
- Understand the role of the jerk upper and lower limit signals originally transmitted from the SCC module, and how they affect the vehicle response

■ Test Overview

- Performed ramp tests between 7 vehicle speed envelopes
 - For each speed level 5 acceleration commands tested
 - For each acceleration command 3 levels of jerk upper and lower limits tested

(*) **Note** : at the time of the test no on-road data for ACC operation was available. The jerk lower limit was set to negative for these exploratory tests. Data collected later on-road showed that both jerk limits actually vary in the positive range

Speed Range	Acceleration Command	Jerk Limits ^(*)
10-20 mph	$\pm 0.5 \text{ mps}^2$	$\pm 0.5 \text{ mps}^3$
		$\pm 1 \text{ mps}^3$
		$\pm 4 \text{ mps}^3$
20-30 mph	$\pm 1 \text{ mps}^2$	$\pm 0.5 \text{ mps}^3$
30-40 mph		$\pm 1 \text{ mps}^3$
		$\pm 4 \text{ mps}^3$
40-50 mph	$\pm 2 \text{ mps}^2$	$\pm 0.5 \text{ mps}^3$
50-60 mph		$\pm 1 \text{ mps}^3$
		$\pm 4 \text{ mps}^3$
60-70 mph	$\pm 3 \text{ mps}^2$	$\pm 0.5 \text{ mps}^3$
		$\pm 1 \text{ mps}^3$
		$\pm 4 \text{ mps}^3$
70-80 mph	$\pm 4 \text{ mps}^2$	$\pm 0.5 \text{ mps}^3$
		$\pm 1 \text{ mps}^3$
		$\pm 4 \text{ mps}^3$

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

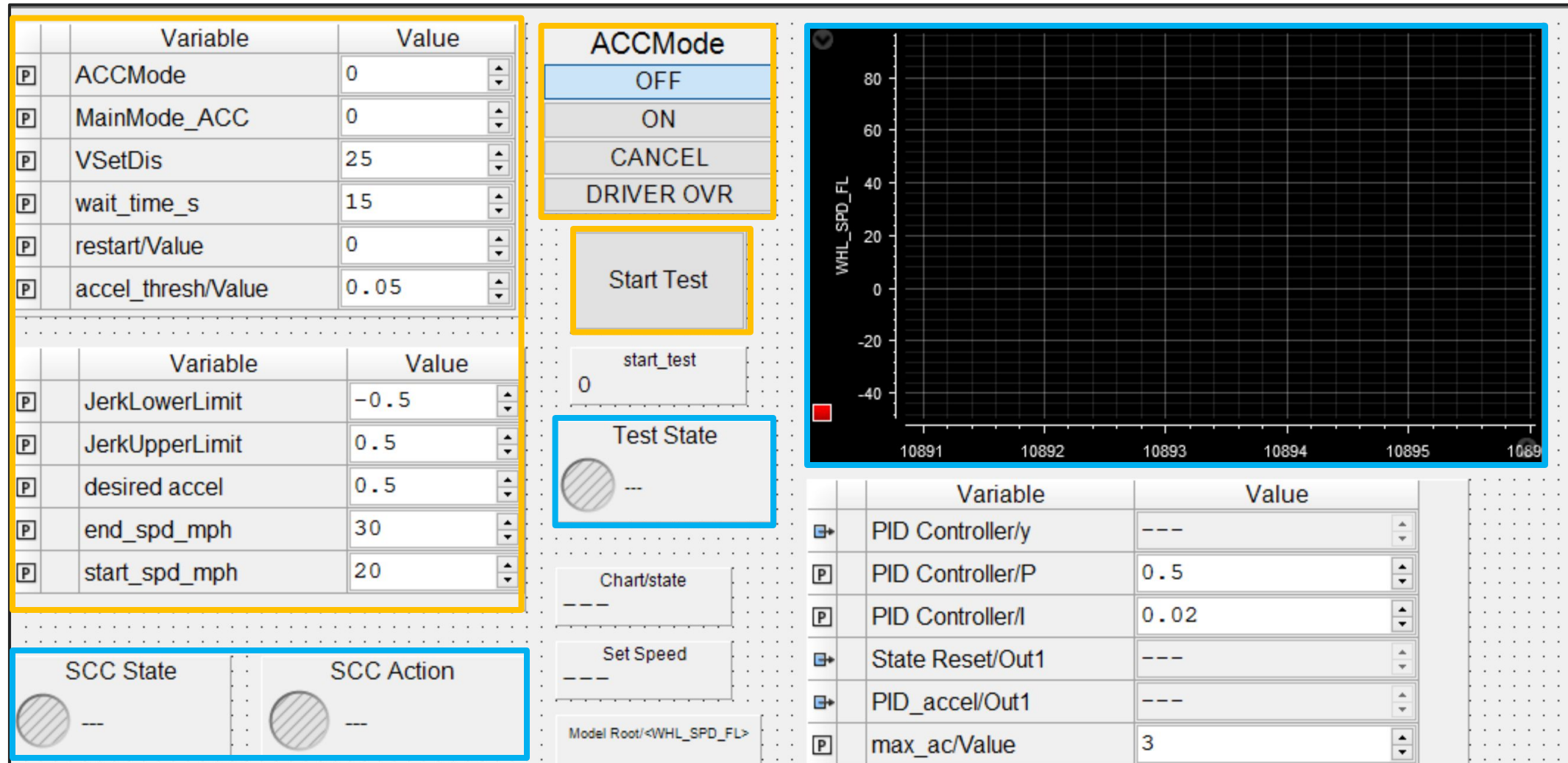
▪ Test Results Summary

- Transmission operations impact the dynamic response of the vehicle
- Variations in the upper jerk limit impact the dynamic response of the vehicle
 - The higher the upper jerk limit, the higher is the actual max acceleration and the lower the acceleration time, for the same acceleration request
 - The higher the upper jerk limit, the higher is the actual max jerk for the same accel request
- Variations in the lower jerk limit did not seem to impact the vehicle dynamic response
 - ACC operational on-road data was not available prior to these exploratory tests
 - On-road data collected after the tests shows that both jerk limits vary in the positive range, therefore a negative jerk limit shows no impact on vehicle dynamic operations
 - With this new information additional testing will be performed on the dyno to explore additional combination of jerk limits/acceleration commands
 - Additional on-road data will also be collected prior to the next round of dyno testing to understand the behavior of these limits for different driving conditions/ powertrain modes (Sport, Eco)
 - This document reports all the results for the sake of completeness

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests HMI

Monitor
Control

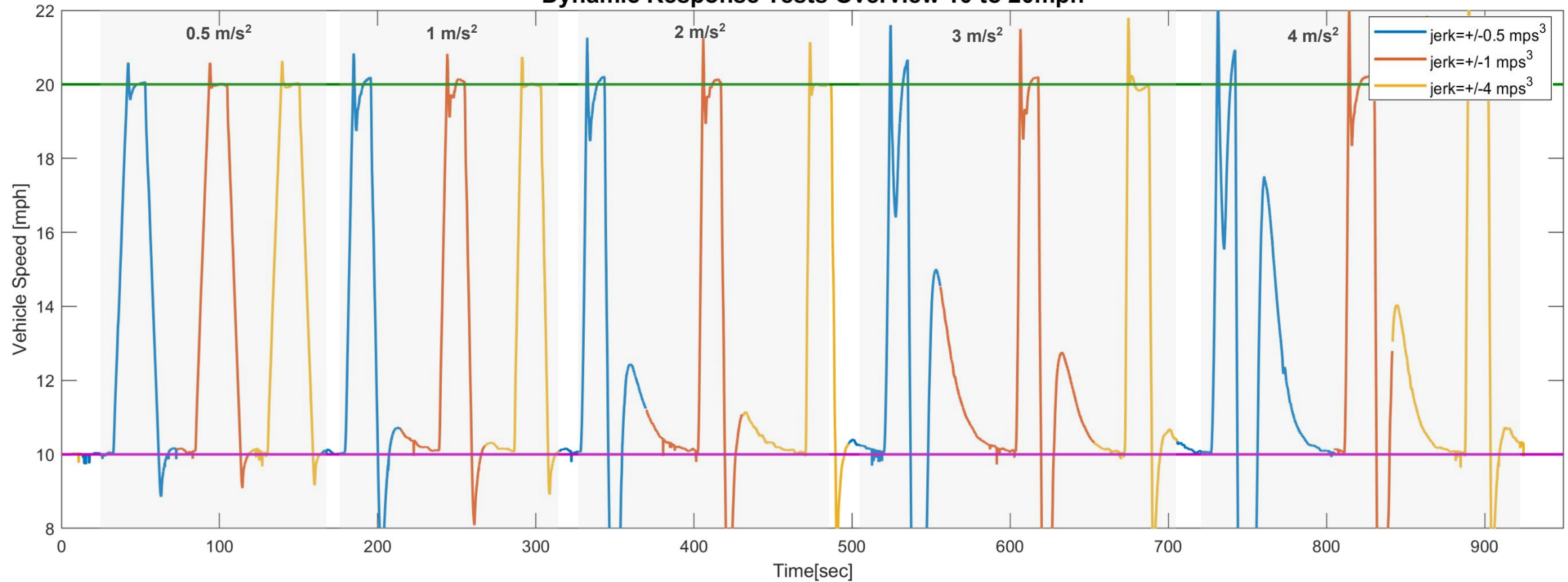


Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

10-20 mph

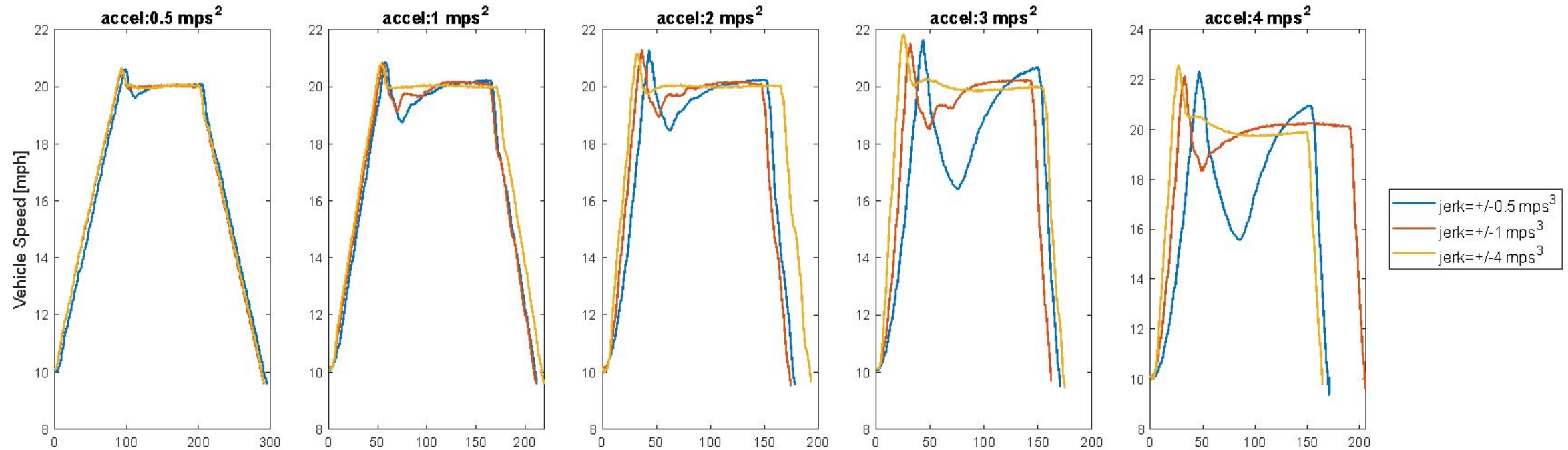
Dynamic Response Tests Overview 10 to 20mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

10-20 mph



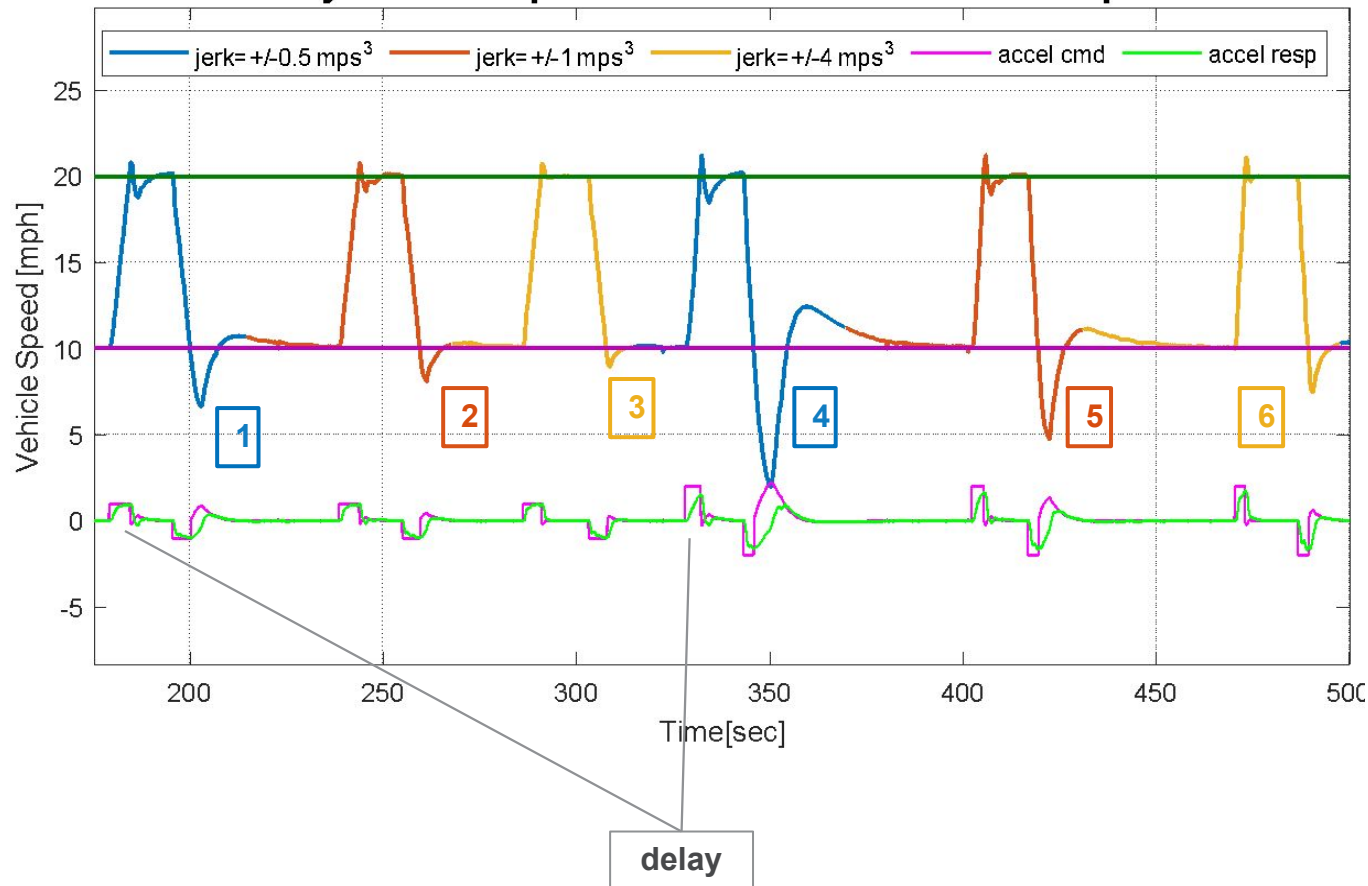
	accel req : 0.5 mps ²		accel req : 1 mps ²		accel req : 2 mps ²		accel req : 3 mps ²		accel req : 4 mps ²	
Jerk Limits	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time
±0.5 mps ³	8.9 s	9.0 s	5.3 s	4.7 s	3.8 s	2.6 s	3.8 s	2.0 s	3.7 s	1.6 s
±1 mps ³	8.6 s	9.0 s	4.8 s	4.7 s	3.0 s	2.7 s	2.6 s	1.9 s	2.7 s	1.5 s
±4 mps ³	8.6 s	9.1 s	4.6 s	4.8 s	2.5 s	2.8 s	2.0 s	2.0 s	2.0 s	1.5 s

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

10-20 mph

Dynamic Response Tests Overview 10 to 20mph



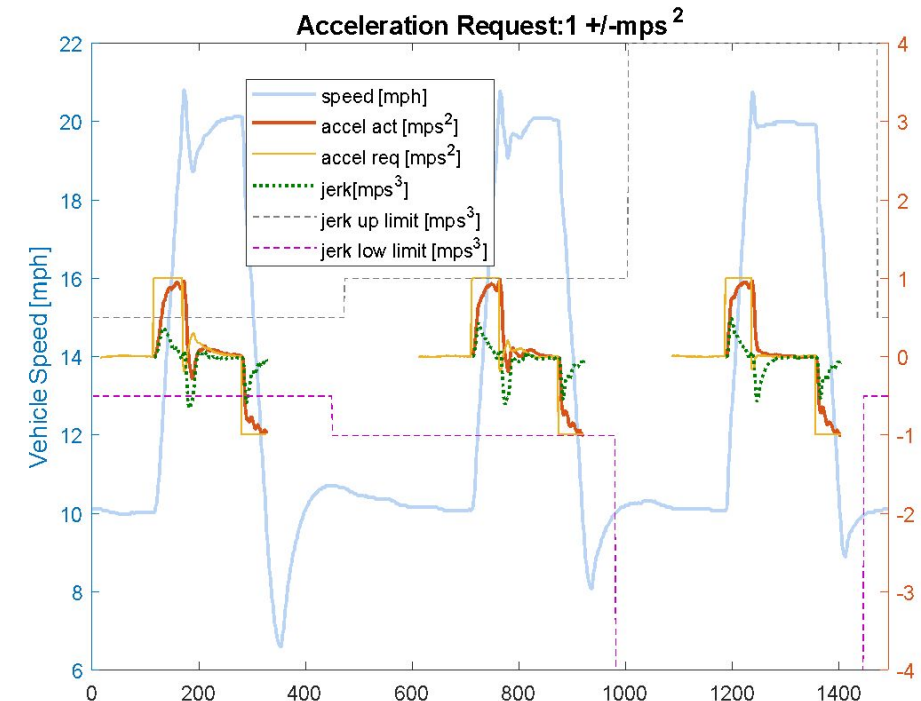
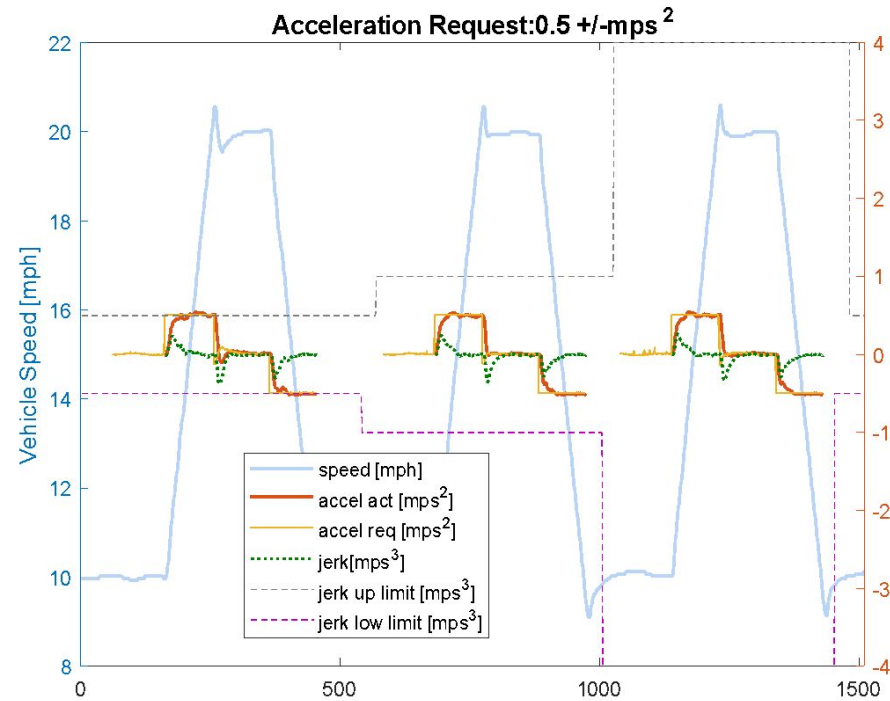
- For the same acceleration request level (1 and 2 mps² in figure) **the lower the jerk limit, the more delay there is in acceleration changes**
- This translates into an undershoot in the vehicle speed which is higher for lower jerk limits (0.5 m/s³)
- The higher the acceleration request, the more evident this behavior is

Point	Accel Req	Speed Undershoot
1	±1 mps ²	34%
2	±1 mps ²	19%
3	±1 mps ²	10%
4	±2 mps ²	80%
5	±2 mps ²	52%
6	±2 mps ²	25%

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

10-20 mph



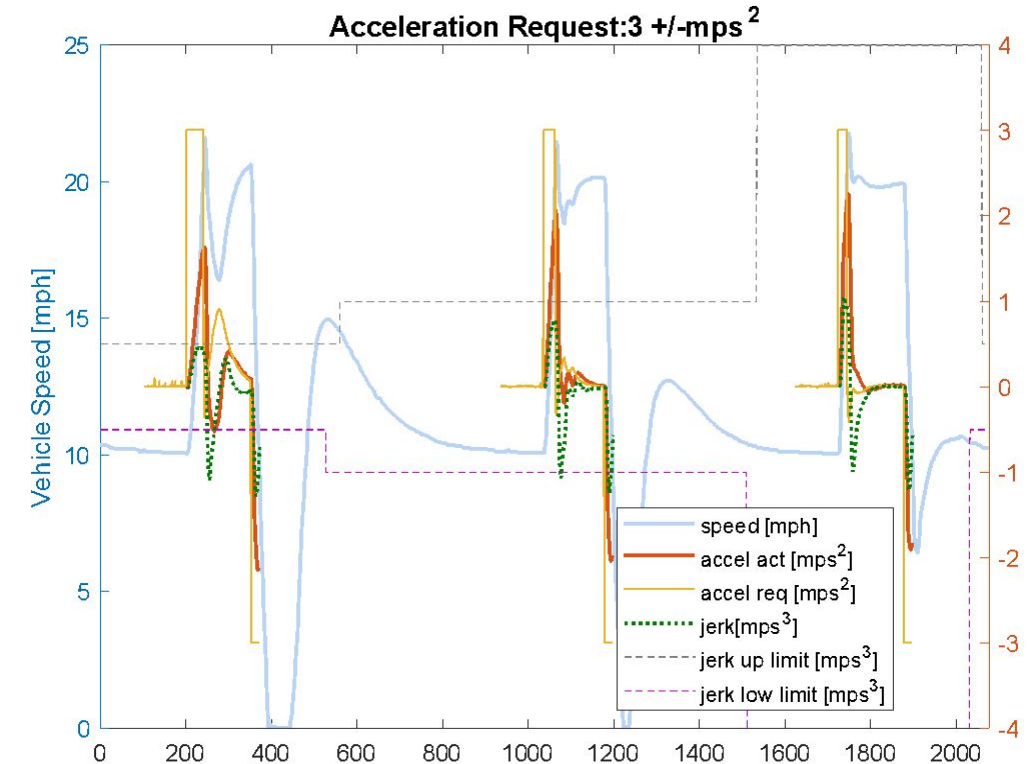
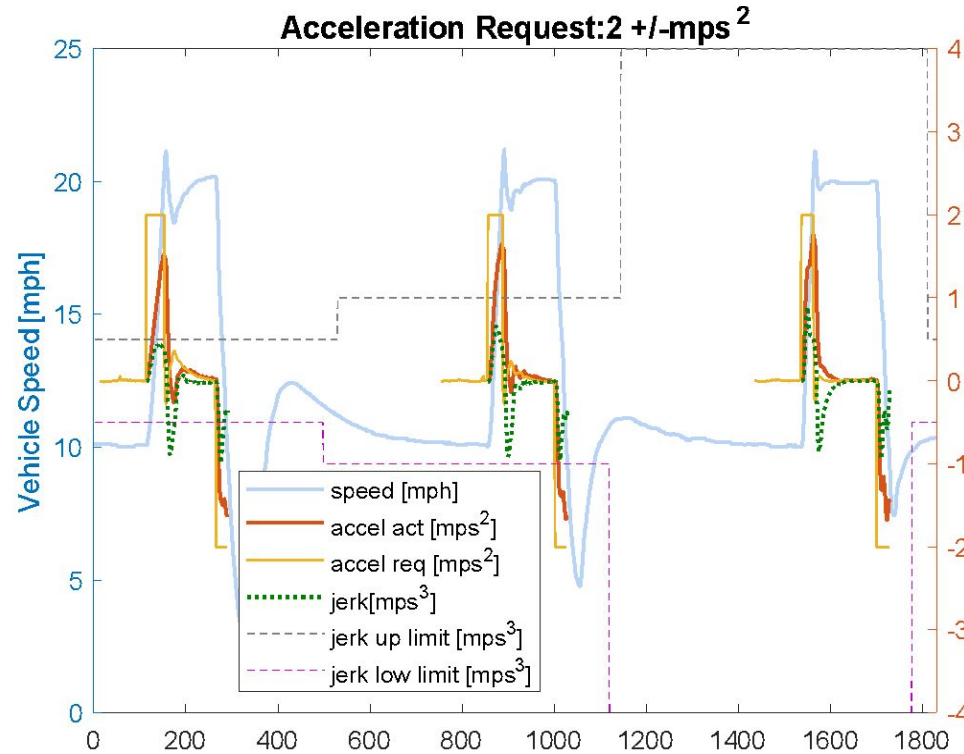
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.54	-0.53	0.24	-0.30
$\pm 1 \text{ mps}^3$	0.53	-0.53	0.25	-0.28
$\pm 4 \text{ mps}^3$	0.55	-0.53	0.26	-0.28

Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.95	-0.99	0.37	-0.59
$\pm 1 \text{ mps}^3$	0.95	-1.00	0.44	-0.54
$\pm 4 \text{ mps}^3$	0.95	-1.00	0.49	-0.54

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

10-20 mph



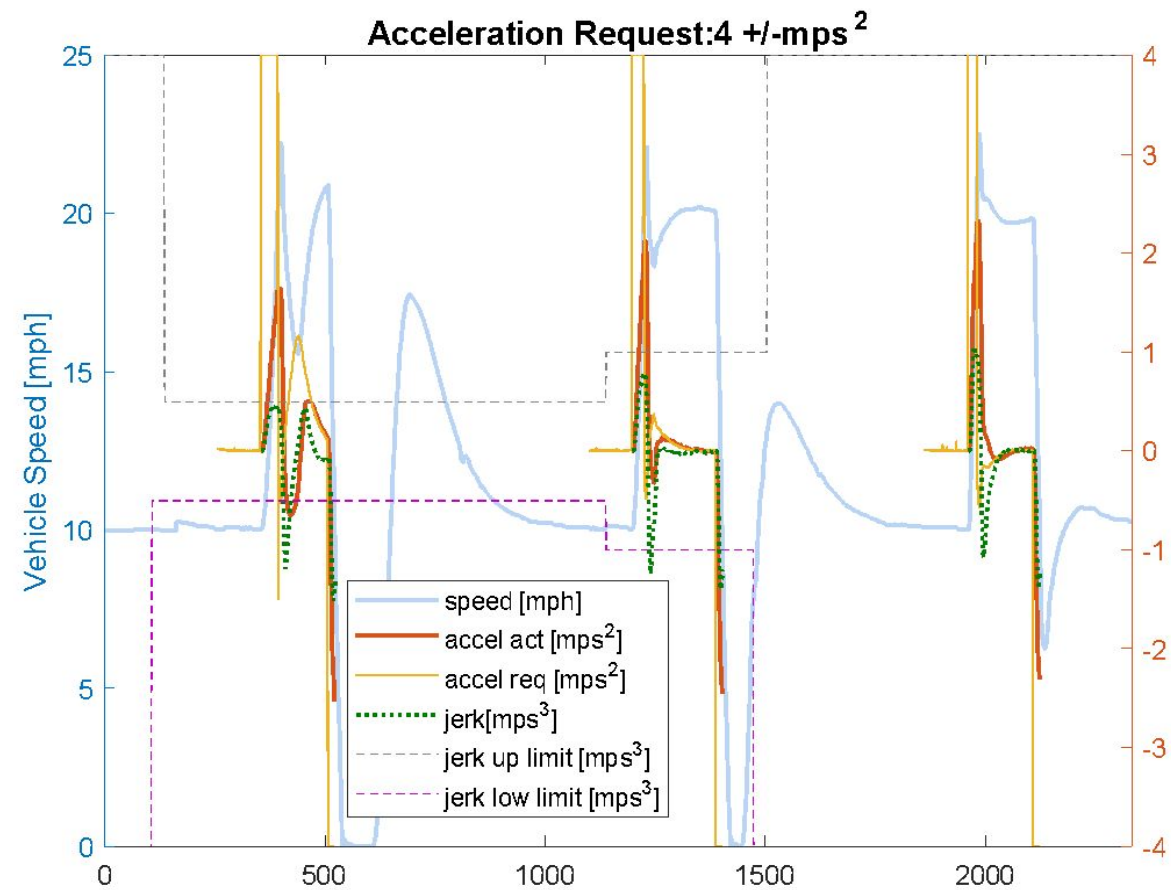
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.51	-1.65	0.45	-0.99
$\pm 1 \text{ mps}^3$	1.64	-1.67	0.65	-0.95
$\pm 4 \text{ mps}^3$	1.76	-1.68	0.89	-0.94

Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.60	-2.15	0.46	-1.28
$\pm 1 \text{ mps}^3$	1.95	-2.04	0.79	-1.24
$\pm 4 \text{ mps}^3$	2.14	-1.91	1.05	-1.21

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

10-20 mph

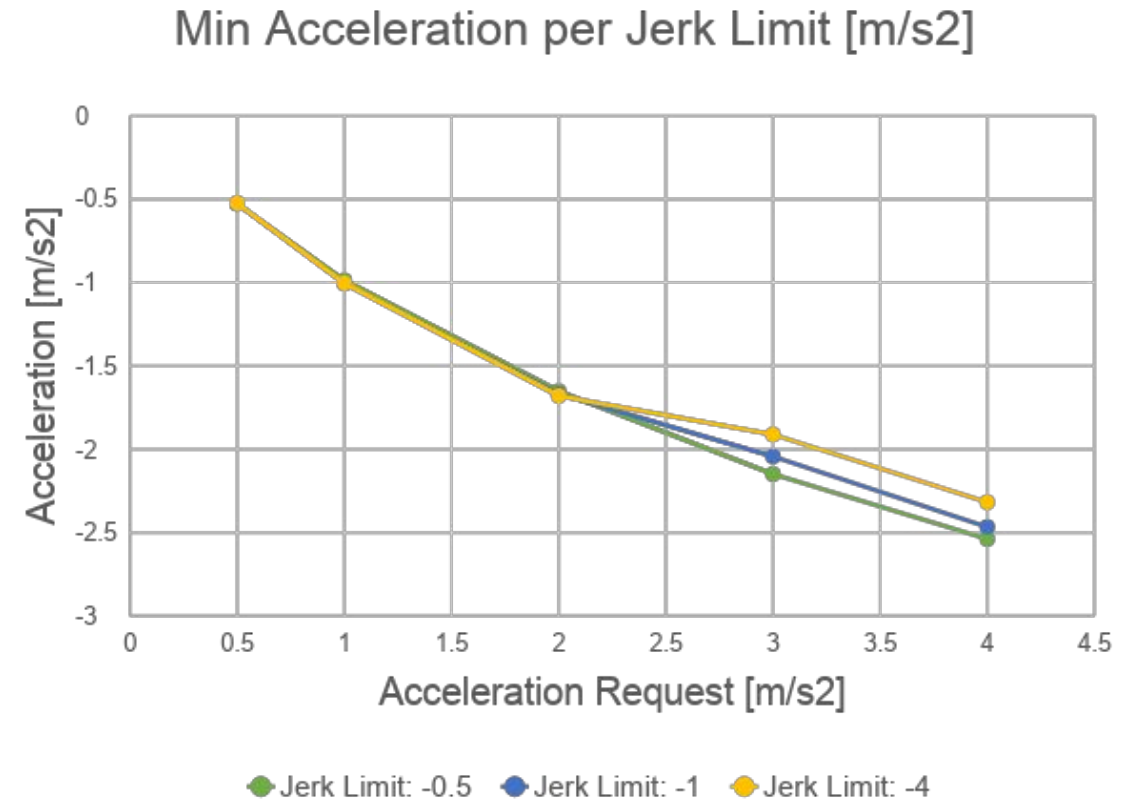
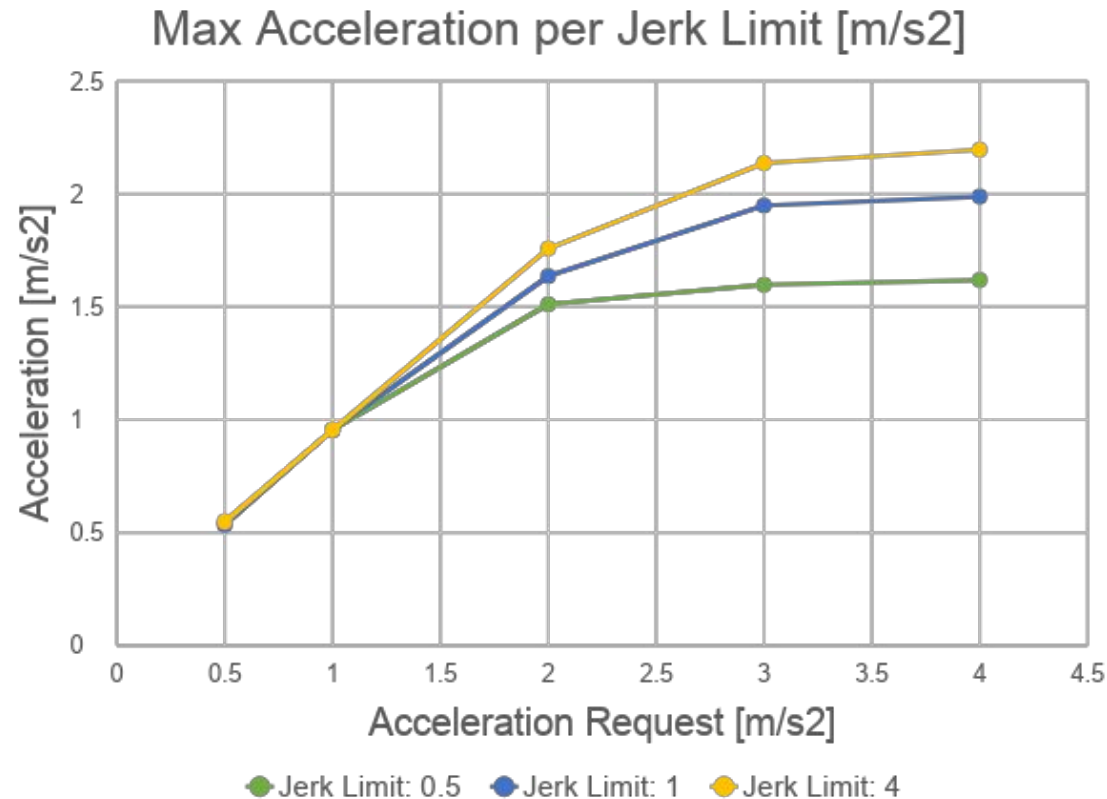


Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.62	-2.54	0.47	-1.51
$\pm 1 \text{ mps}^3$	1.99	-2.47	0.79	-1.40
$\pm 4 \text{ mps}^3$	2.20	-2.32	1.06	-1.39

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

10-20 mph

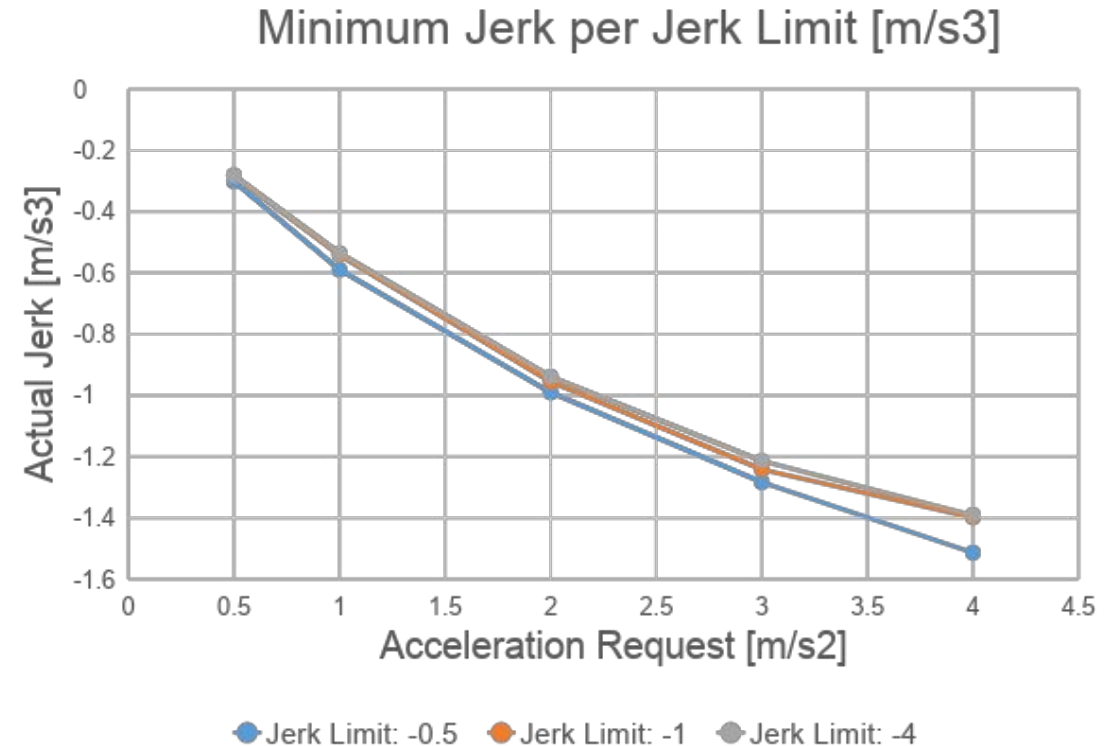
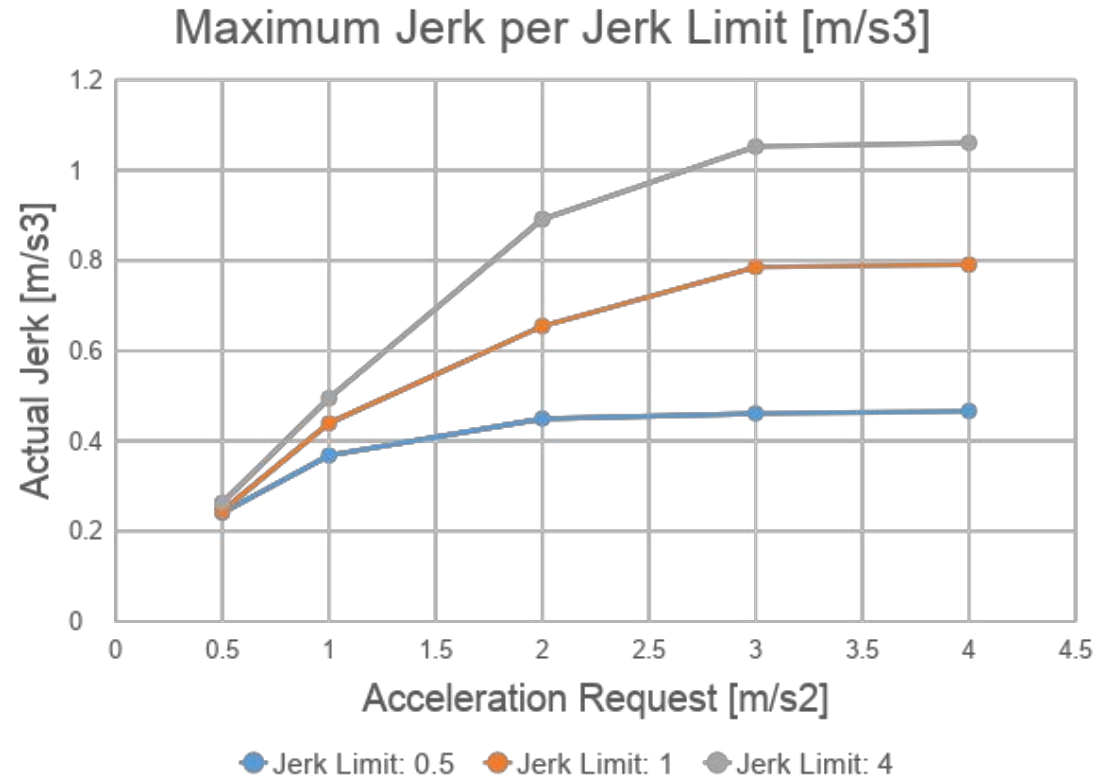


- The higher the jerk upper limit (the more “relaxed”), the higher is the maximum acceleration reached by the vehicle for higher acceleration requests (effect is not visible on lower acceleration requests)

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

10-20 mph

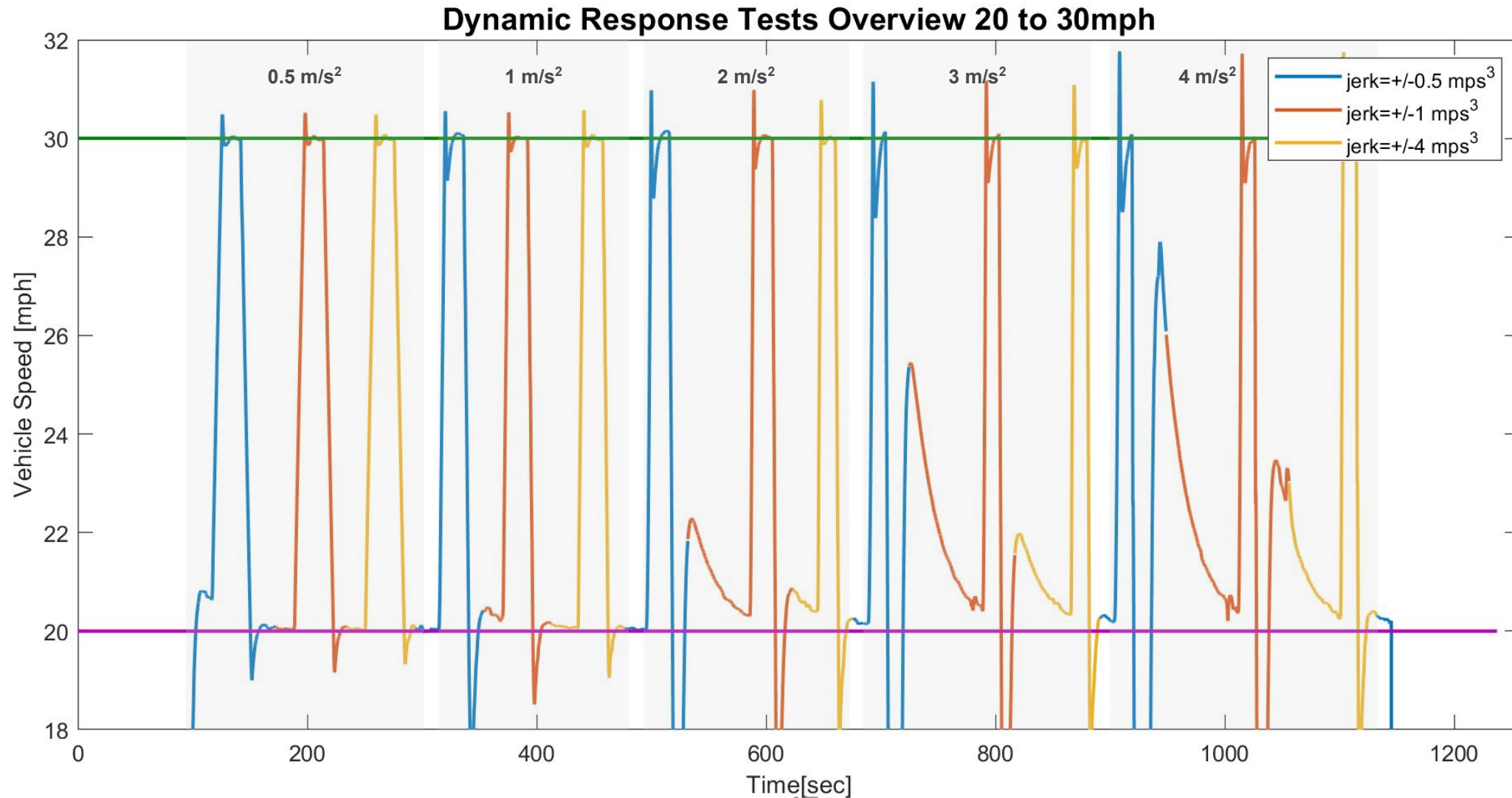


- On the maximum jerk the effect of the jerk limit is visible : the higher the jerk upper limit and the higher the acceleration request, the higher the actual jerk
- The effect of the jerk limit is not as visible on the minimum jerk

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

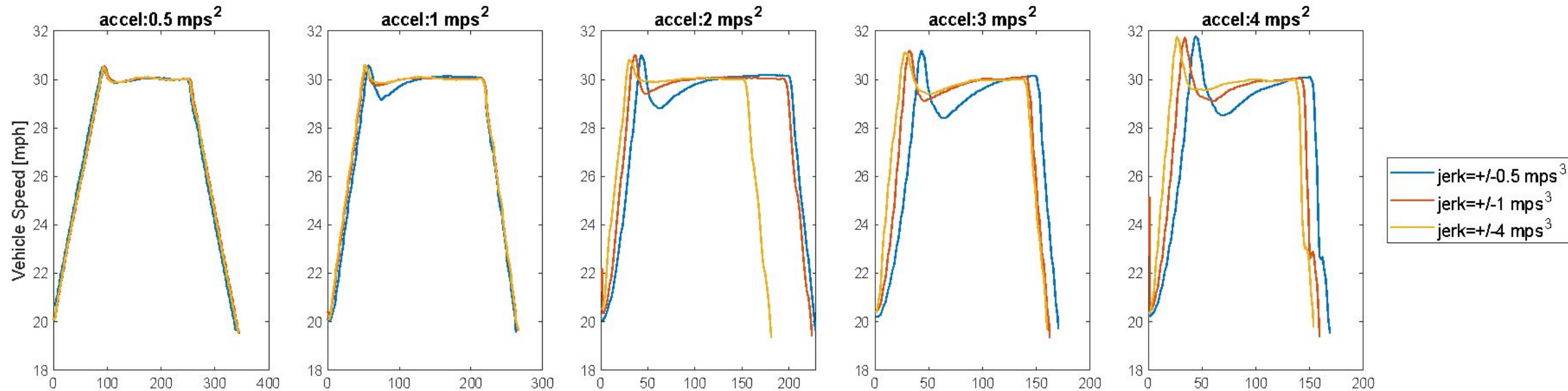
20-30 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

20-30 mph

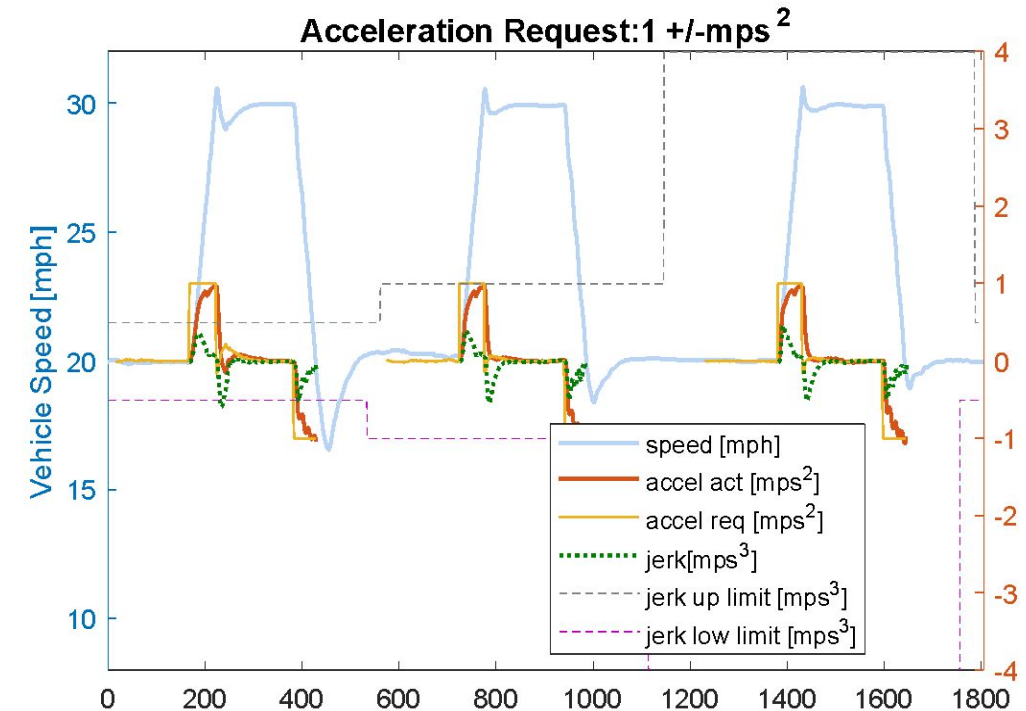
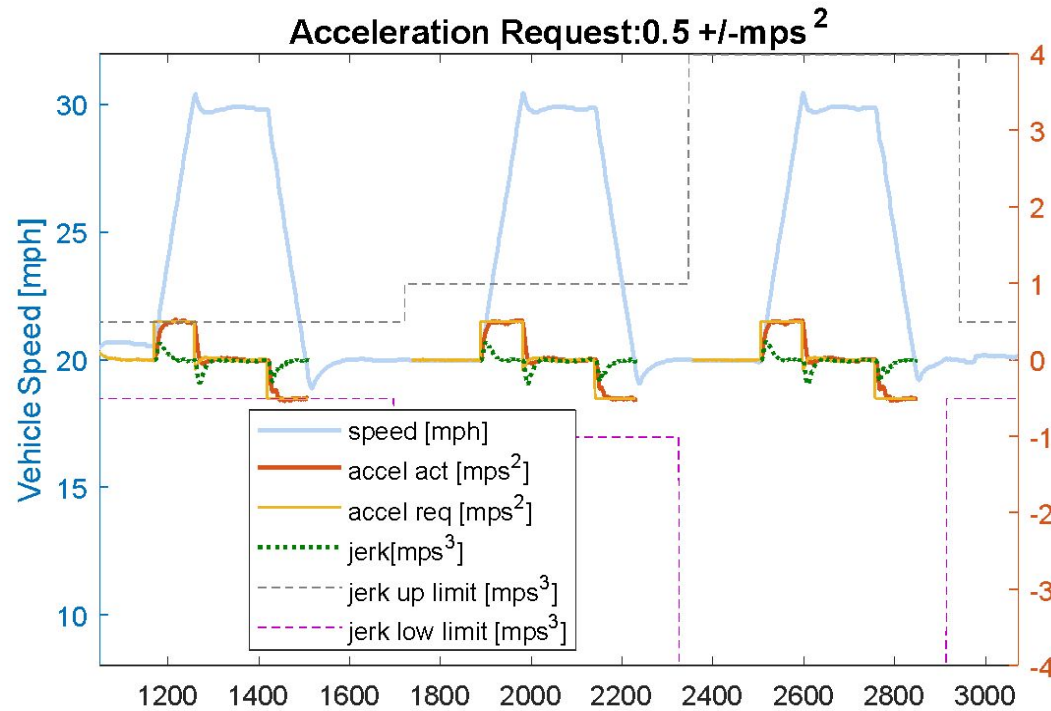


	accel req : 0.5 mps^2		accel req : 1 mps^2		accel req : 2 mps^2		accel req : 3 mps^2		accel req : 4 mps^2	
Jerk Limits	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time
$\pm 0.5 \text{ mps}^3$	8.5 s	8.9 s	5.3 s	4.6 s	3.9 s	2.6 s	3.8 s	1.9 s	3.9 s	1.6 s
$\pm 1 \text{ mps}^3$	8.8 s	9.1 s	4.9 s	4.7 s	3.1 s	2.7 s	2.7 s	2.0 s	2.8 s	1.5 s
$\pm 4 \text{ mps}^3$	8.7 s	9.1 s	4.7 s	4.8 s	2.7 s	2.7 s	2.4 s	2.0 s	2.1 s	1.5 s

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

20-30 mph



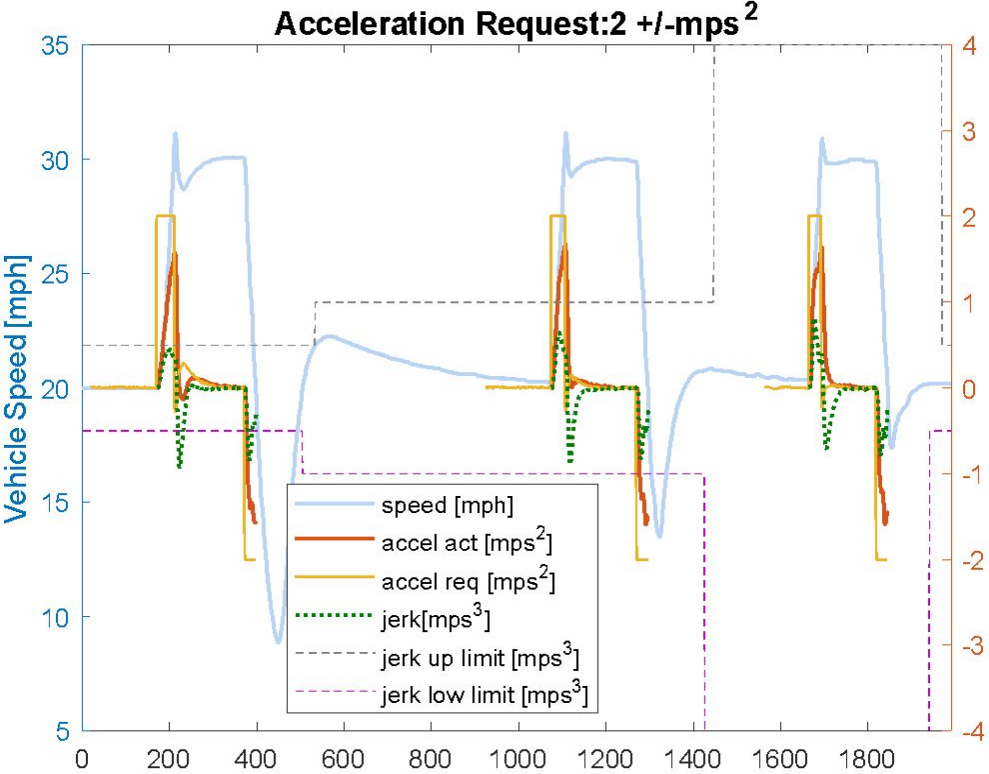
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.53	-0.54	0.28	-0.27
$\pm 1 \text{ mps}^3$	0.51	-0.52	0.24	-0.28
$\pm 4 \text{ mps}^3$	0.51	-0.54	0.25	-0.28

Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.96	-1.05	0.36	-0.52
$\pm 1 \text{ mps}^3$	0.96	-1.05	0.40	-0.50
$\pm 4 \text{ mps}^3$	0.96	-1.07	0.46	-0.49

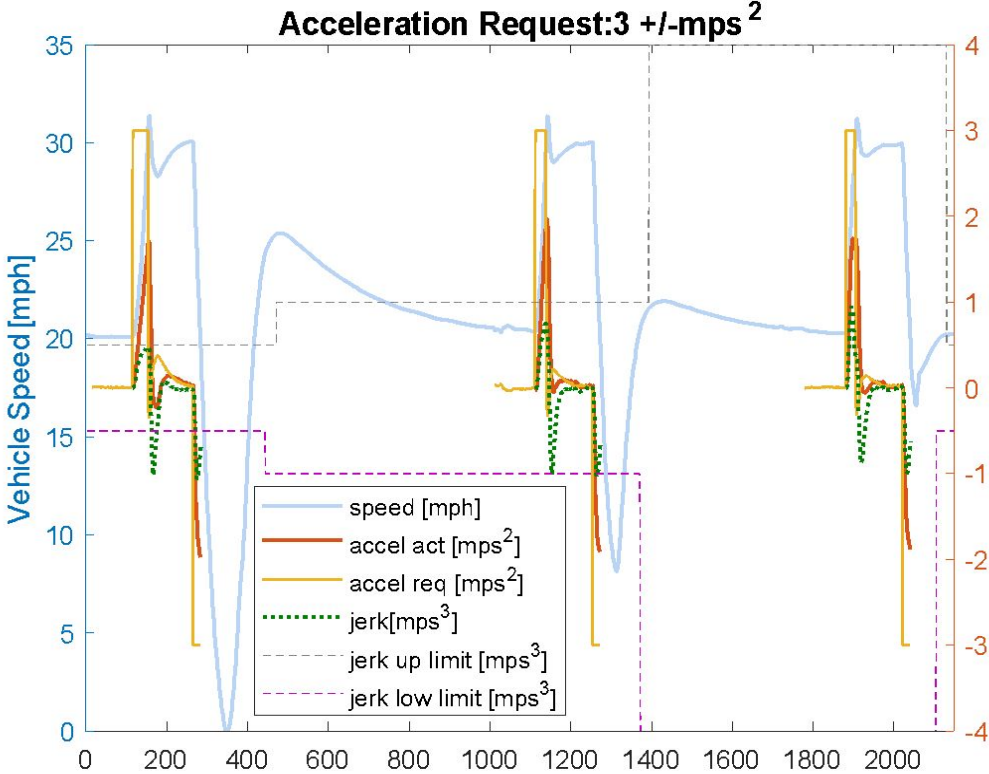
Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

20-30 mph



Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.49	-1.58	0.45	-0.85
$\pm 1 \text{ mps}^3$	1.61	-1.60	0.64	-0.83
$\pm 4 \text{ mps}^3$	1.56	-1.60	0.82	-0.79

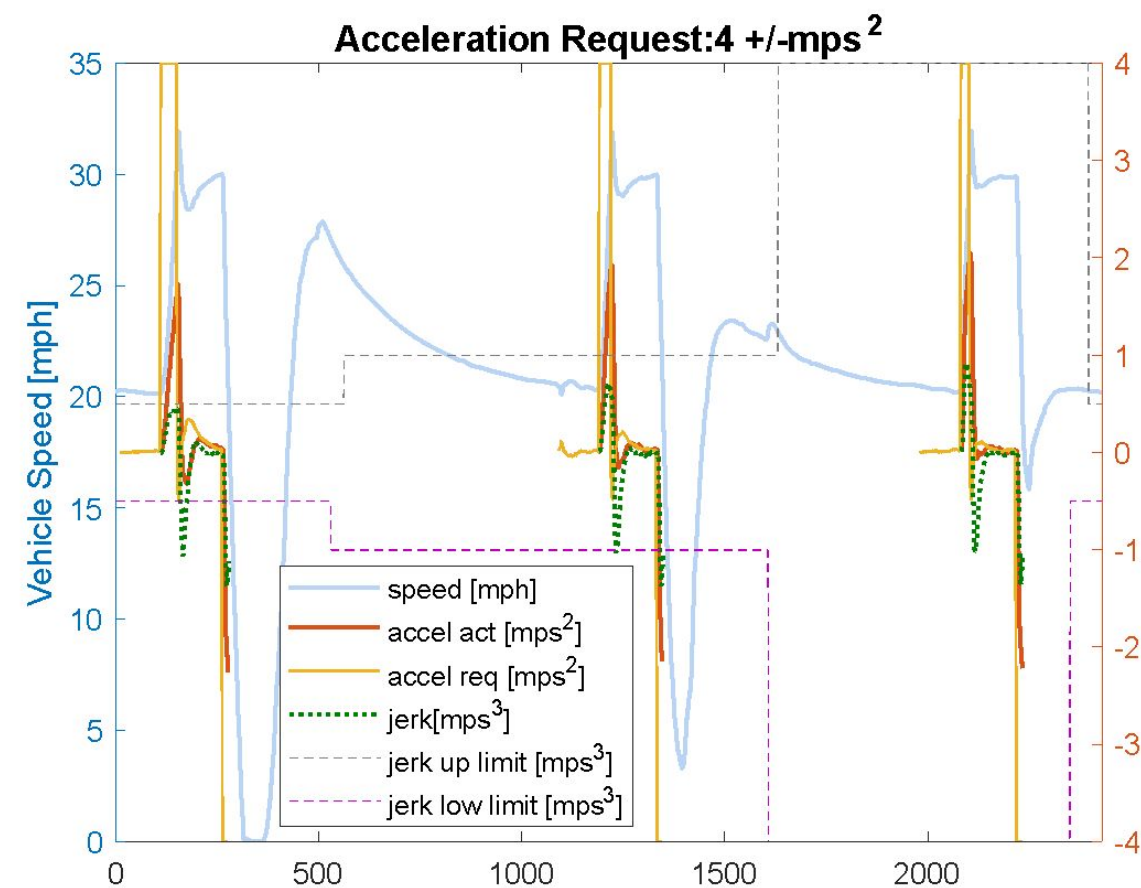


Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.58	-1.98	0.47	-1.06
$\pm 1 \text{ mps}^3$	1.84	-1.91	0.75	-1.03
$\pm 4 \text{ mps}^3$	1.74	-1.89	0.95	-0.99

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

20-30 mph

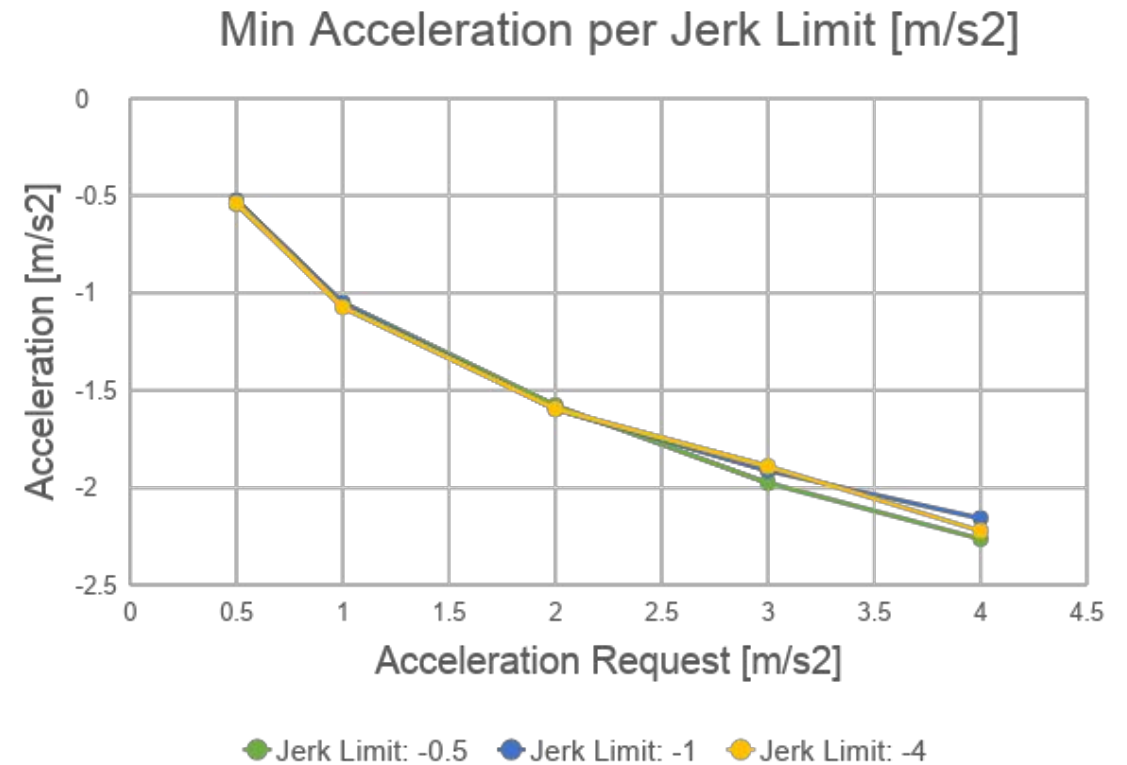
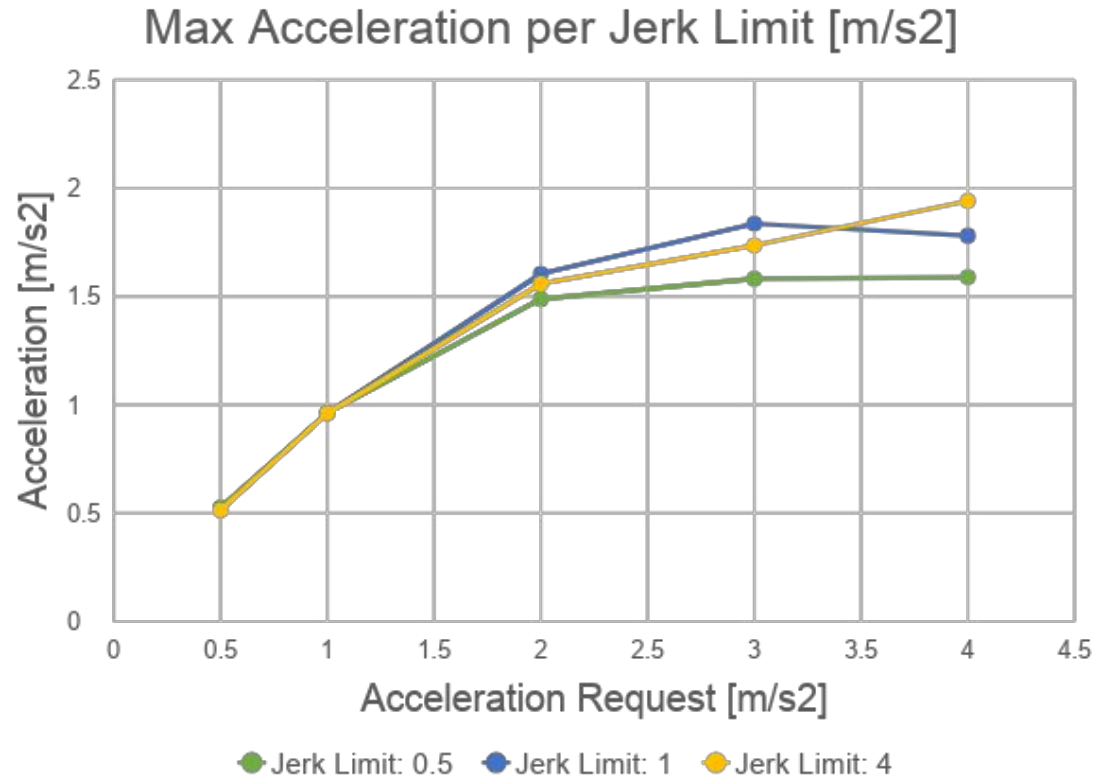


Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.59	-2.26	0.46	-1.37
$\pm 1 \text{ mps}^3$	1.78	-2.16	0.71	-1.37
$\pm 4 \text{ mps}^3$	1.94	-2.22	0.92	-1.36

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

20-30 mph

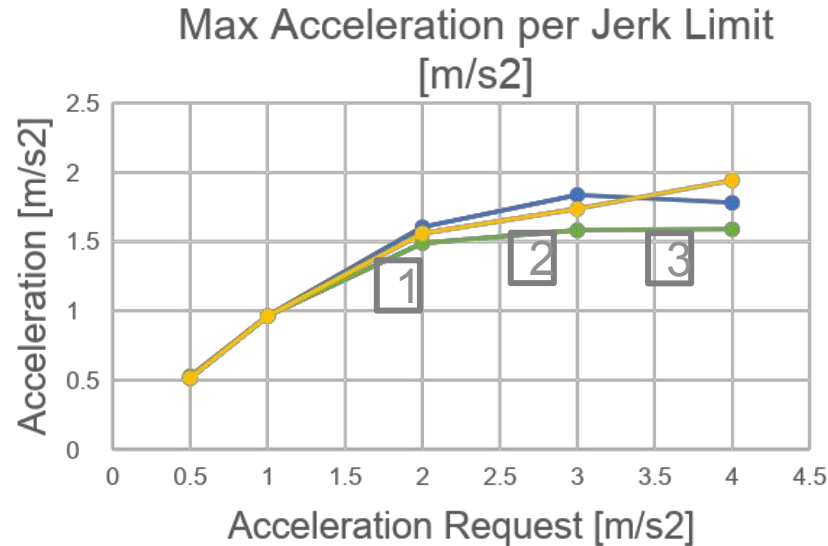


- For the max acceleration the behavior is slightly different from the 10-20 mph case. This is due to the transmission shifting behavior, which limits the max acceleration achievable (discussed in next slide)
- No significant effect of jerk lower limit is observable on the minimum acceleration

Task 1- Expanding the XIL Research Fleet

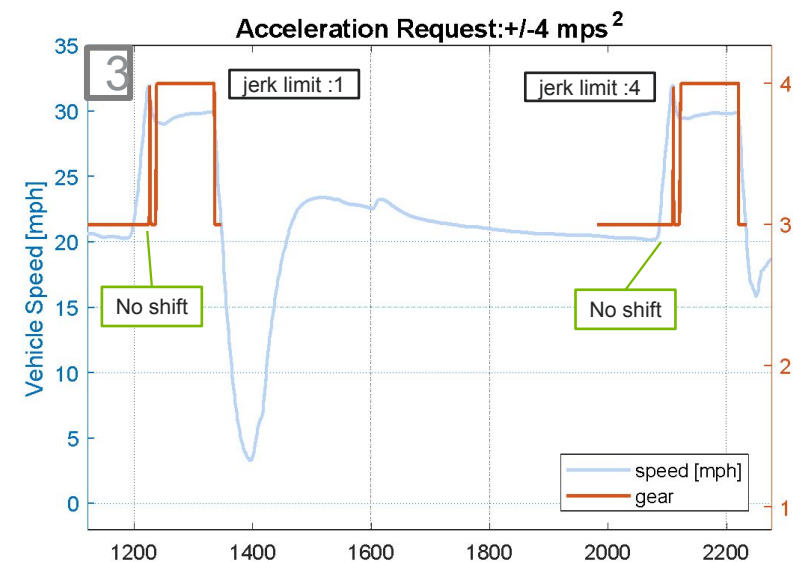
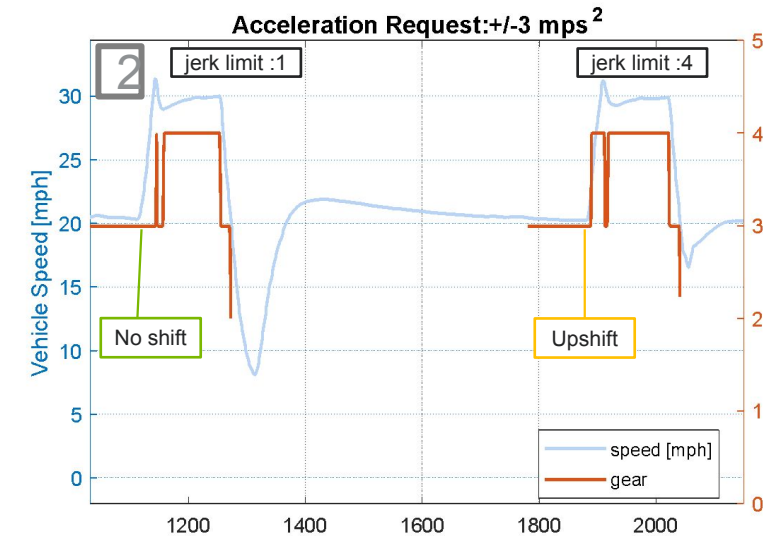
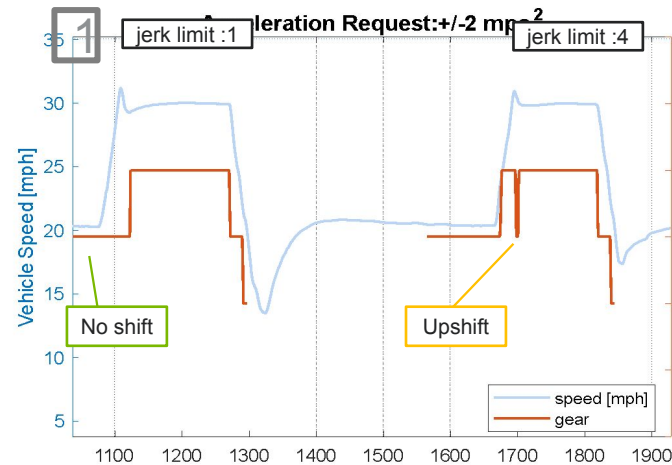
2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

20-30 mph



● Jerk Limit: 0.5 ● Jerk Limit: 1 ● Jerk Limit: 4

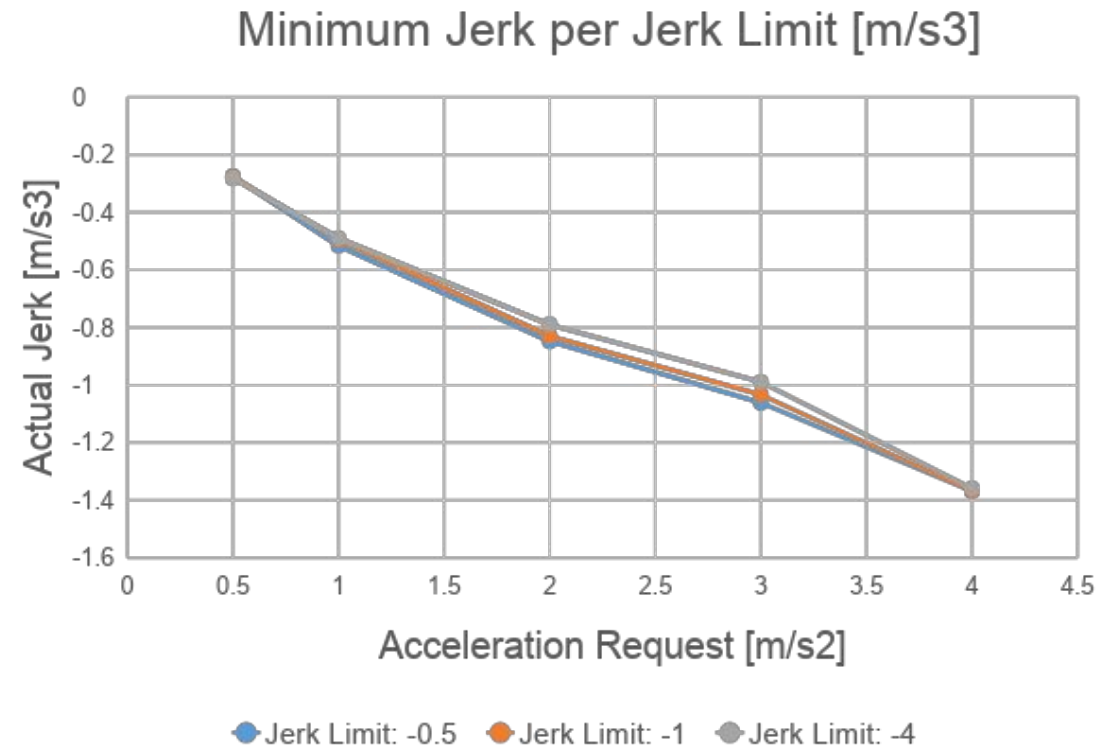
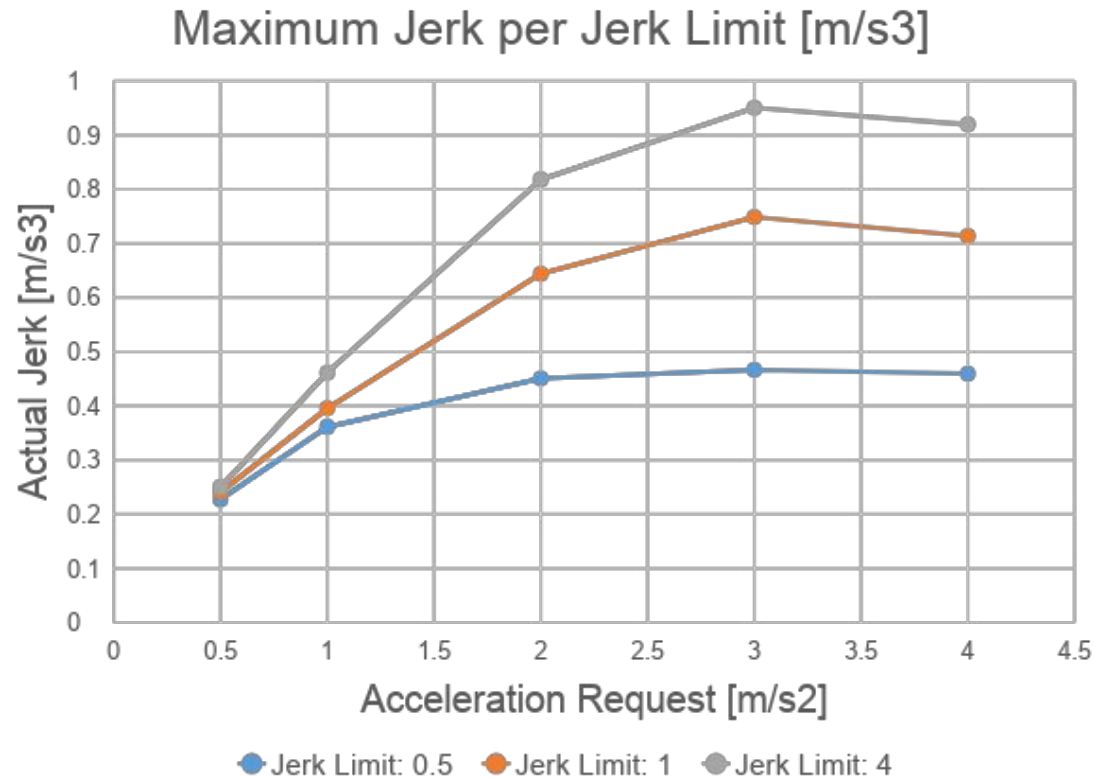
- For [1] and [2] max accel jerk 1 > max accel jerk 4
 - For jerk limit 1 there is no shifting during the acceleration from 20 to 30 mph
 - For jerk limit 4 the transmission upshifts : output torque is lower, this limits max acceleration
- For [3] max accel jerk 1 < max accel jerk 4
 - No shifting in both jerk limit cases



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

20-30 mph

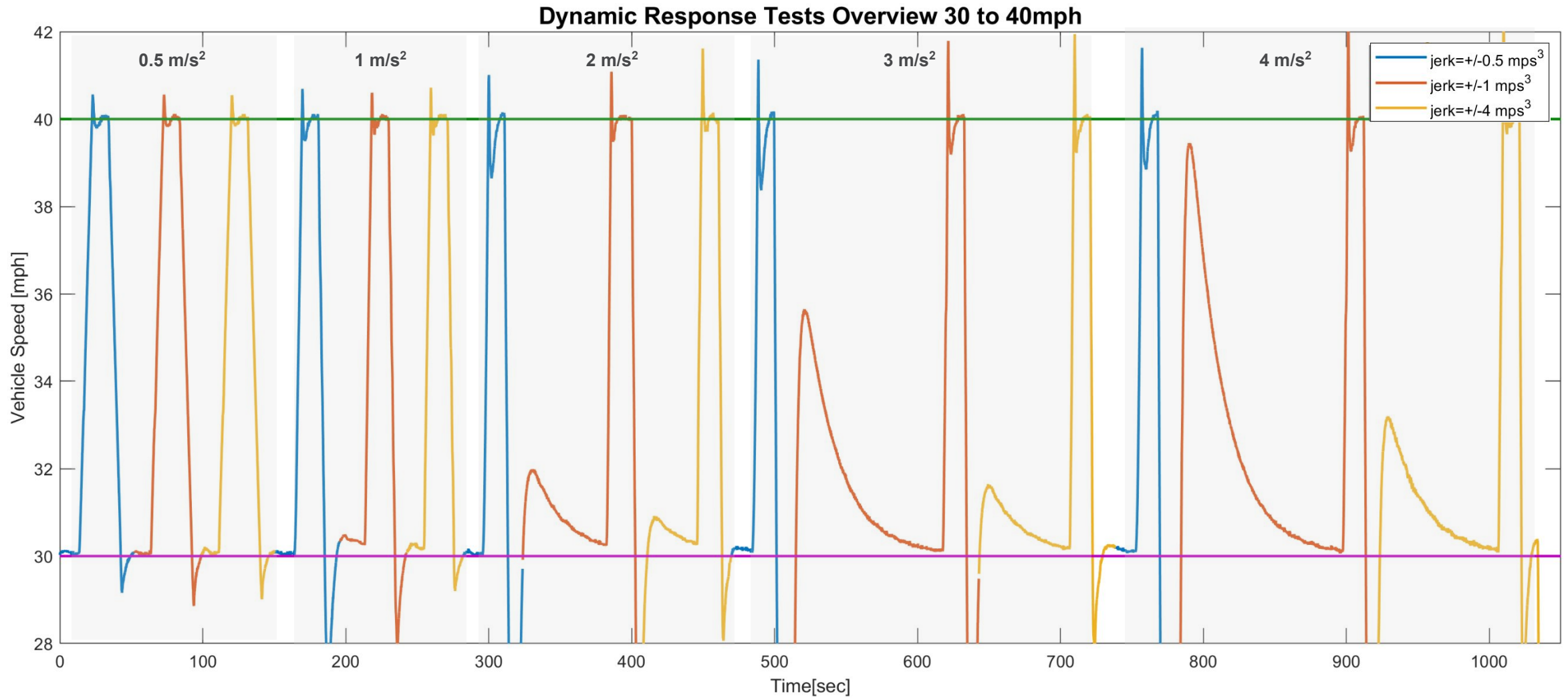


- On the maximum jerk the effect of the jerk limit is visible : the higher the limit the higher the actual jerk, and the difference between the jerk for different limits increases with the acceleration request
- The effect of the jerk limit is not as visible on the minimum jerk

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

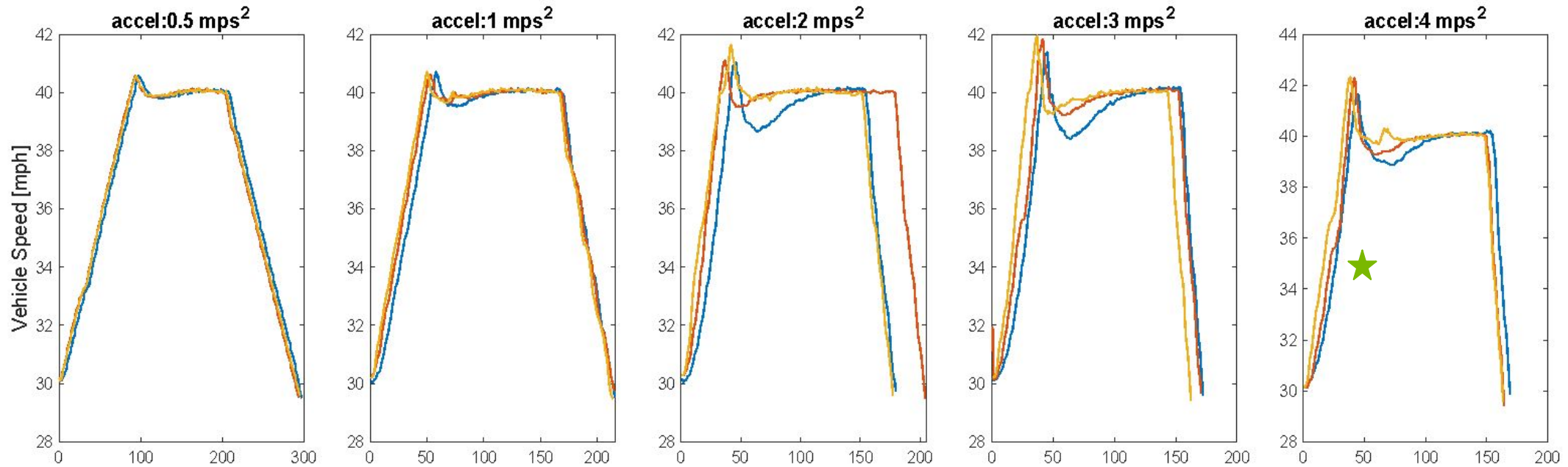
30-40 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

30-40 mph

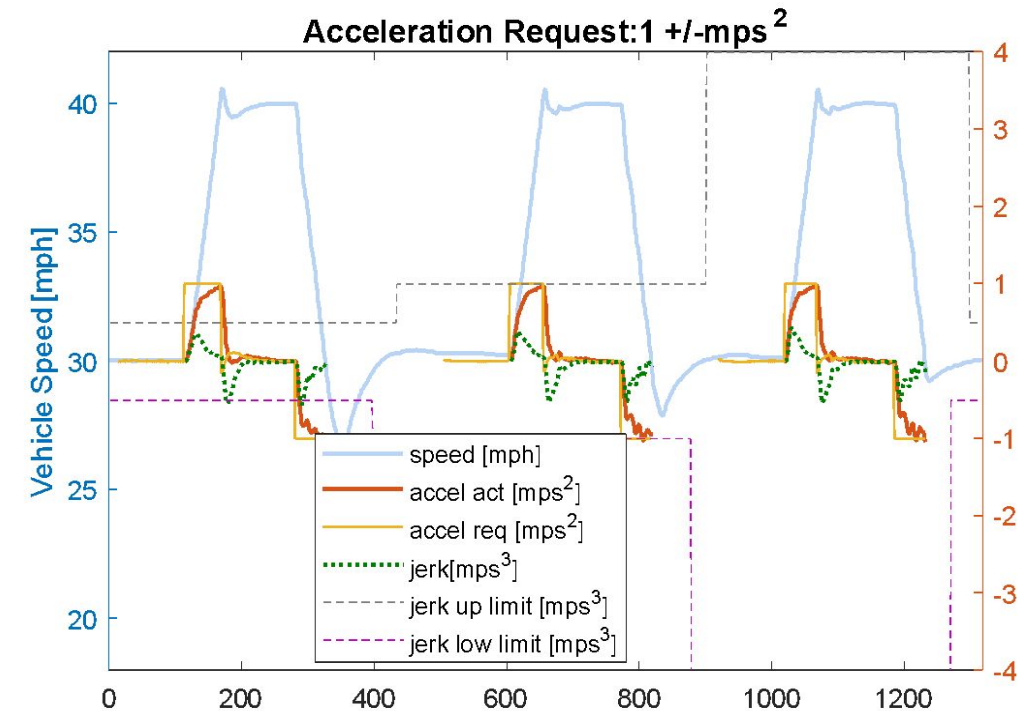
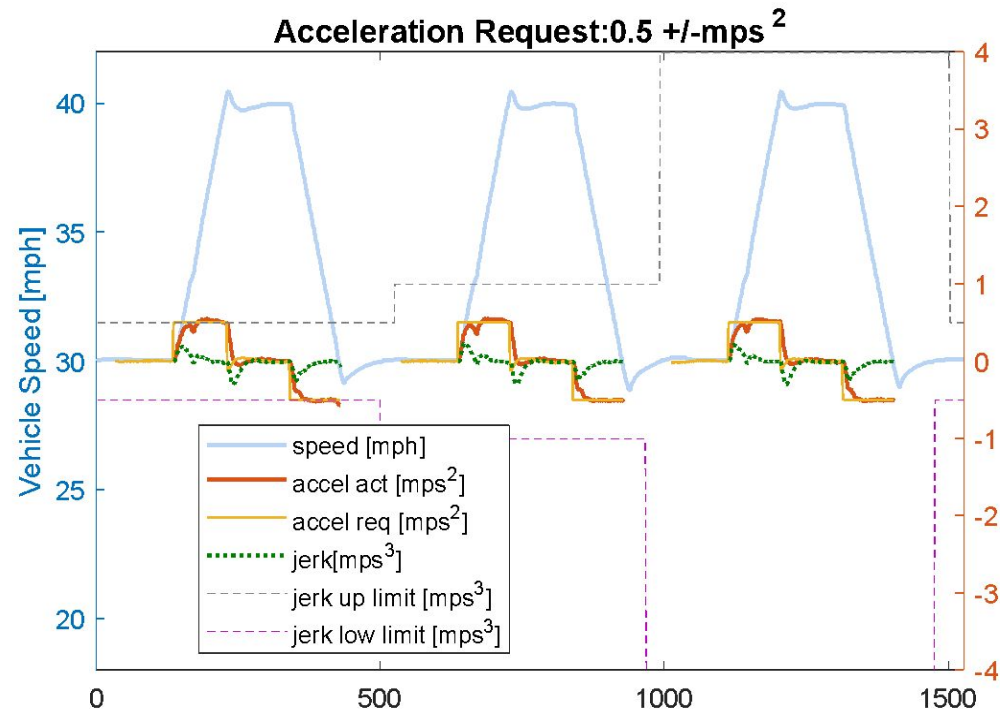


	accel req : 0.5 mps ²		accel req : 1 mps ²		accel req : 2 mps ²		accel req : 3 mps ²		accel req : 4 mps ²	
Jerk Limits	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time
±0.5 mps ³	9.0 s	9.0 s	5.1 s	4.6 s	3.9 s	2.6 s	3.8 s	1.9 s	3.8 s	1.5 s
±1 mps ³	8.7 s	9.1 s	4.7 s	4.6 s	3.1 s	2.6 s	3.5 s	1.9 s	3.3 s	1.5 s
±4 mps ³	8.5 s	9.0 s	4.6 s	4.8 s	3.2 s	2.6 s	2.9 s	1.9 s	3.2 s	1.5 s

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

30-40 mph



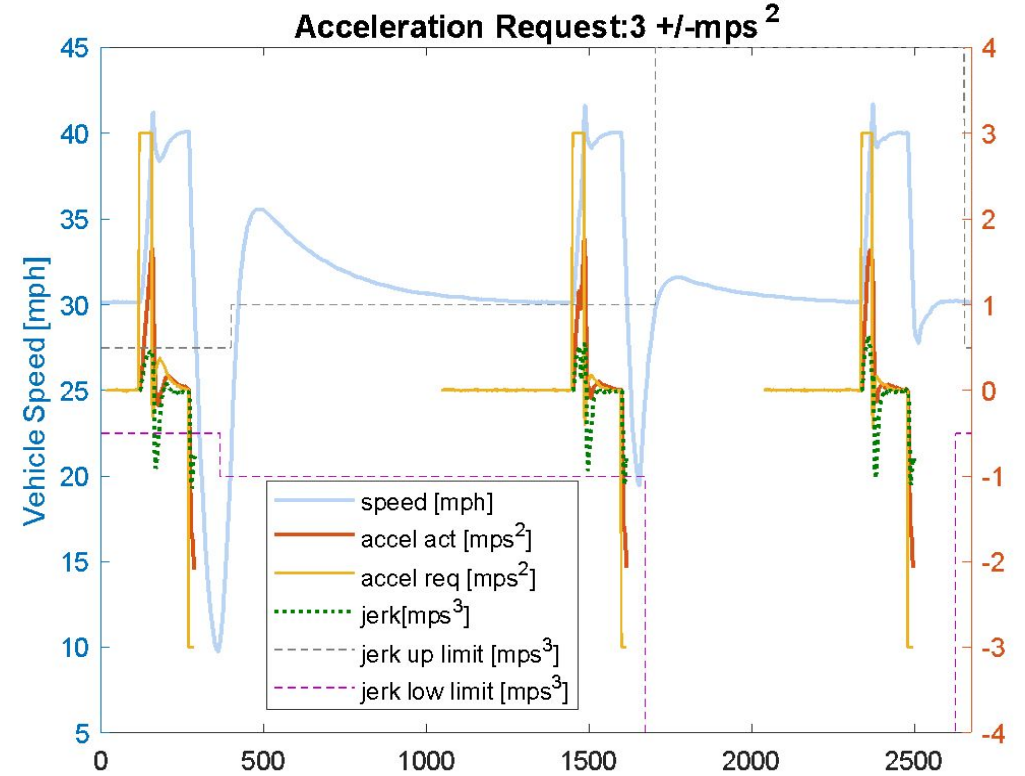
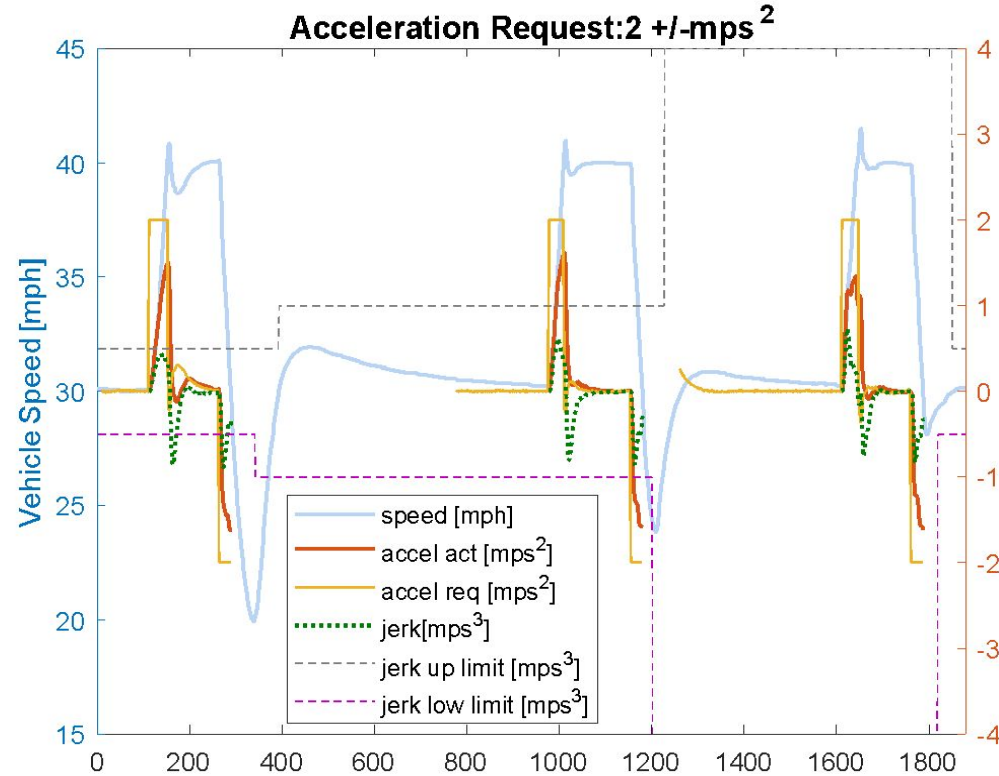
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.55	-0.57	0.20	-0.26
$\pm 1 \text{ mps}^3$	0.55	-0.52	0.24	-0.25
$\pm 4 \text{ mps}^3$	0.54	-0.52	0.22	-0.25

Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.96	-0.99	0.36	-0.56
$\pm 1 \text{ mps}^3$	0.97	-1.03	0.39	-0.52
$\pm 4 \text{ mps}^3$	0.98	-1.04	0.44	-0.51

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

30-40 mph



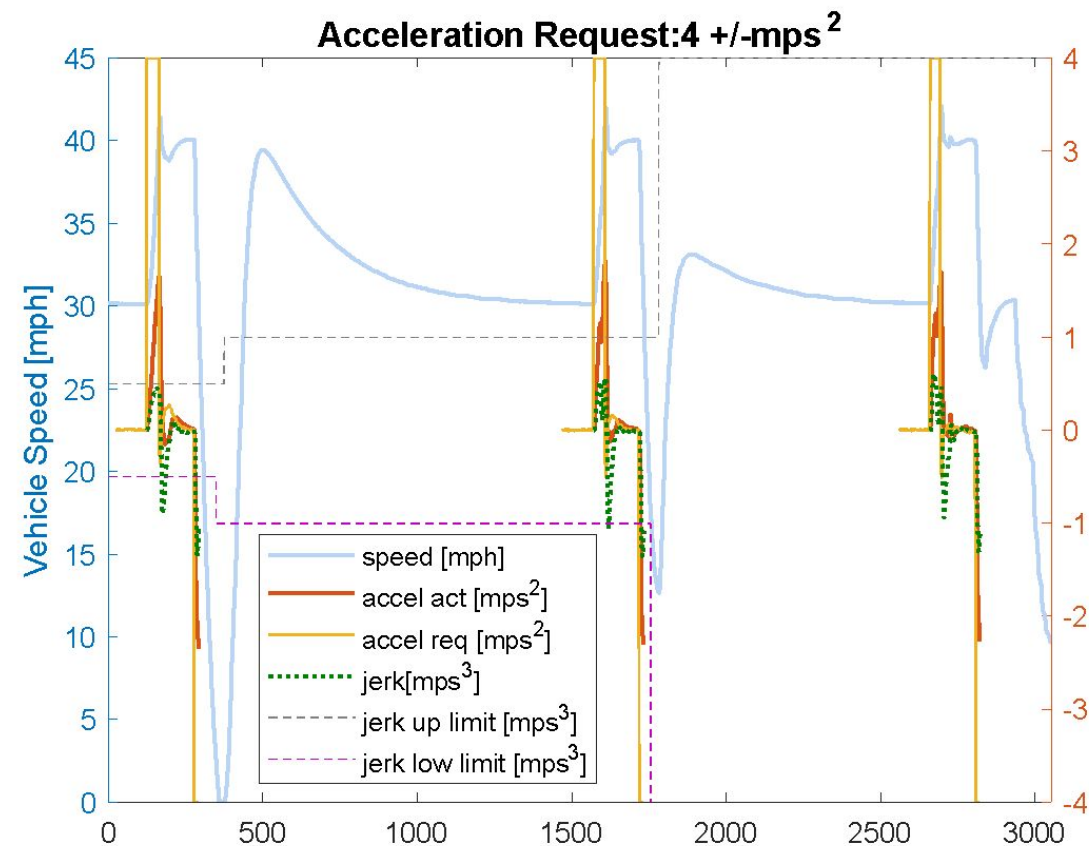
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.48	-1.63	0.43	-0.88
$\pm 1 \text{ mps}^3$	1.57	-1.58	0.62	-0.85
$\pm 4 \text{ mps}^3$	1.34	-1.62	0.74	-0.83

Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.58	-2.10	0.46	-1.14
$\pm 1 \text{ mps}^3$	1.61	-2.08	0.52	-1.10
$\pm 4 \text{ mps}^3$	1.64	-2.08	0.64	-1.10

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

30-40 mph



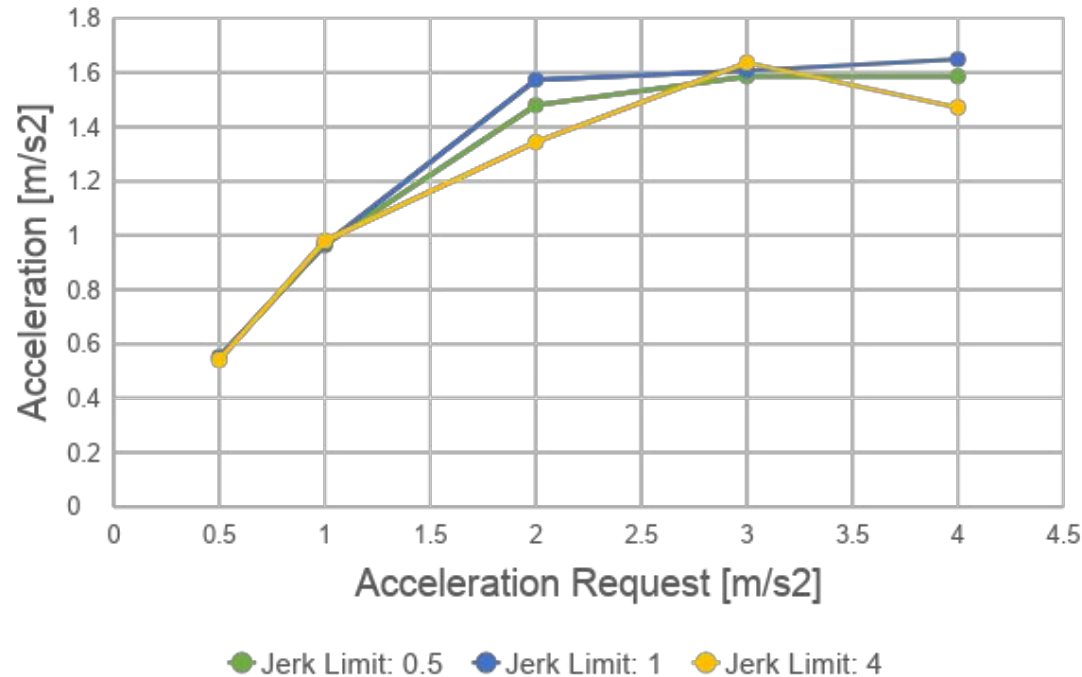
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.59	-2.36	0.45	-1.34
$\pm 1 \text{ mps}^3$	1.65	-2.32	0.57	-1.36
$\pm 4 \text{ mps}^3$	1.47	-2.28	0.61	-1.30

Task 1- Expanding the XIL Research Fleet

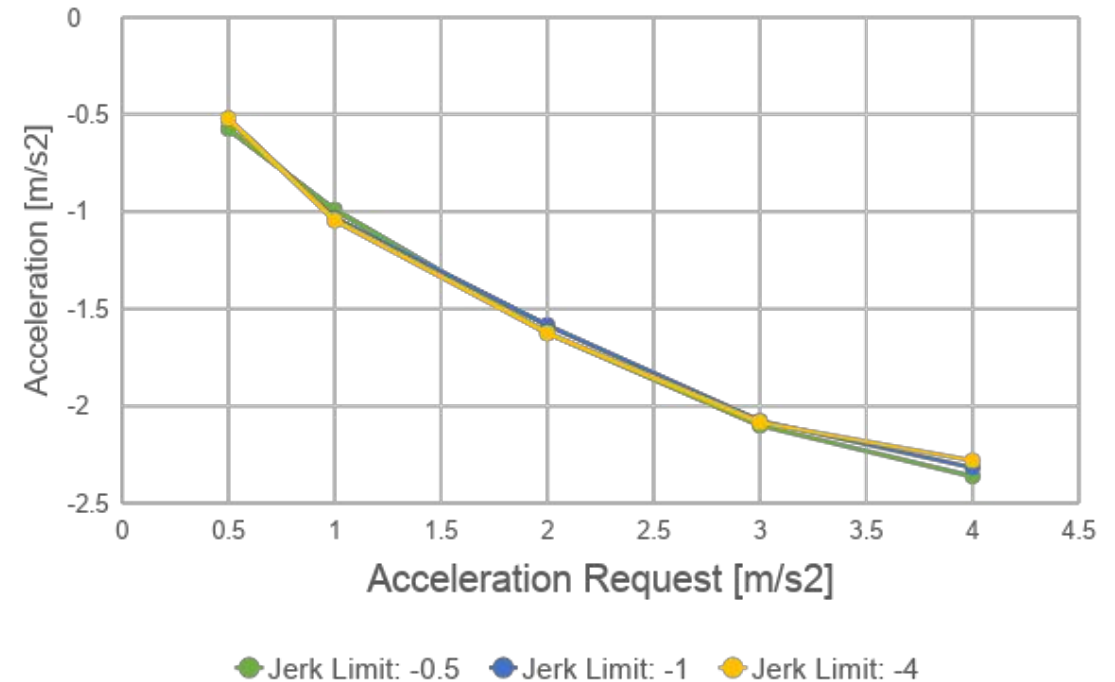
2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

30- 40 mph

Max Acceleration per Jerk Limit [m/s²]



Min Acceleration per Jerk Limit [m/s²]

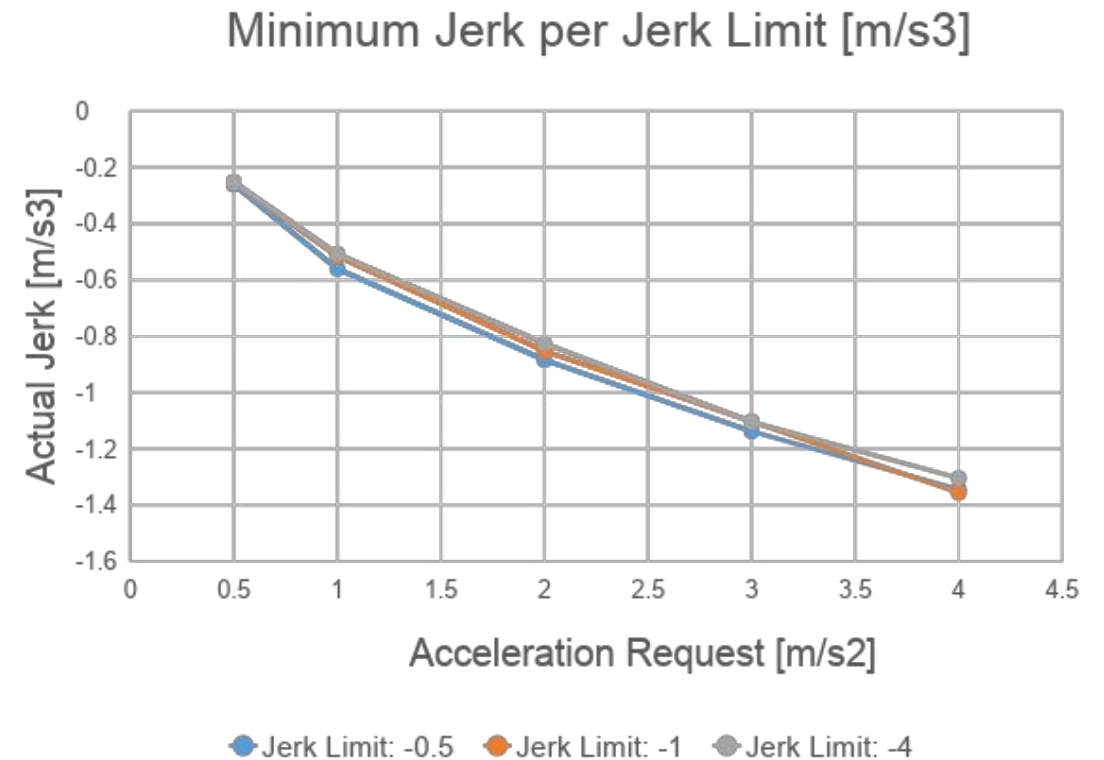
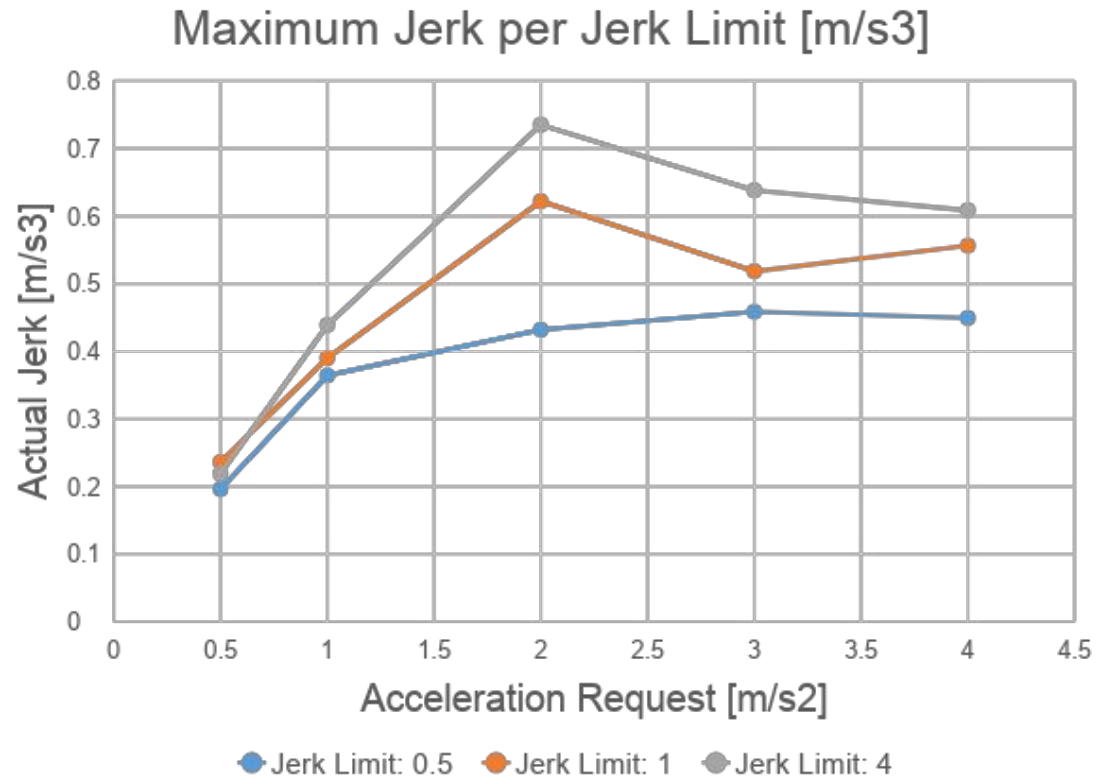


- For acceleration requests 0.5, 1 and 2 mps², the max acceleration for jerk limit 4 is lower than in the other two cases due to trans upshifting
- For acceleration request 3 mps² the max acceleration is similar for all cases since in every case the transmission downshifts at the beginning of the acceleration maneuver
- For acceleration request 4 mps² the max acceleration for the case of jerk 4 is lower because after the downshift happens when the vehicle speed is closer to the target speed and the vehicle does not need to accelerate much more

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

30- 40 mph

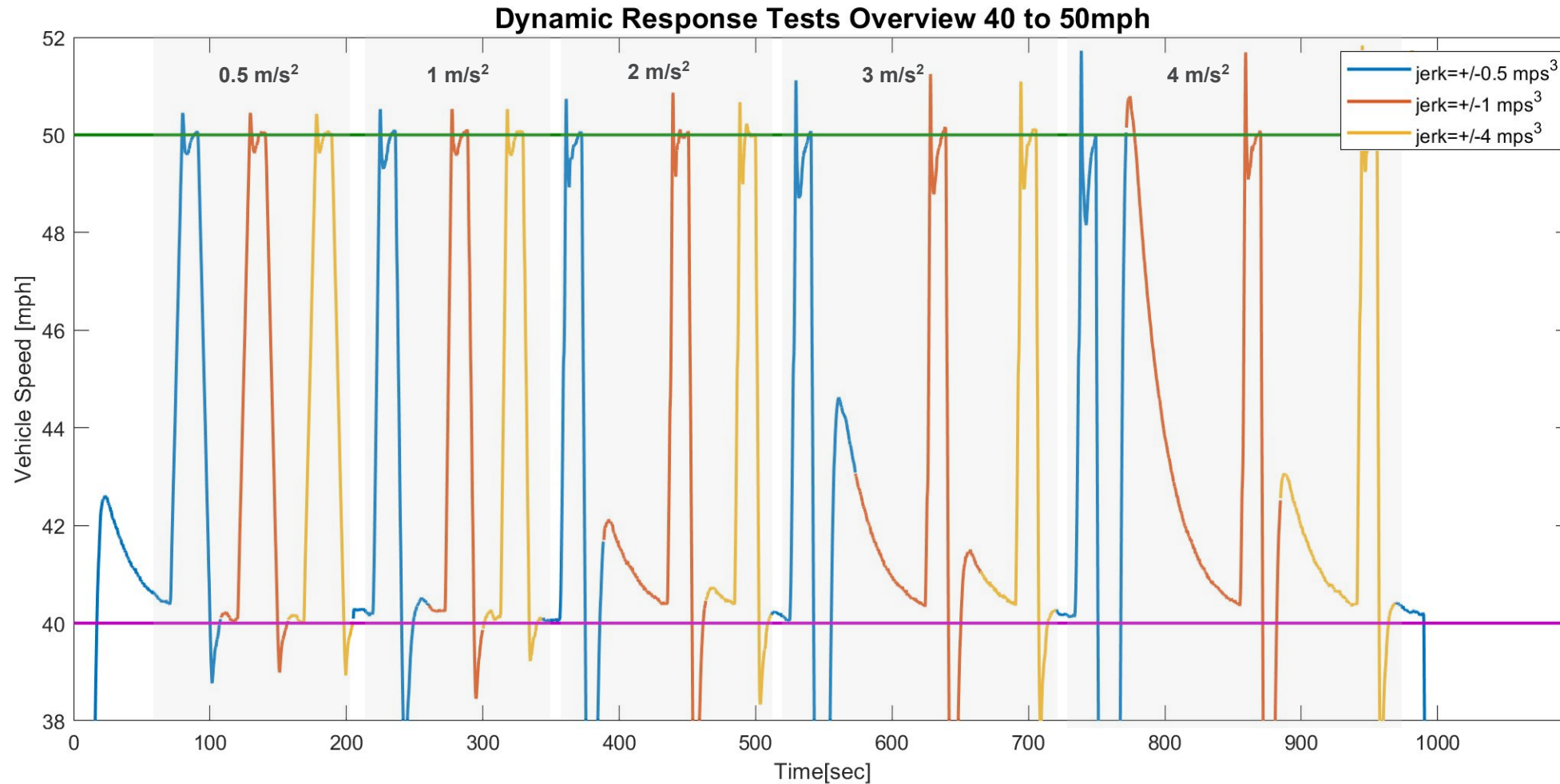


- For the maximum jerk, the downshift at 3 mps² causes an inversion in the trend of the actual jerk
 - The maximum actual jerk is lower than in the previous acceleration case, and decreases more for the jerk limit 1 mps³, in that case the transmission shifts from 5th to 3rd gear, while in the case of jerk limit 4 mps³ the transmission shifts from 4th to 3rd

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

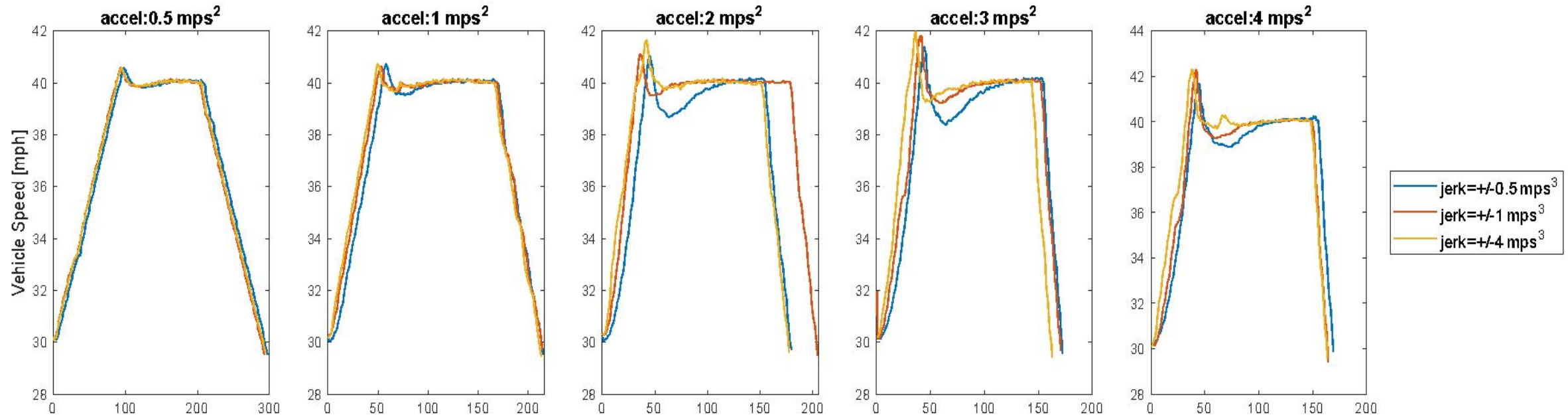
40-50 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

40-50 mph

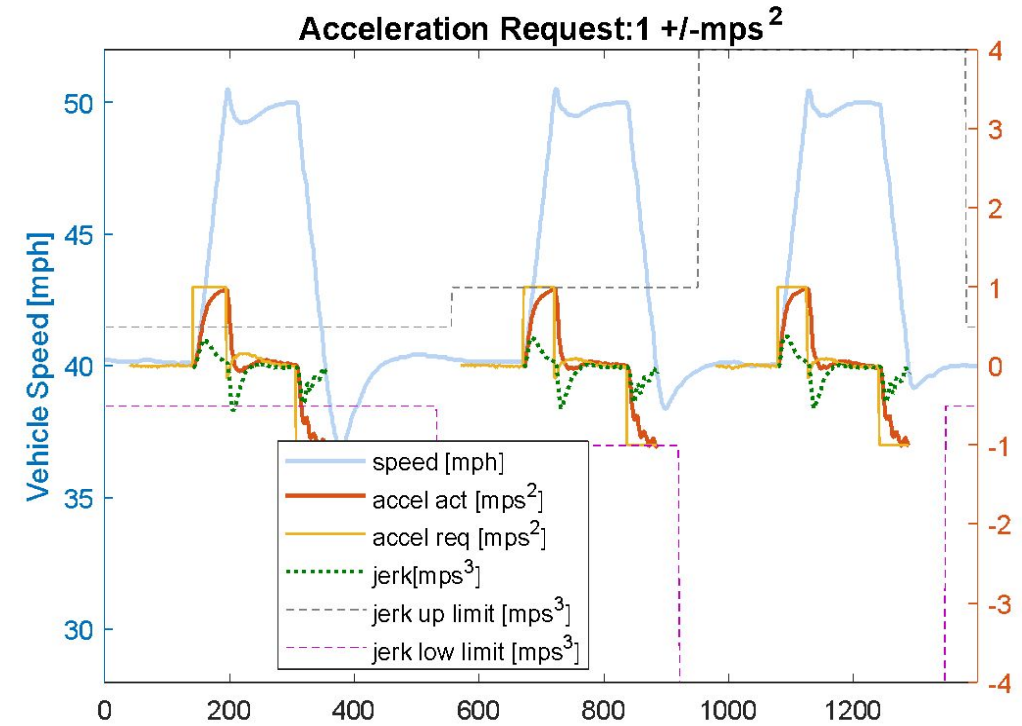
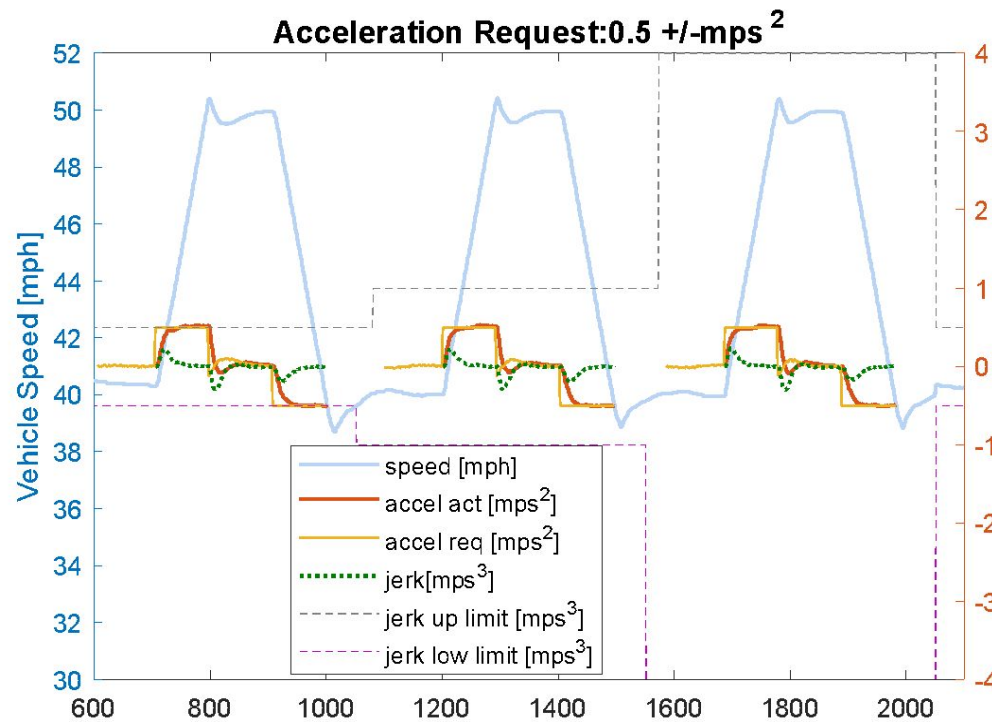


	accel req : 0.5 mps^2		accel req : 1 mps^2		accel req : 2 mps^2		accel req : 3 mps^2		accel req : 4 mps^2	
Jerk Limits	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time
$\pm 0.5 \text{ mps}^3$	8.8 s	9.5 s	5.3 s	4.8 s	4.5 s	2.7 s	4.4 s	1.9 s	4.4 s	1.4 s
$\pm 1 \text{ mps}^3$	8.7 s	9.3 s	4.9 s	4.8 s	3.9 s	2.7 s	3.7 s	1.9 s	3.7 s	1.4 s
$\pm 4 \text{ mps}^3$	8.8 s	9.5 s	4.6 s	4.9 s	3.7 s	2.7 s	3.7 s	1.9 s	3.5 s	1.4 s

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

40-50 mph



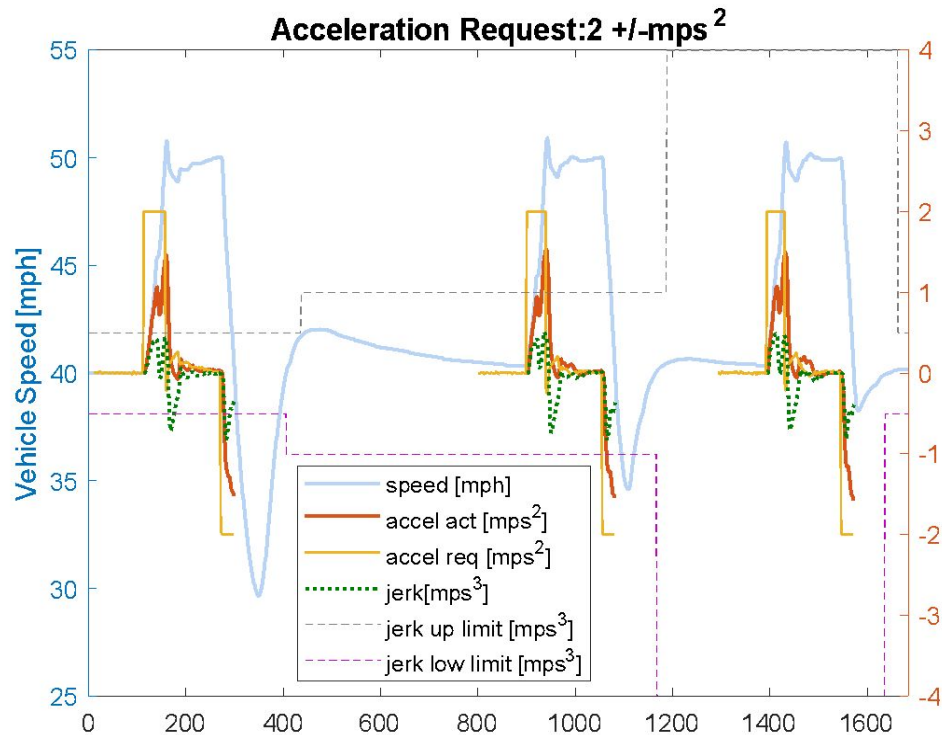
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.52	-0.52	0.22	-0.21
$\pm 1 \text{ mps}^3$	0.52	-0.52	0.22	-0.20
$\pm 4 \text{ mps}^3$	0.52	-0.52	0.23	-0.18

Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.96	-1.02	0.33	-0.47
$\pm 1 \text{ mps}^3$	0.97	-1.02	0.37	-0.45
$\pm 4 \text{ mps}^3$	0.98	-1.01	0.38	-0.45

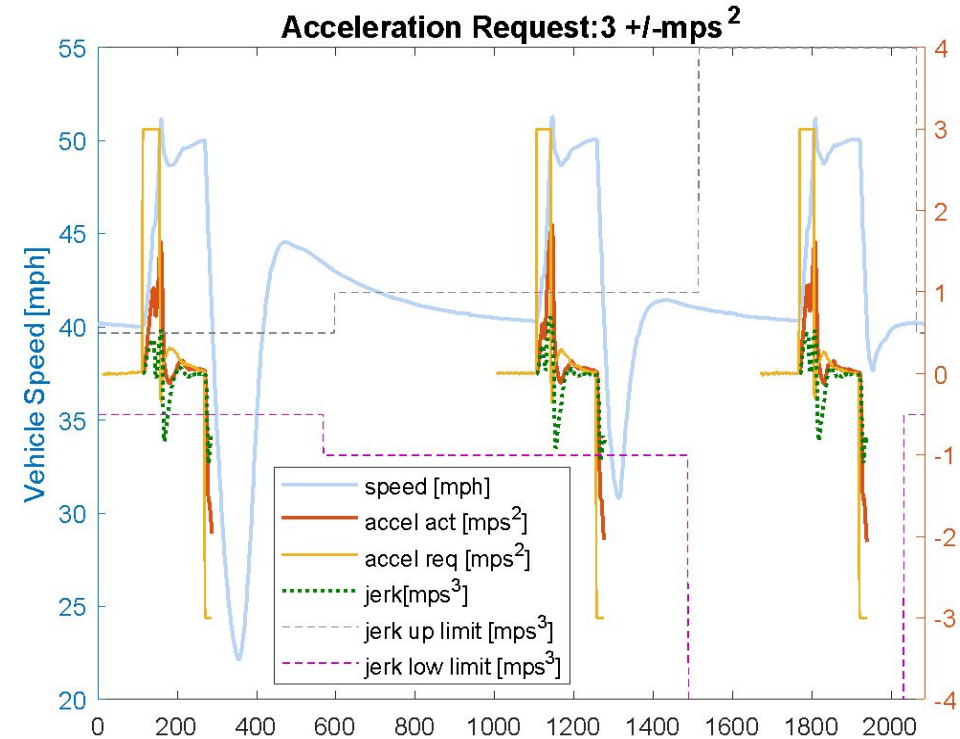
Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

40-50 mph



Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.39	-1.53	0.44	-0.82
$\pm 1 \text{ mps}^3$	1.48	-1.55	0.51	-0.83
$\pm 4 \text{ mps}^3$	1.41	-1.58	0.49	-0.81

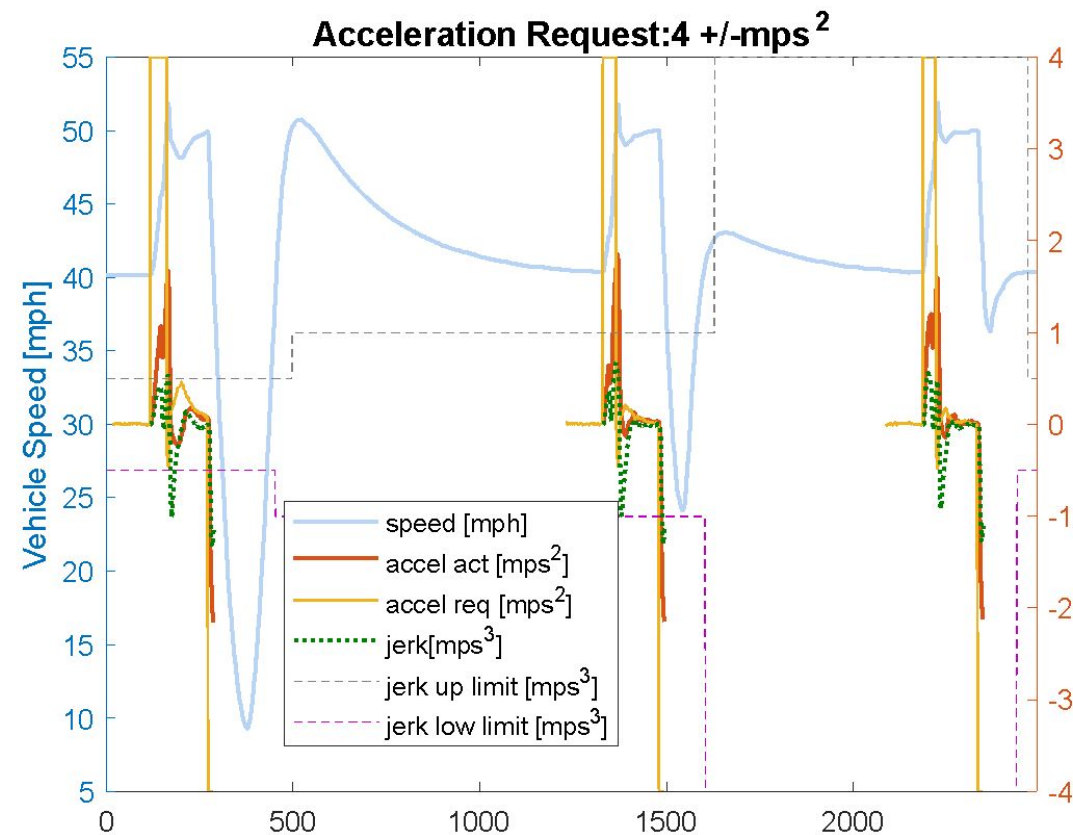


Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.40	-1.97	0.48	-1.10
$\pm 1 \text{ mps}^3$	1.68	-2.05	0.69	-1.01
$\pm 4 \text{ mps}^3$	1.39	-2.07	0.49	-1.10

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

40-50 mph

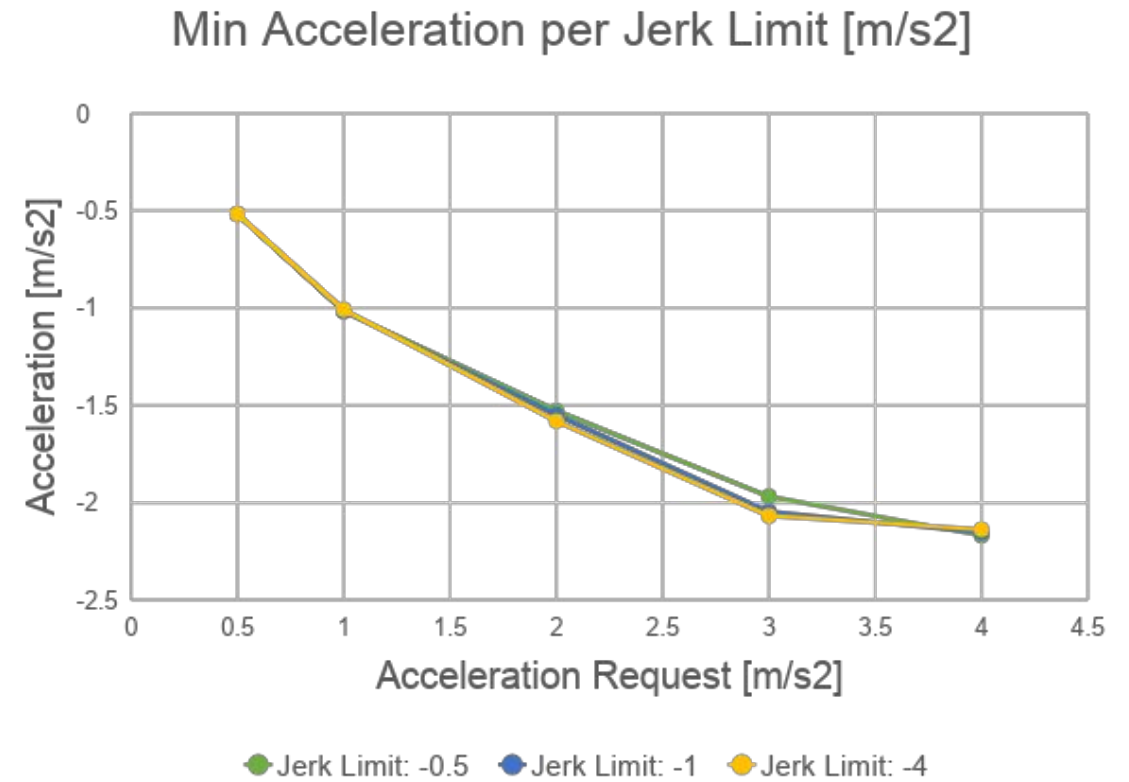
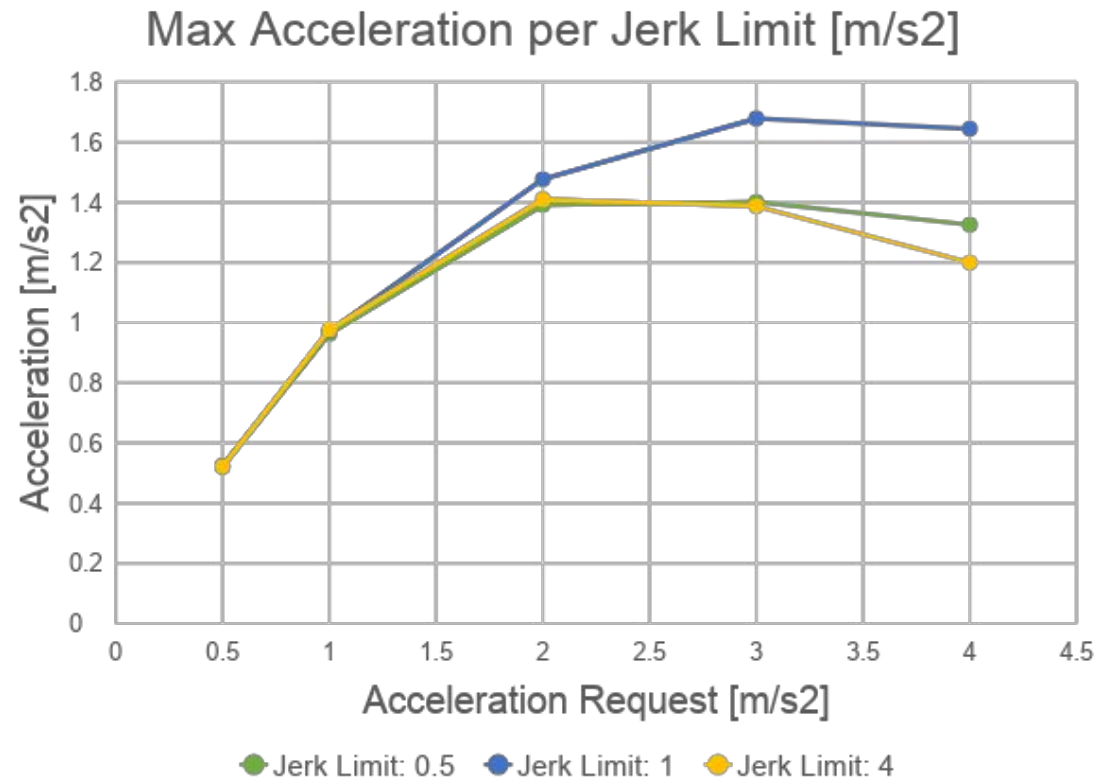


Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.33	-2.17	0.43	-1.33
$\pm 1 \text{ mps}^3$	1.64	-2.15	0.67	-1.30
$\pm 4 \text{ mps}^3$	1.20	-2.14	0.56	-1.28

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

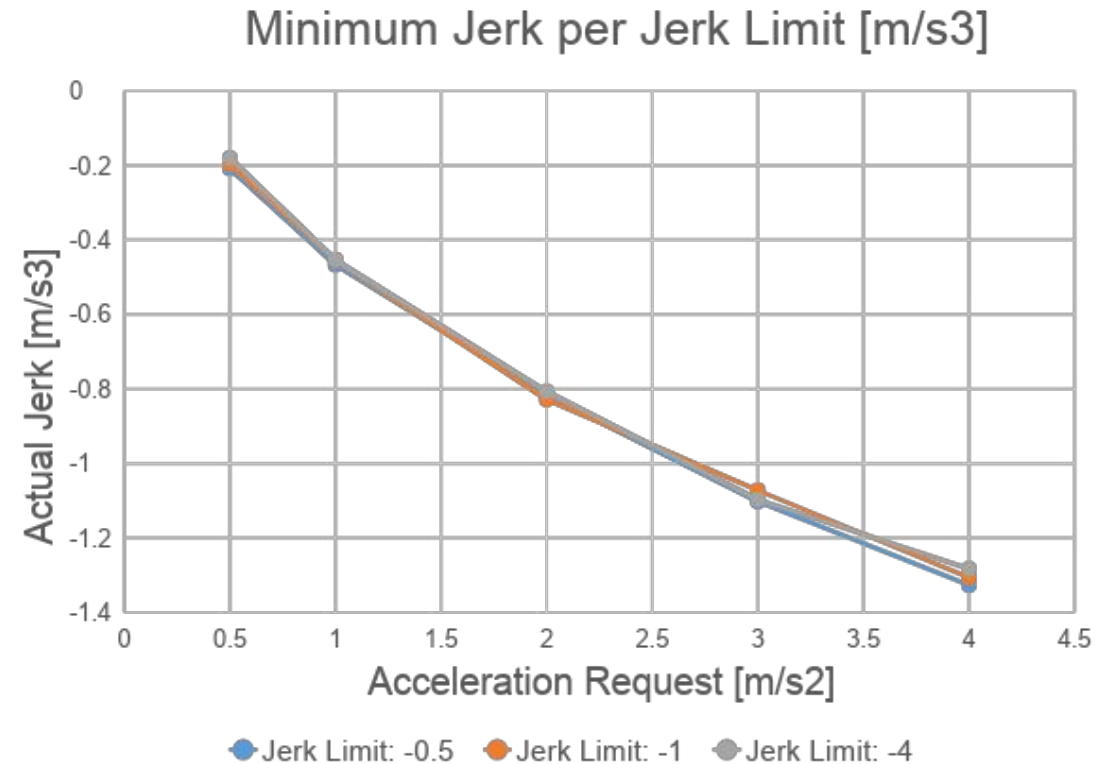
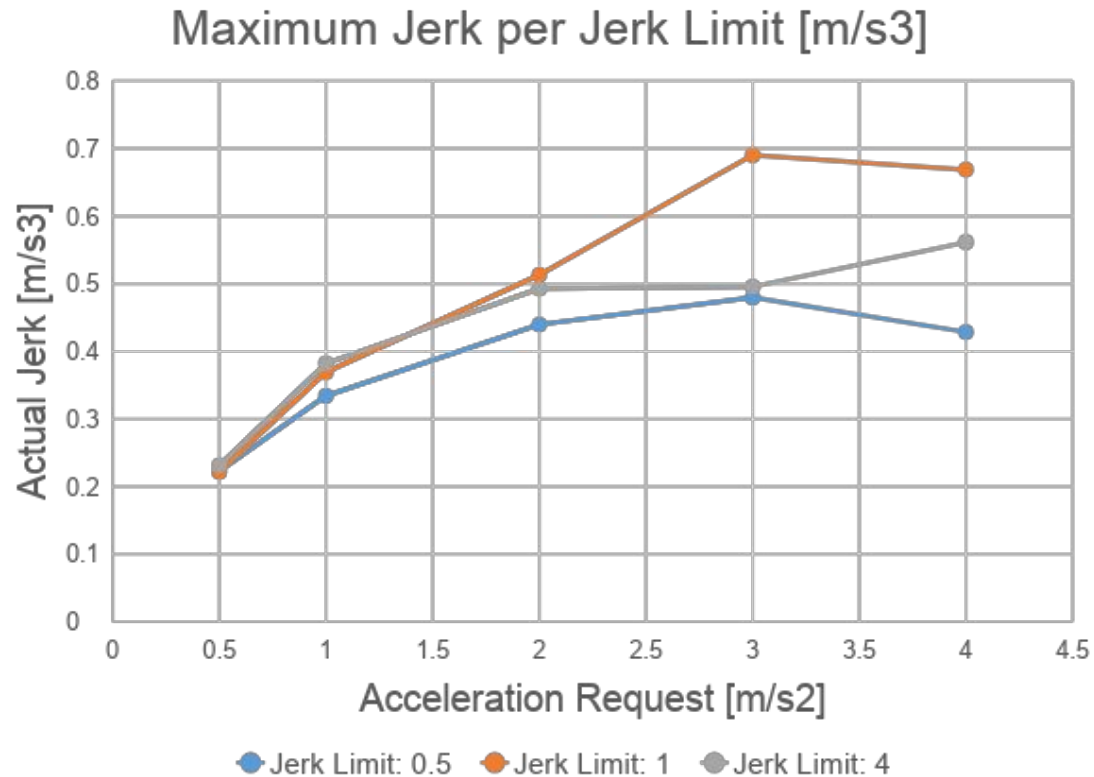
40-50 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

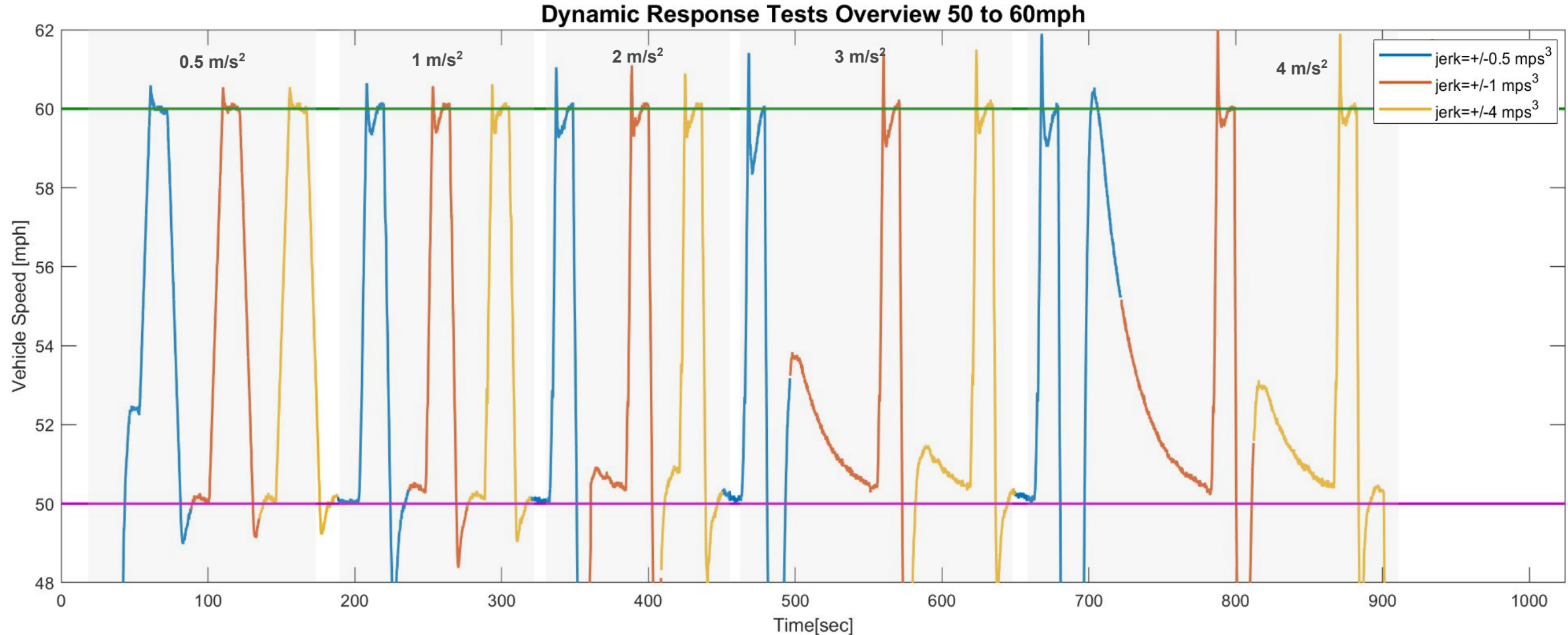
40-50 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

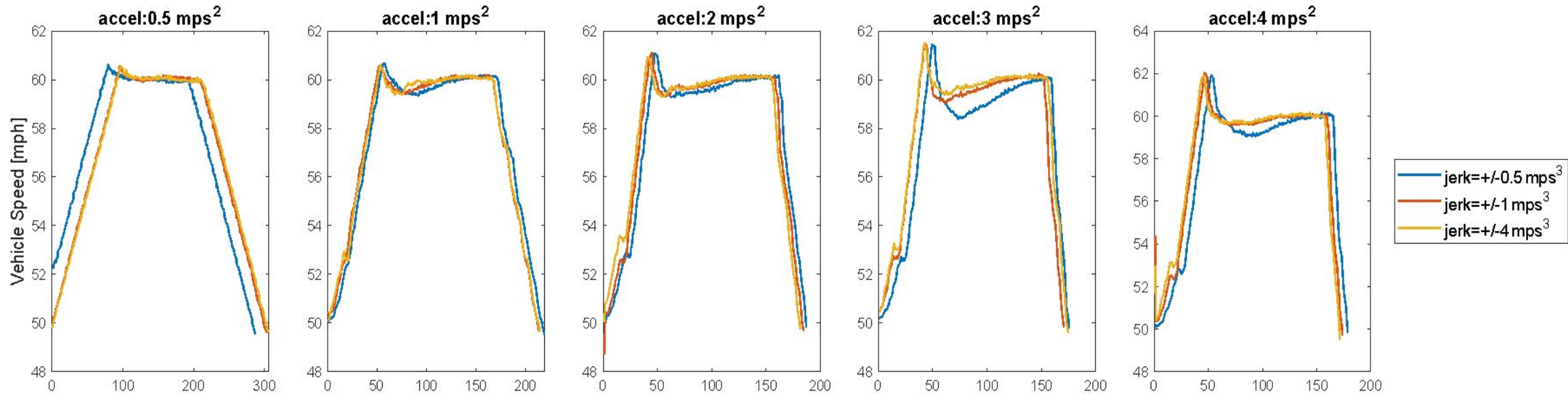
50-60 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

50-60 mph



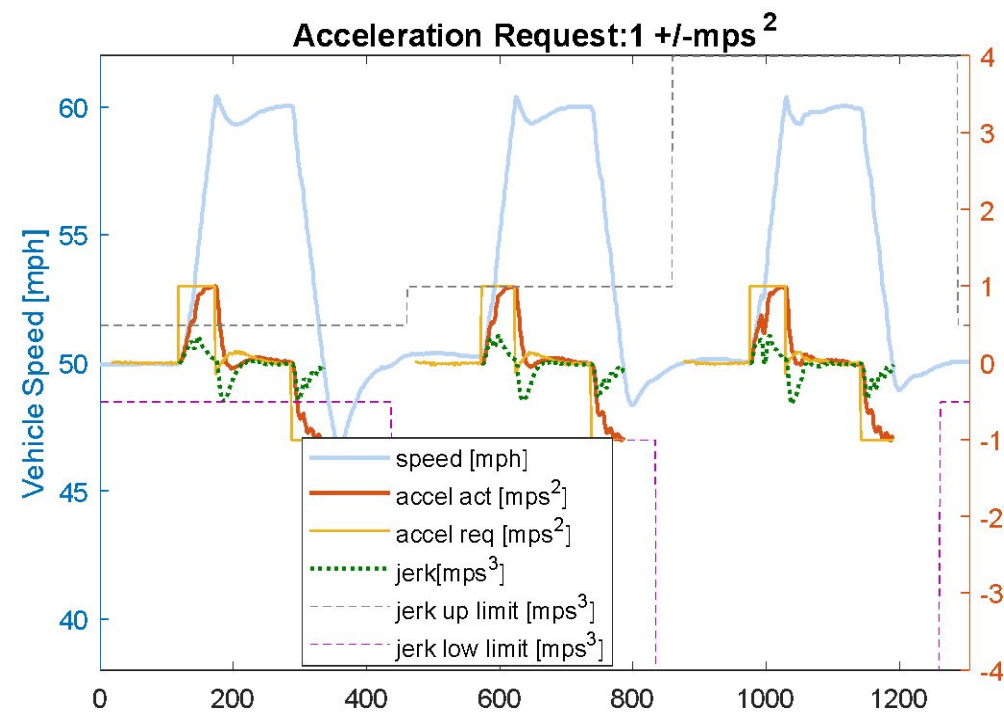
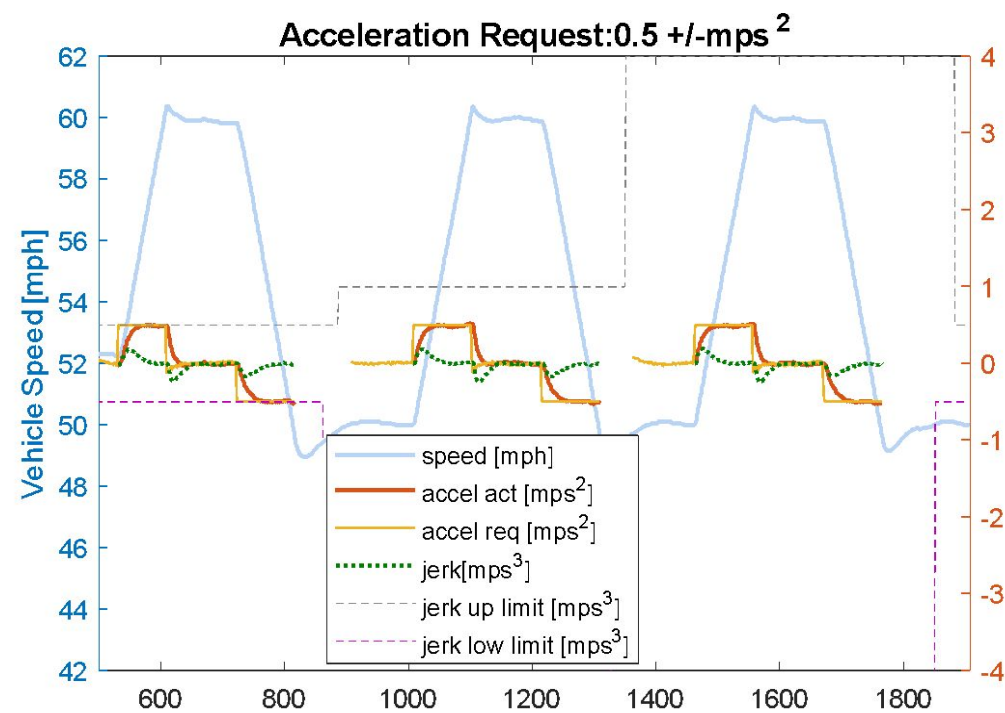
	accel req : 0.5 mps^2		accel req : 1 mps^2		accel req : 2 mps^2		accel req : 3 mps^2		accel req : 4 mps^2	
Jerk Limits	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time
$\pm 0.5 \text{ mps}^3$	7.4 s*	9.4 s	5.4 s	4.9 s	4.6 s	2.6 s	4.6 s	1.7 s	4.6 s	1.5 s
$\pm 1 \text{ mps}^3$	9.1 s	9.4 s	4.8 s	4.8 s	4.0 s	2.6 s	4.0 s	1.9 s	4.0 s	1.5 s
$\pm 4 \text{ mps}^3$	9.1 s	9.4 s	5.1 s	4.9 s	3.7 s	2.6 s	3.8 s	1.9 s	3.8 s	1.5 s

*this result is affected by the fact that the test has been started at 52 mph instead of 50 mph

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

50-60 mph



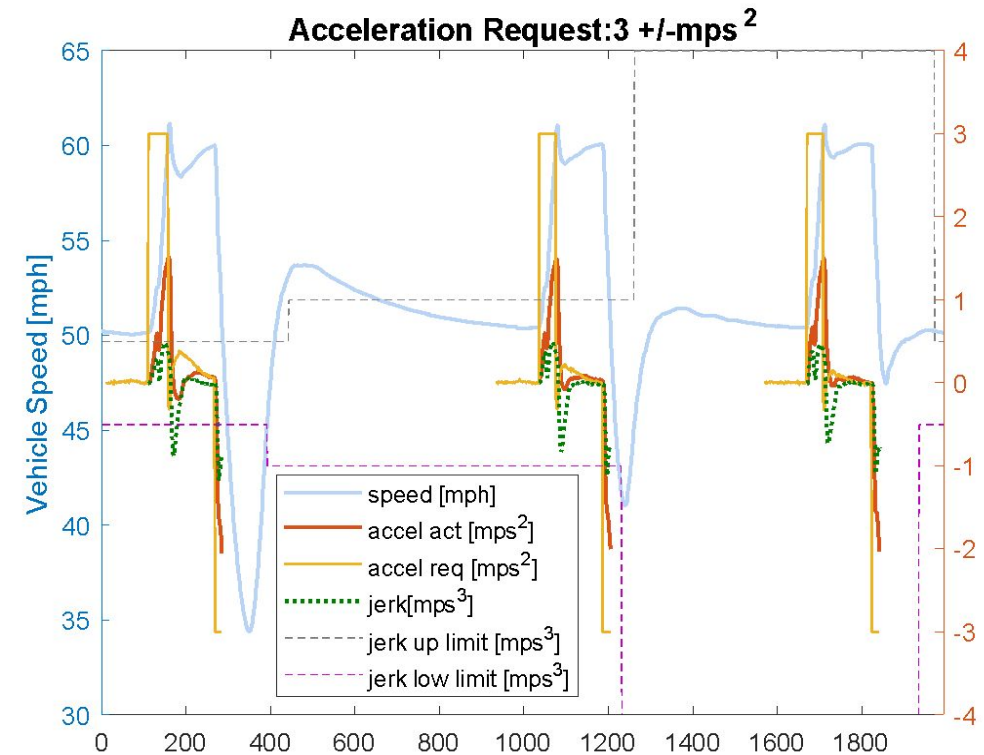
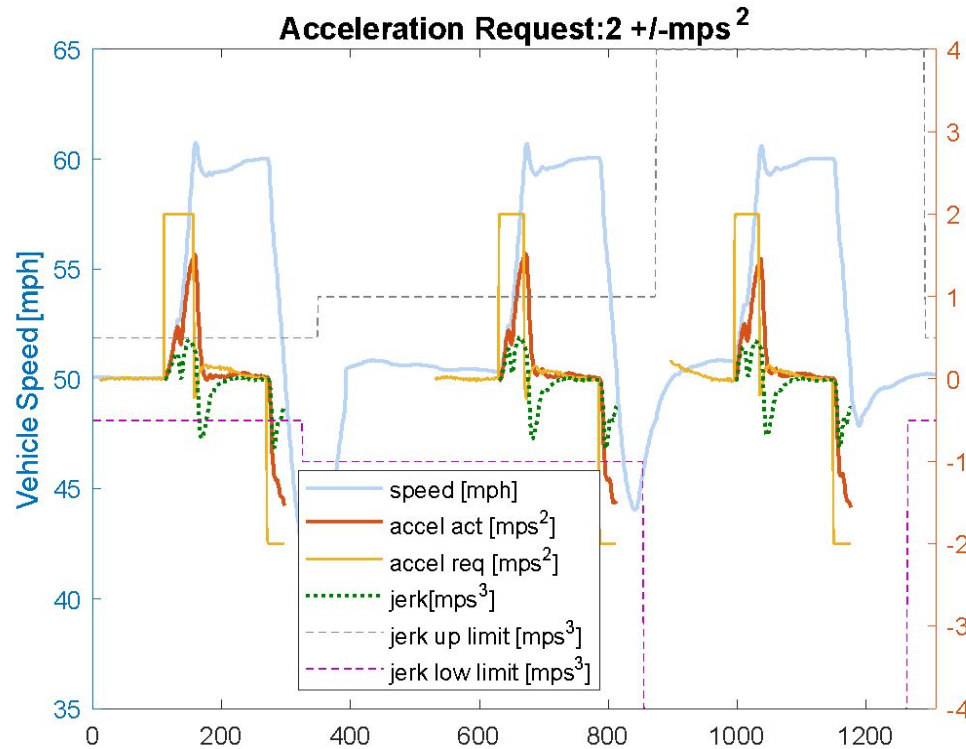
Jerk Limits	Max Accel [mps ²]	Min Accel [mps ²]	Max Jerk [mps ³]	Min Jerk [mps ³]
±0.5 mps ³	0.50	-0.54	0.19	-0.18
±1 mps ³	0.52	-0.53	0.19	-0.17
±4 mps ³	0.51	-0.53	0.20	-0.16

Jerk Limits	Max Accel [mps ²]	Min Accel [mps ²]	Max Jerk [mps ³]	Min Jerk [mps ³]
±0.5 mps ³	1.00	-1.01	0.35	-0.47
±1 mps ³	1.00	-1.01	0.37	-0.45
±4 mps ³	1.00	-1.01	0.36	-0.46

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

50-60 mph



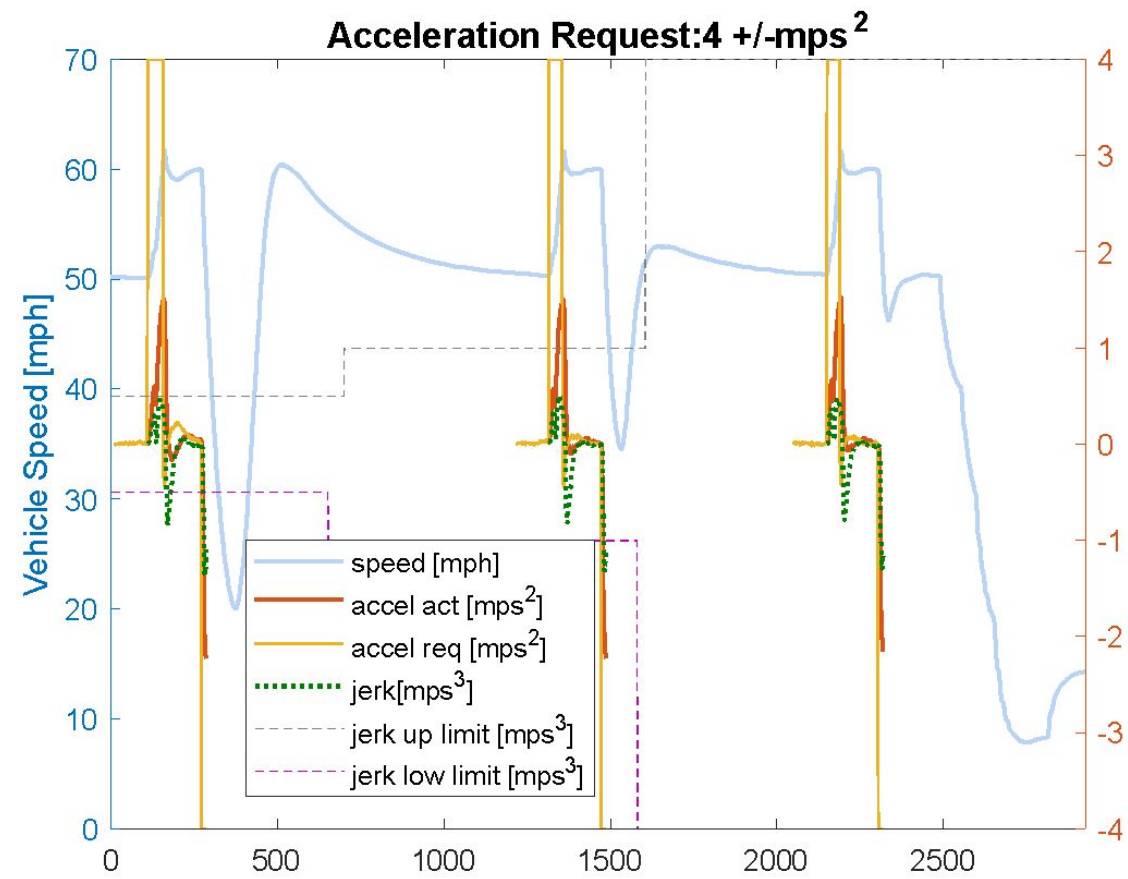
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.45	-1.53	0.47	-0.84
$\pm 1 \text{ mps}^3$	1.48	-1.52	0.511	-0.85
$\pm 4 \text{ mps}^3$	1.38	-1.56	0.46	-0.83

Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.45	-2.06	0.49	-1.17
$\pm 1 \text{ mps}^3$	1.42	-2.00	0.49	-1.12
$\pm 4 \text{ mps}^3$	1.44	-2.04	0.47	-1.10

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

50-60 mph

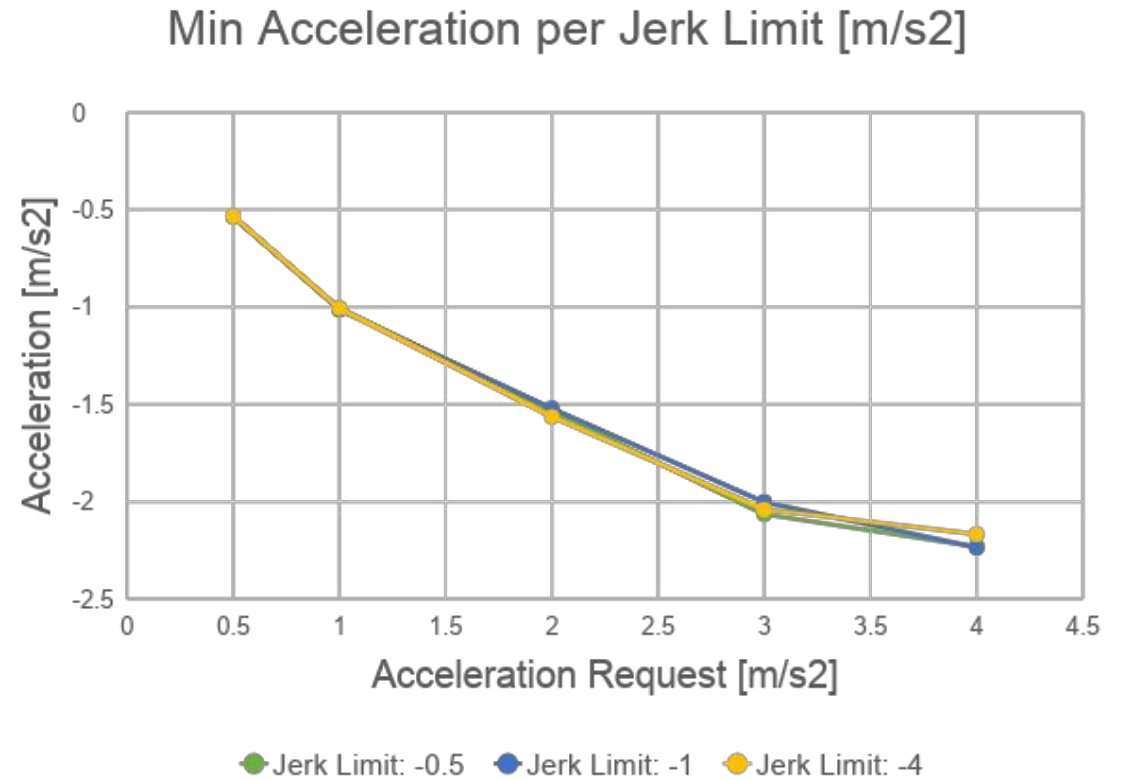
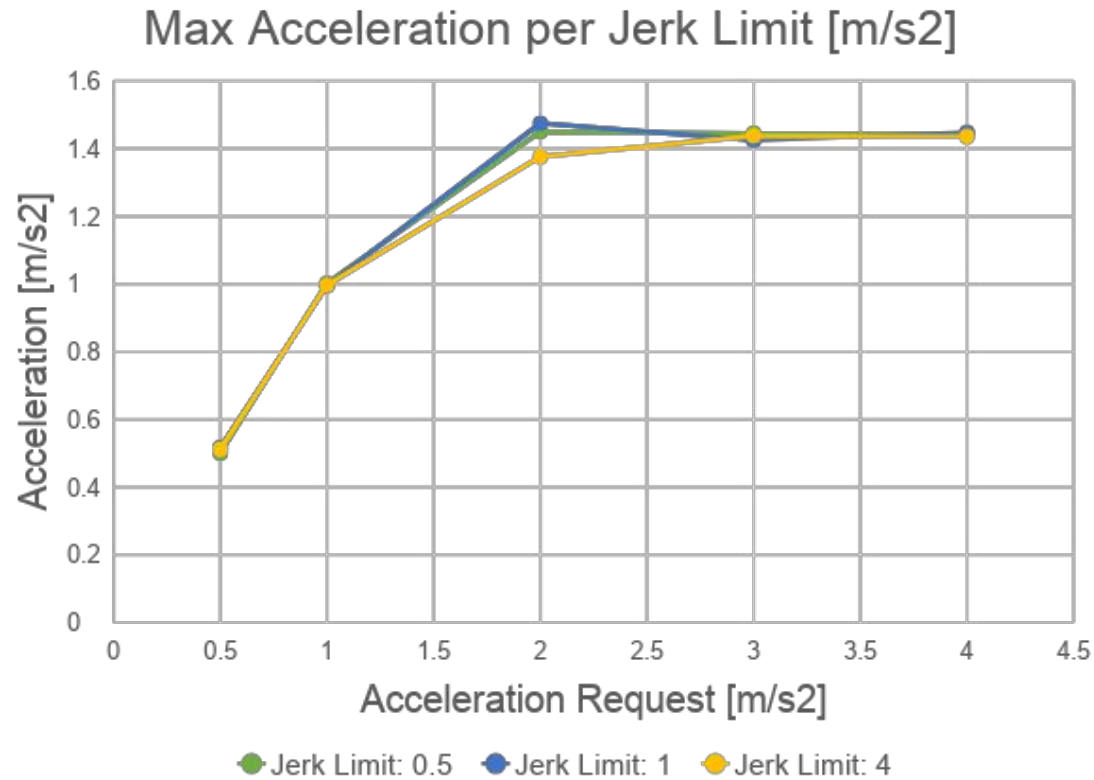


Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.44	-2.23	0.47	-1.36
$\pm 1 \text{ mps}^3$	1.45	-2.24	0.51	-1.32
$\pm 4 \text{ mps}^3$	1.44	-2.17	0.48	-1.30

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

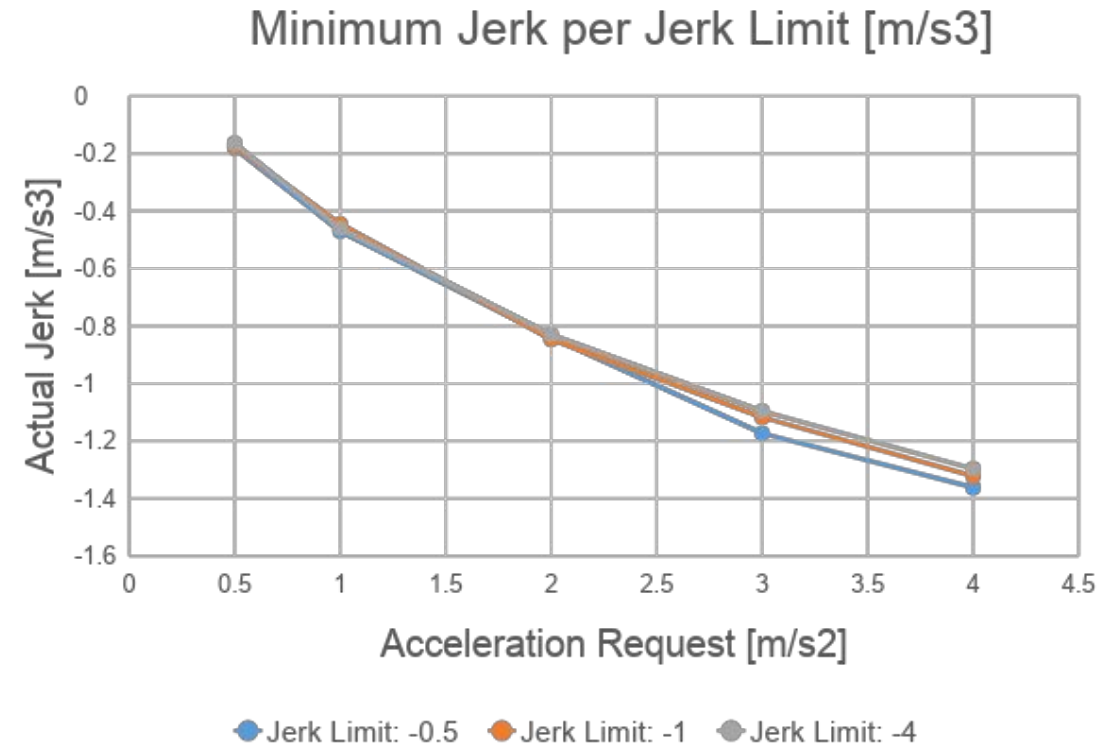
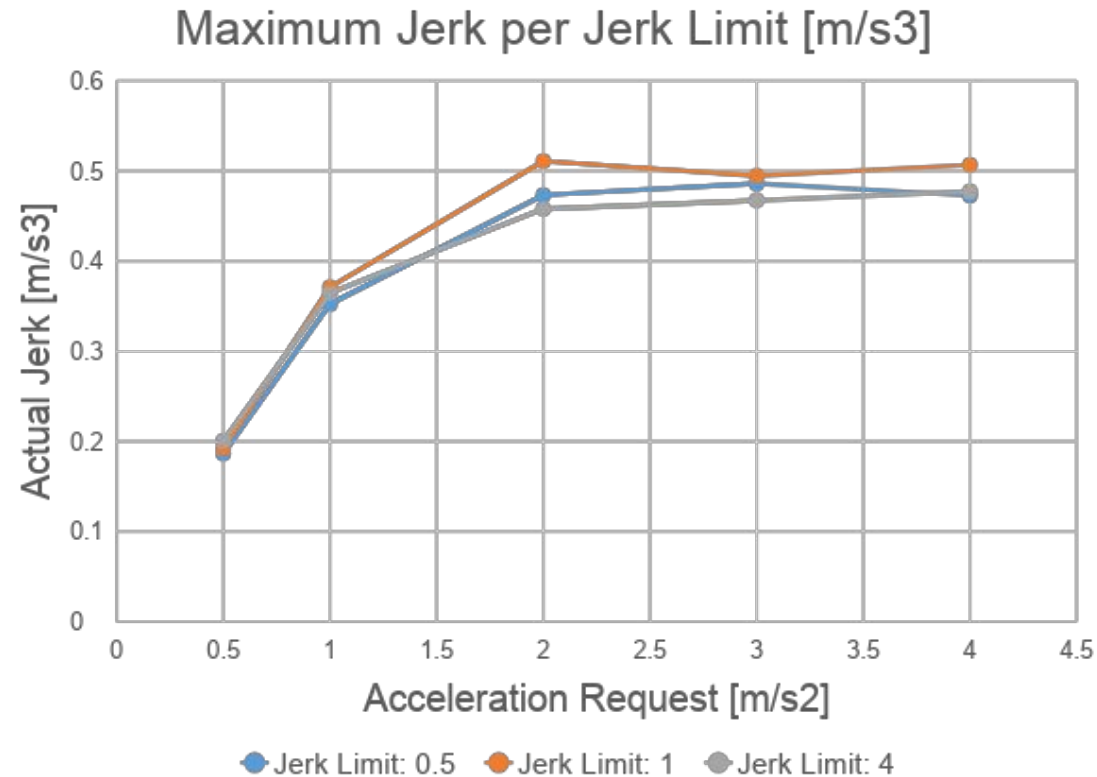
50-60 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

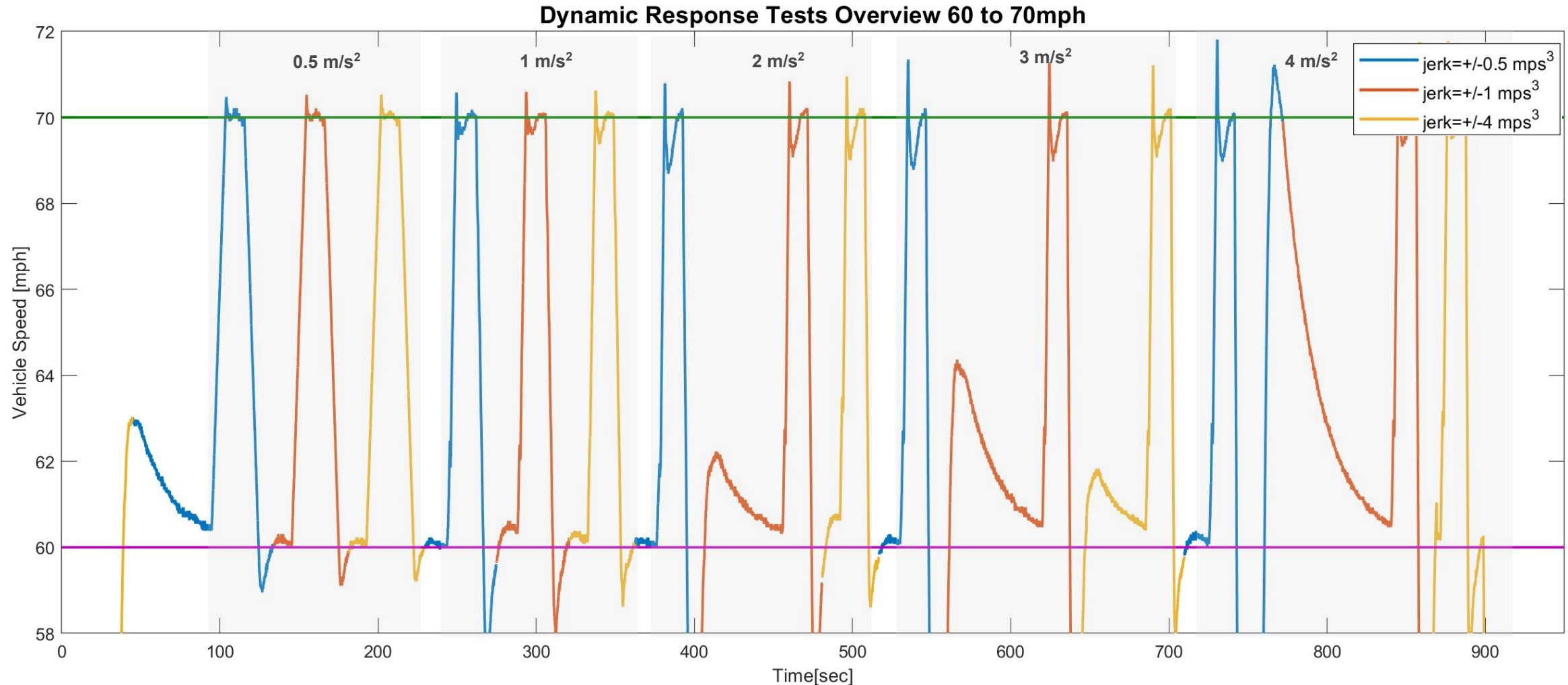
50-60 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

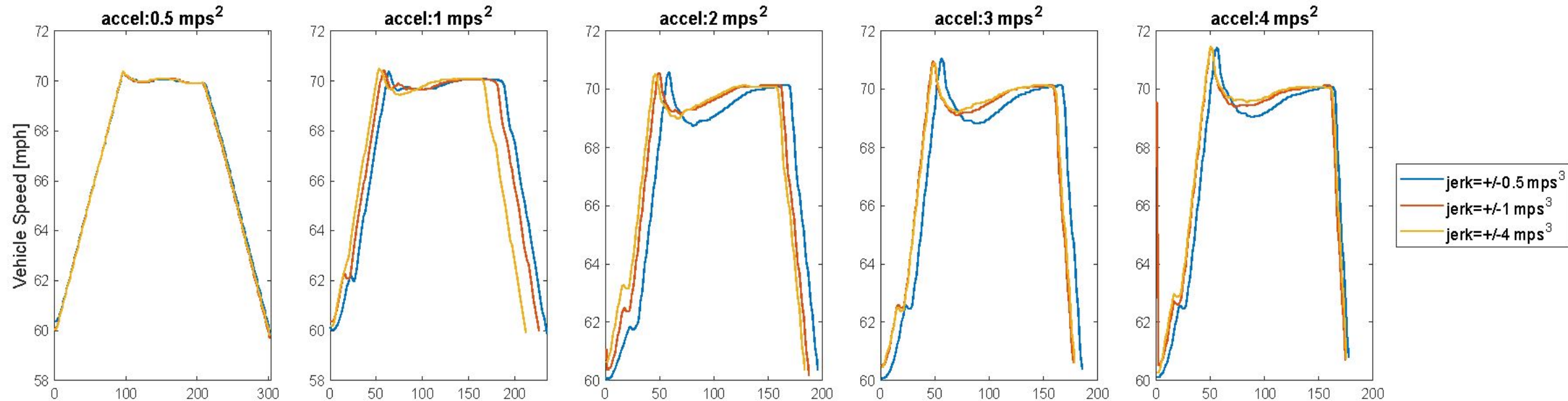
60-70 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

60-70 mph

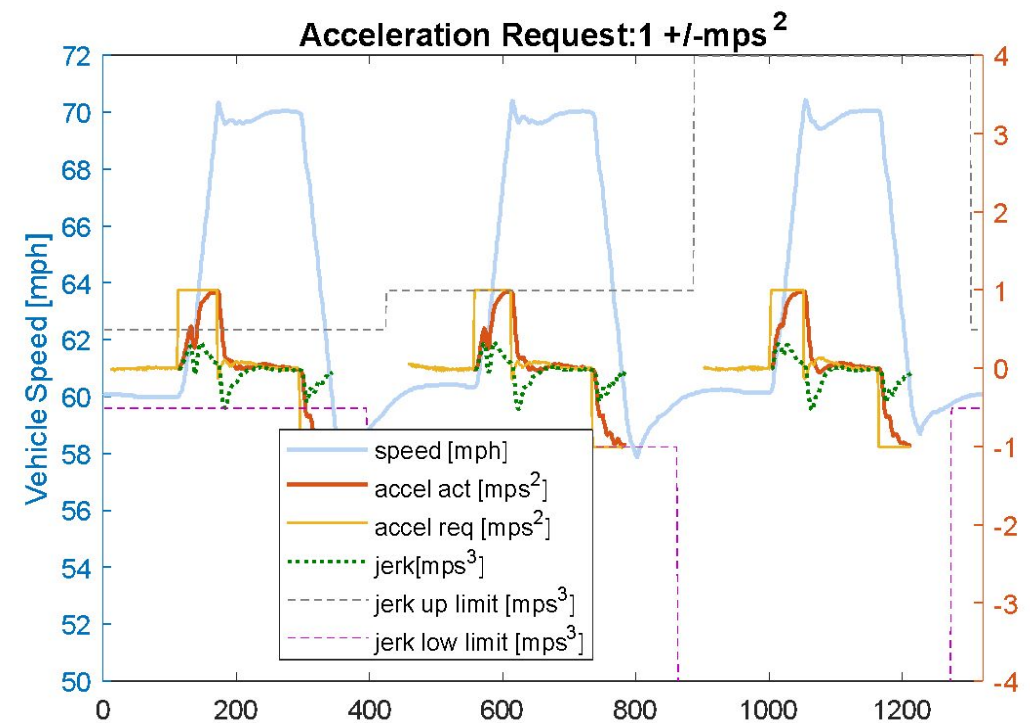
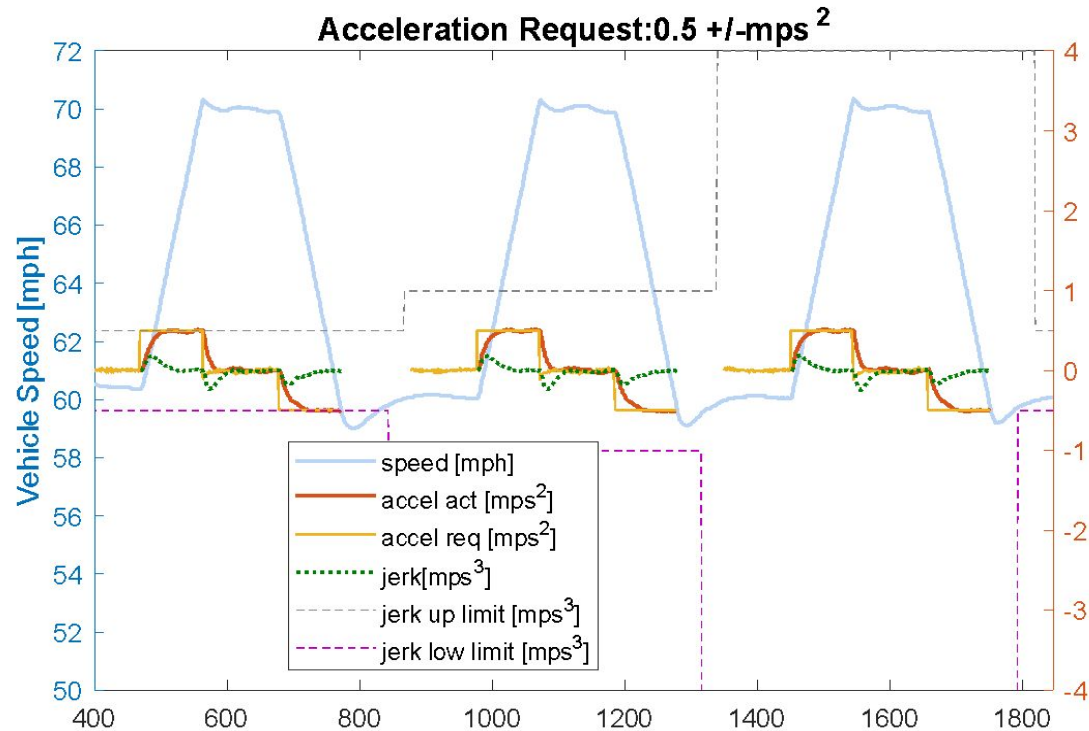


	accel req : 0.5 mps^2		accel req : 1 mps^2		accel req : 2 mps^2		accel req : 3 mps^2		accel req : 4 mps^2	
Jerk Limits	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time
$\pm 0.5 \text{ mps}^3$	8.8 s	9.2 s	5.6 s	4.9 s	5.1 s	2.8 s	4.7 s	1.9 s	4.7 s	1.5 s
$\pm 1 \text{ mps}^3$	8.8 s	9.4 s	5.2 s	4.9 s	4.1 s	2.7 s	4.2 s	1.9 s	4.1 s	1.5 s
$\pm 4 \text{ mps}^3$	8.7 s	9.4 s	4.6 s	4.9 s	4.0 s	2.6 s	4.3 s	1.9 s	4.1 s	1.5 s

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

60-70 mph



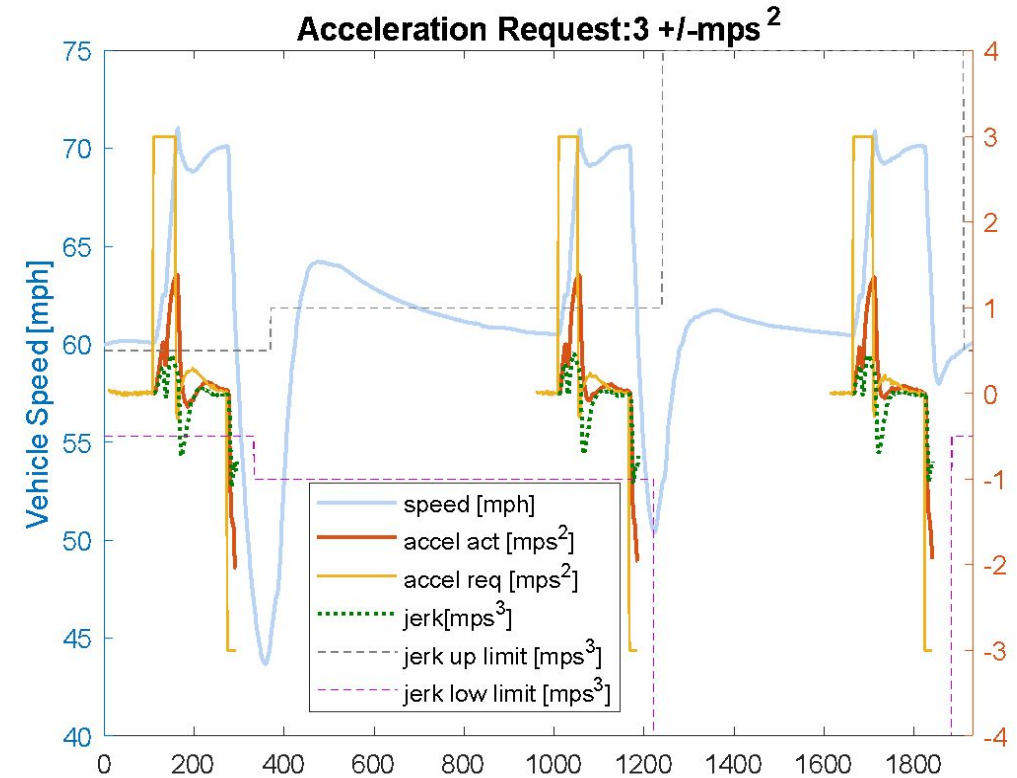
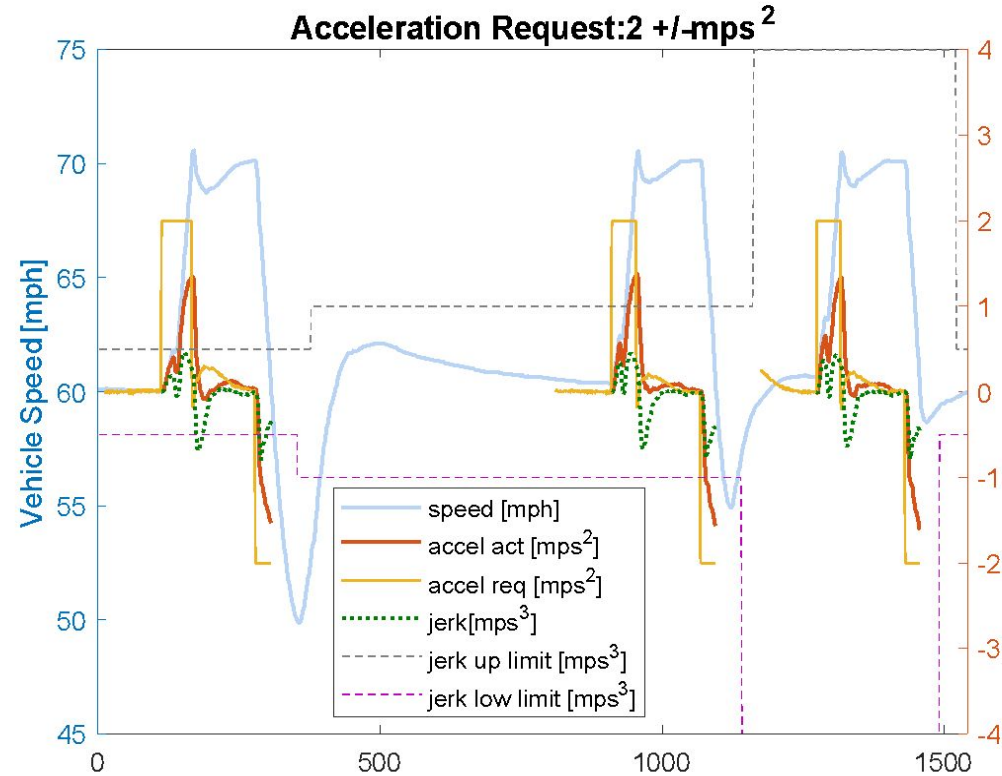
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.51	-0.52	0.18	-0.17
$\pm 1 \text{ mps}^3$	0.51	-0.52	0.19	-0.17
$\pm 4 \text{ mps}^3$	0.51	-0.52	0.19	-0.17

Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.98	-0.99	0.33	-0.43
$\pm 1 \text{ mps}^3$	0.99	-1.00	0.34	-0.41
$\pm 4 \text{ mps}^3$	0.99	-0.99	0.32	-0.43

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

60-70 mph



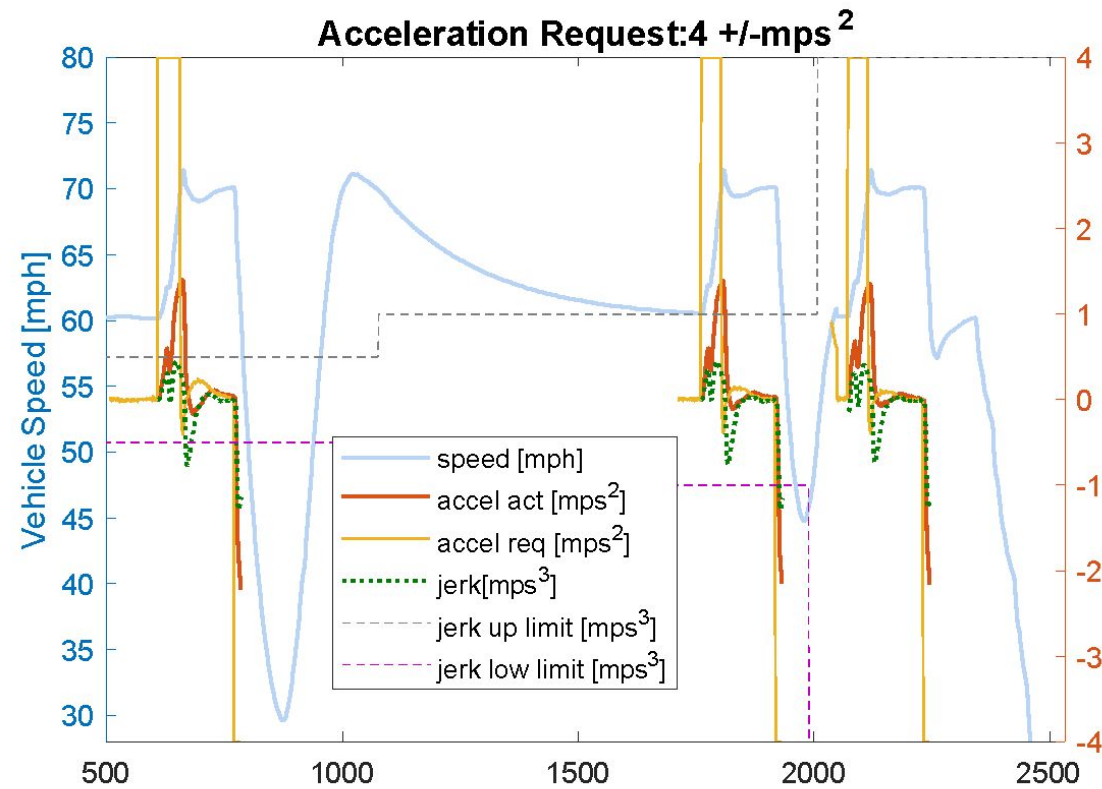
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.32	-1.53	0.46	-0.80
$\pm 1 \text{ mps}^3$	1.35	-1.57	0.45	-0.78
$\pm 4 \text{ mps}^3$	1.30	-1.62	0.43	-0.78

Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.34	-2.05	0.45	-1.09
$\pm 1 \text{ mps}^3$	1.35	-1.97	0.46	-1.05
$\pm 4 \text{ mps}^3$	1.31	-1.95	0.45	-1.04

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

60-70 mph

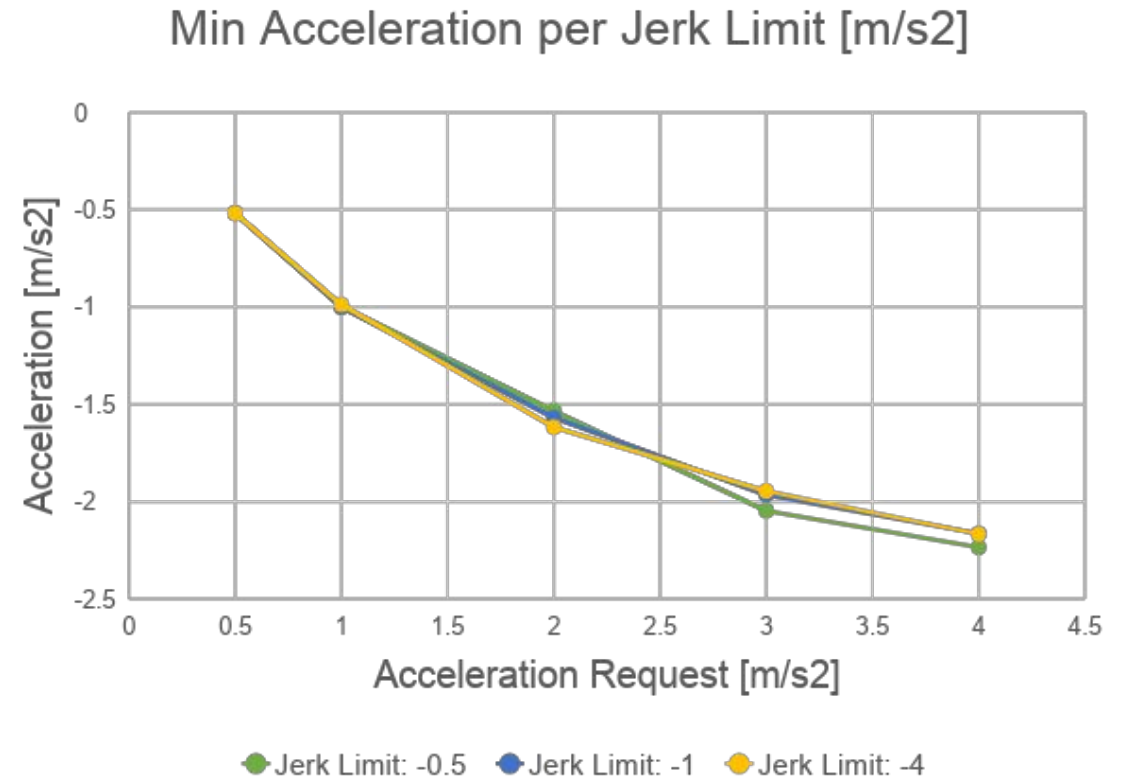
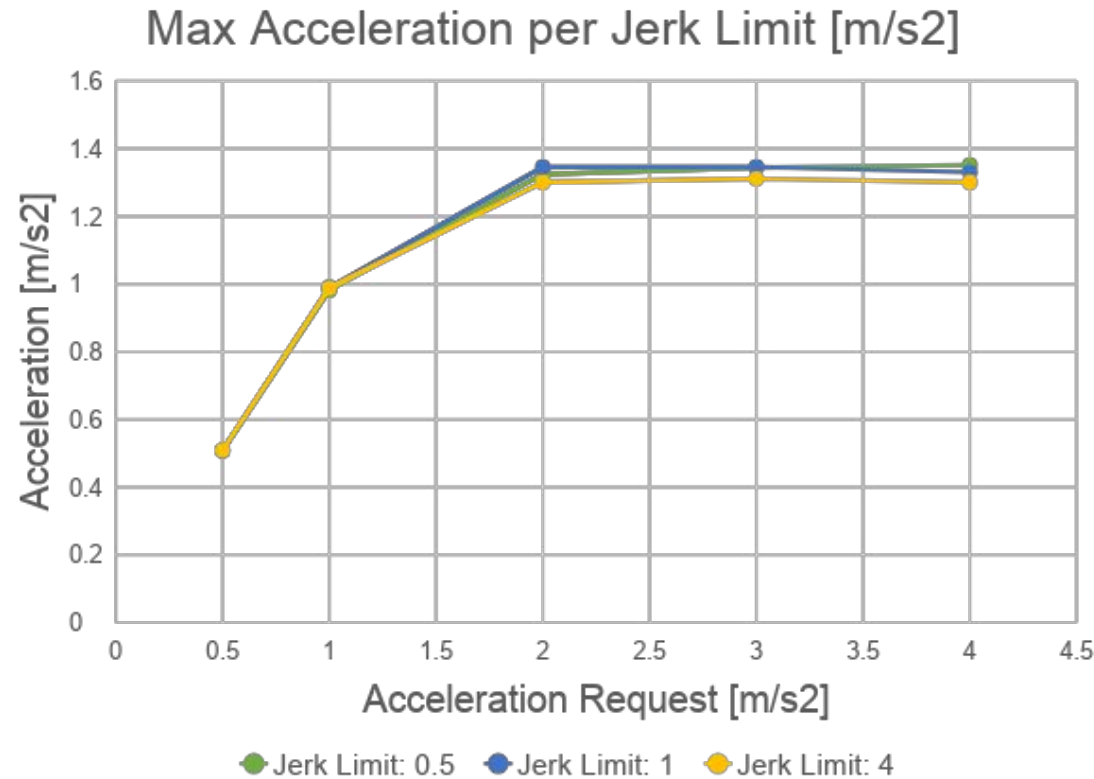


Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.35	-2.24	0.45	-1.29
$\pm 1 \text{ mps}^3$	1.33	-2.16	0.45	-1.28
$\pm 4 \text{ mps}^3$	1.30	-2.17	0.41	-1.26

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

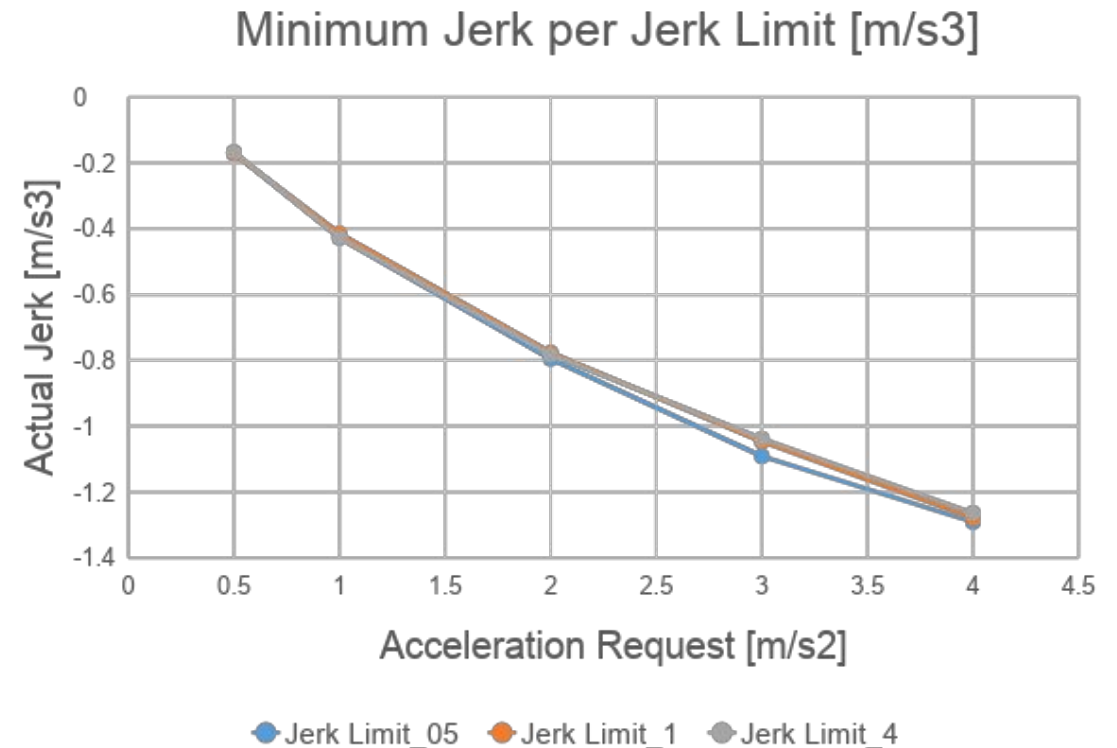
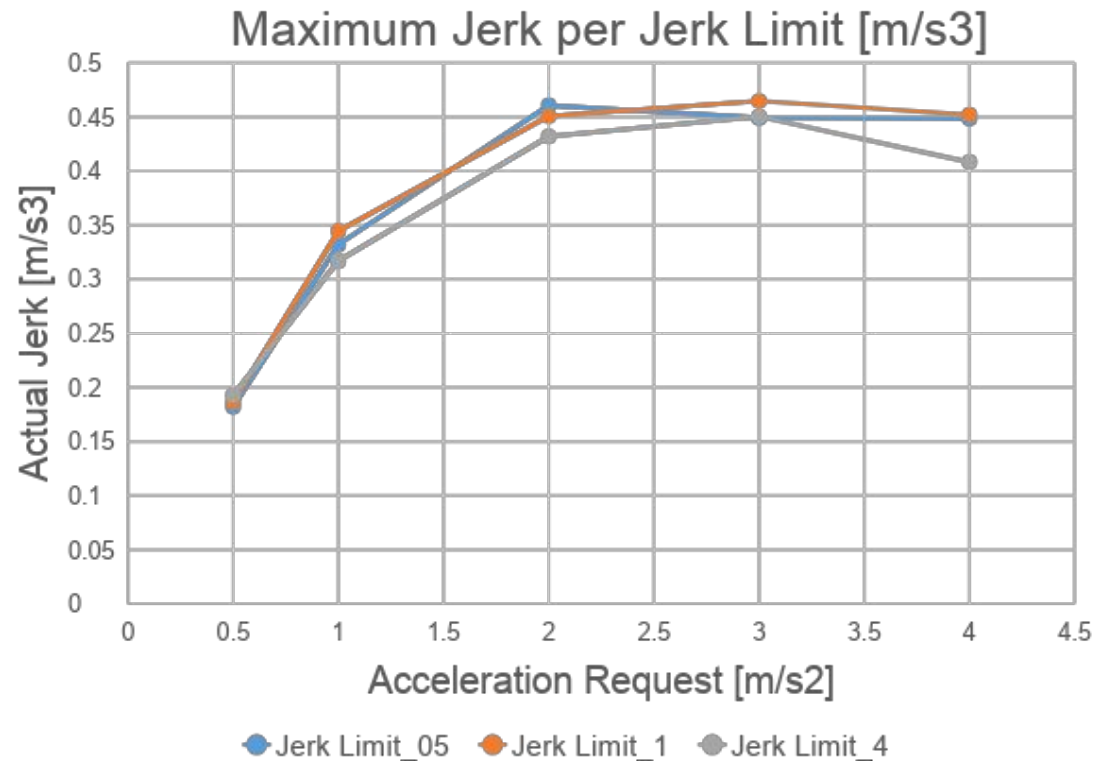
60-70 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

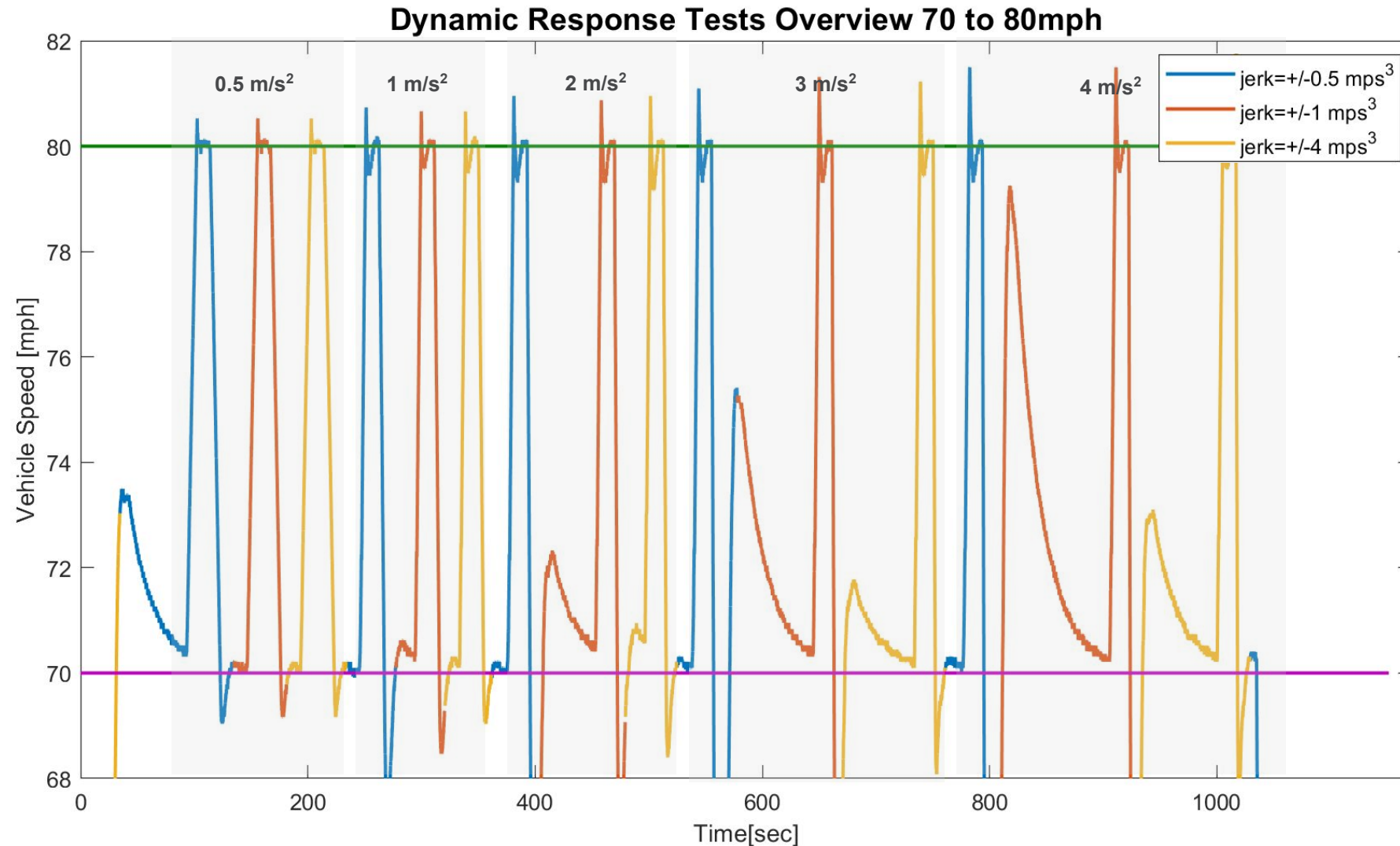
60-70 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

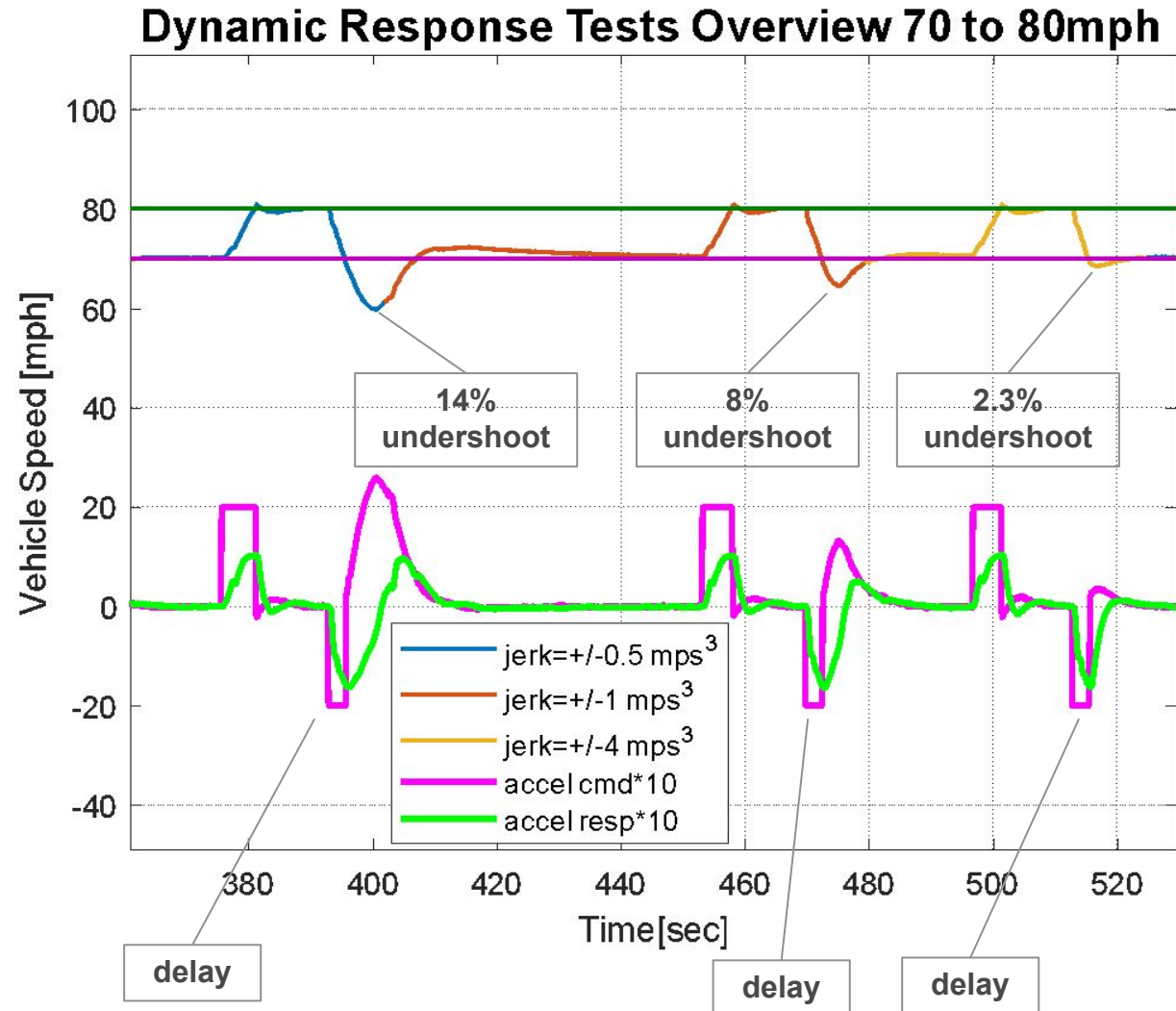
70-80 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

70-80 mph

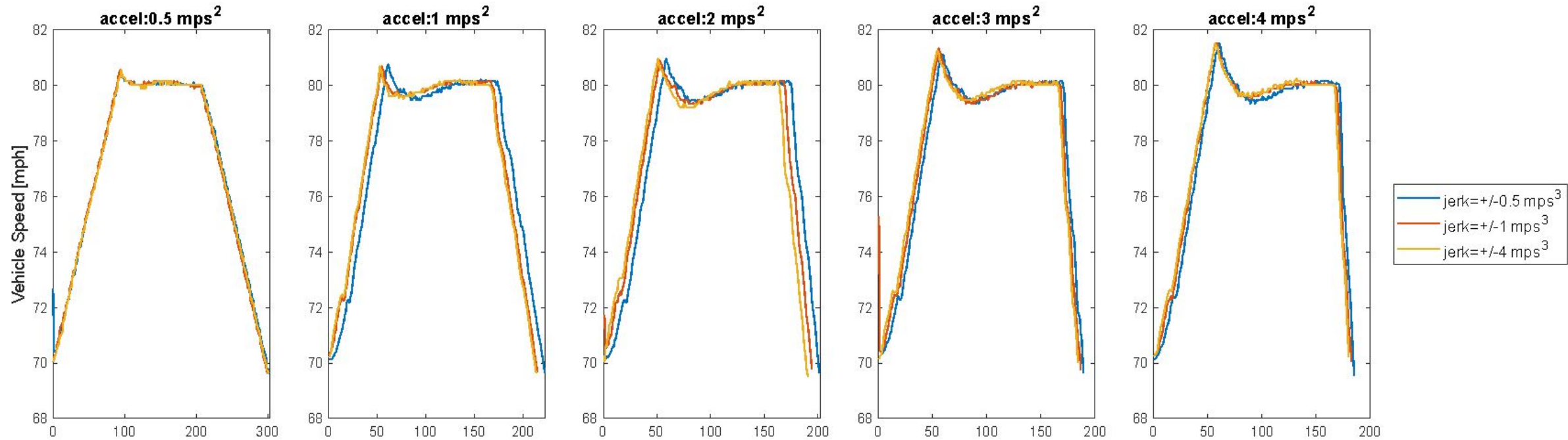


- For the same acceleration request level (2 mps^2 in figure) **the lower the jerk limit is, the more delay there is in acceleration changes**
- This translates into an undershoot in the vehicle speed which is higher for lower jerk limits (0.5 m/s^3)

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

70-80 mph

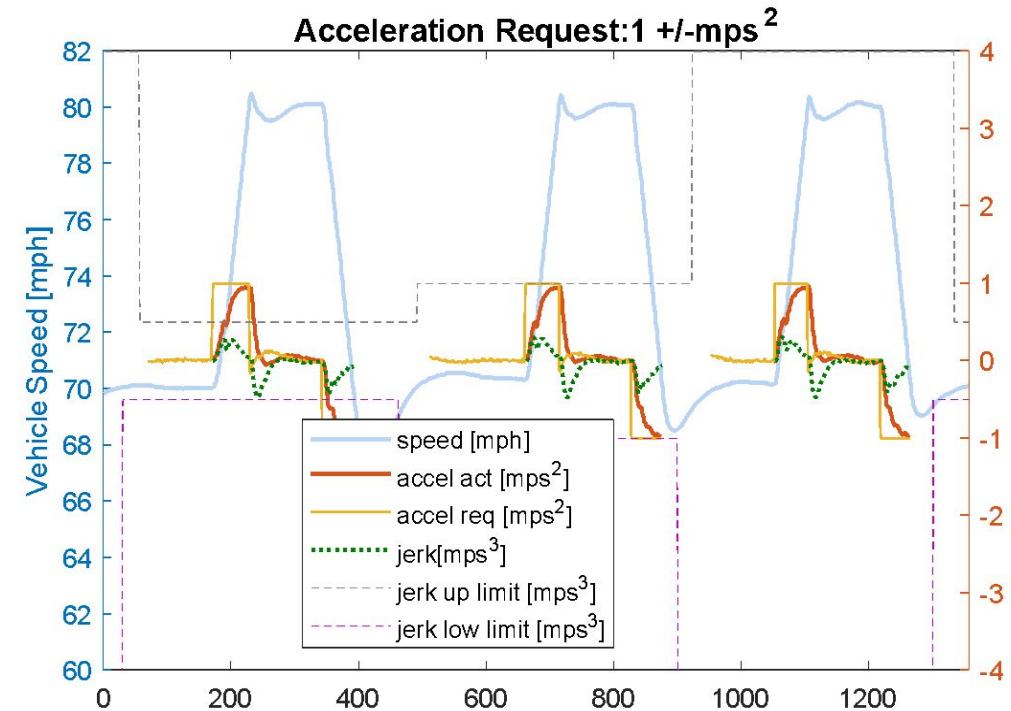
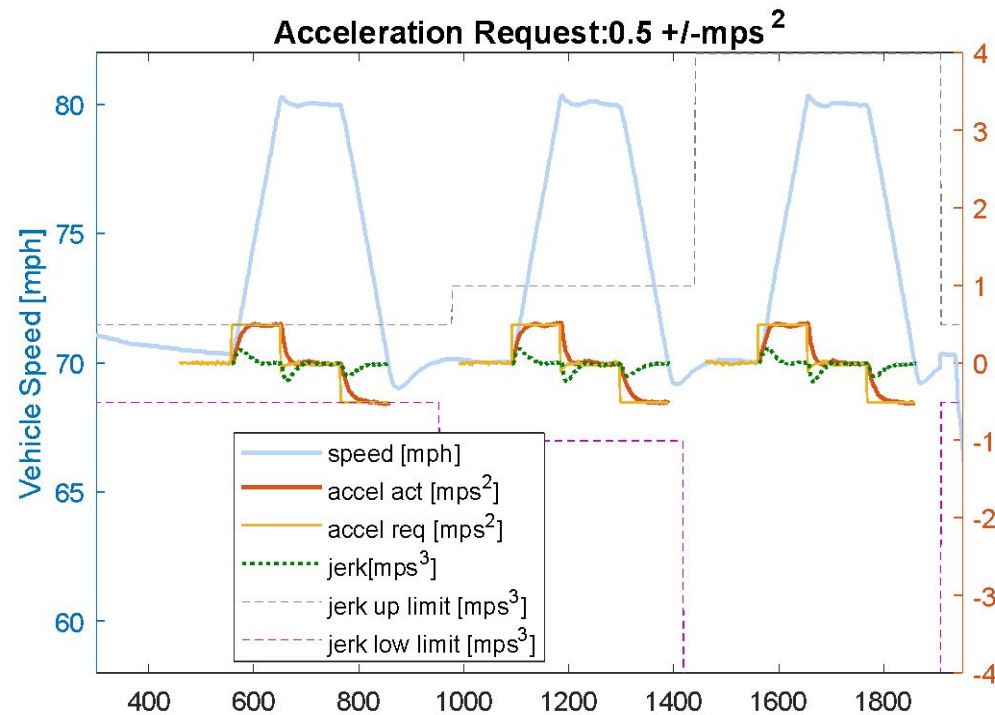


	accel req : 0.5 mps^2		accel req : 1 mps^2		accel req : 2 mps^2		accel req : 3 mps^2		accel req : 4 mps^2	
Jerk Limits	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time
$\pm 0.5 \text{ mps}^3$	9.0 s	9.3 s	5.7 s	4.9 s	5.3 s	2.7 s	5.4 s	1.9 s	5.4 s	1.5 s
$\pm 1 \text{ mps}^3$	9.0 s	9.4 s	5.1 s	4.8 s	4.8 s	2.6 s	4.9 s	1.9 s	5.0 s	1.4 s
$\pm 4 \text{ mps}^3$	9.0 s	9.3 s	5.1 s	4.8 s	4.6 s	2.8 s	4.8 s	1.9 s	5.0 s	1.4 s

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

70-80 mph



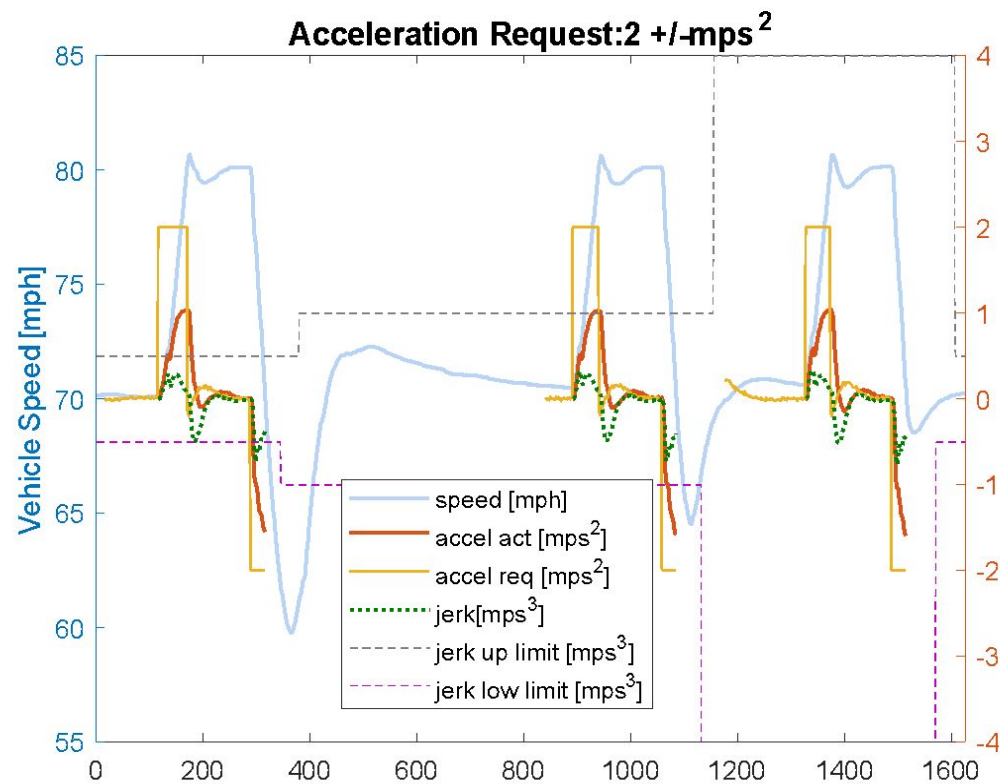
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.51	-0.53	0.18	-0.17
$\pm 1 \text{ mps}^3$	0.51	-0.52	0.19	-0.16
$\pm 4 \text{ mps}^3$	0.51	-0.53	0.20	-0.16

Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.95	-0.99	0.28	-0.43
$\pm 1 \text{ mps}^3$	0.94	-0.99	0.30	-0.41
$\pm 4 \text{ mps}^3$	0.94	-0.99	0.31	-0.40

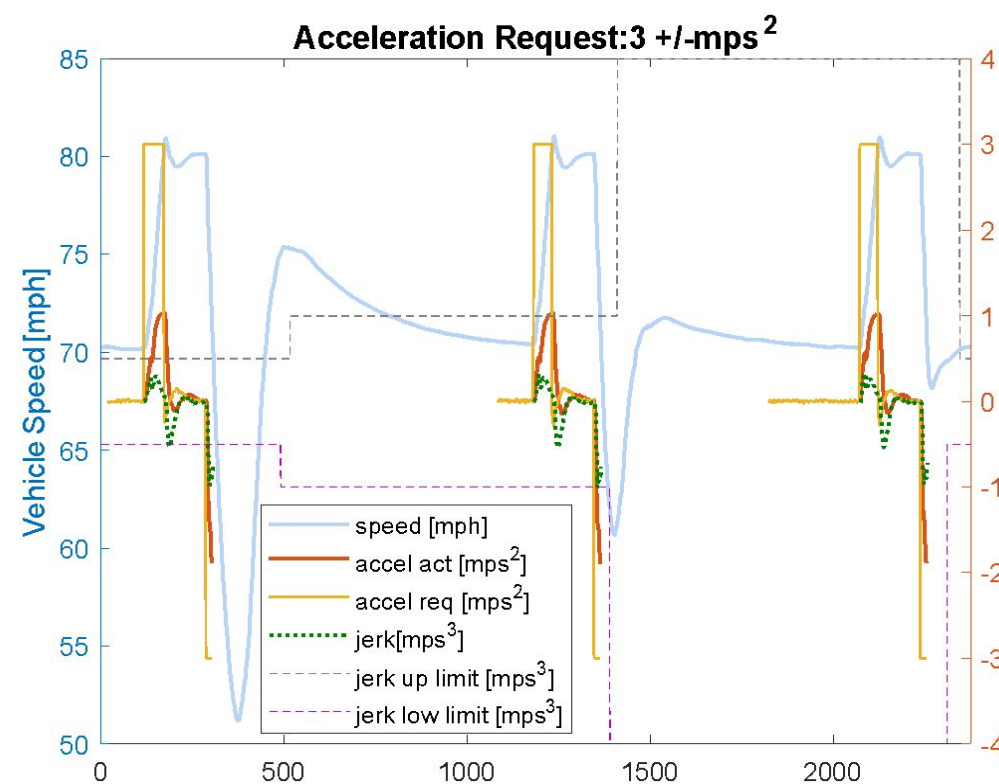
Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

70-80 mph



Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.03	-1.55	0.29	-0.73
$\pm 1 \text{ mps}^3$	1.03	-1.60	0.32	-0.72
$\pm 4 \text{ mps}^3$	1.03	-1.60	0.34	-0.76

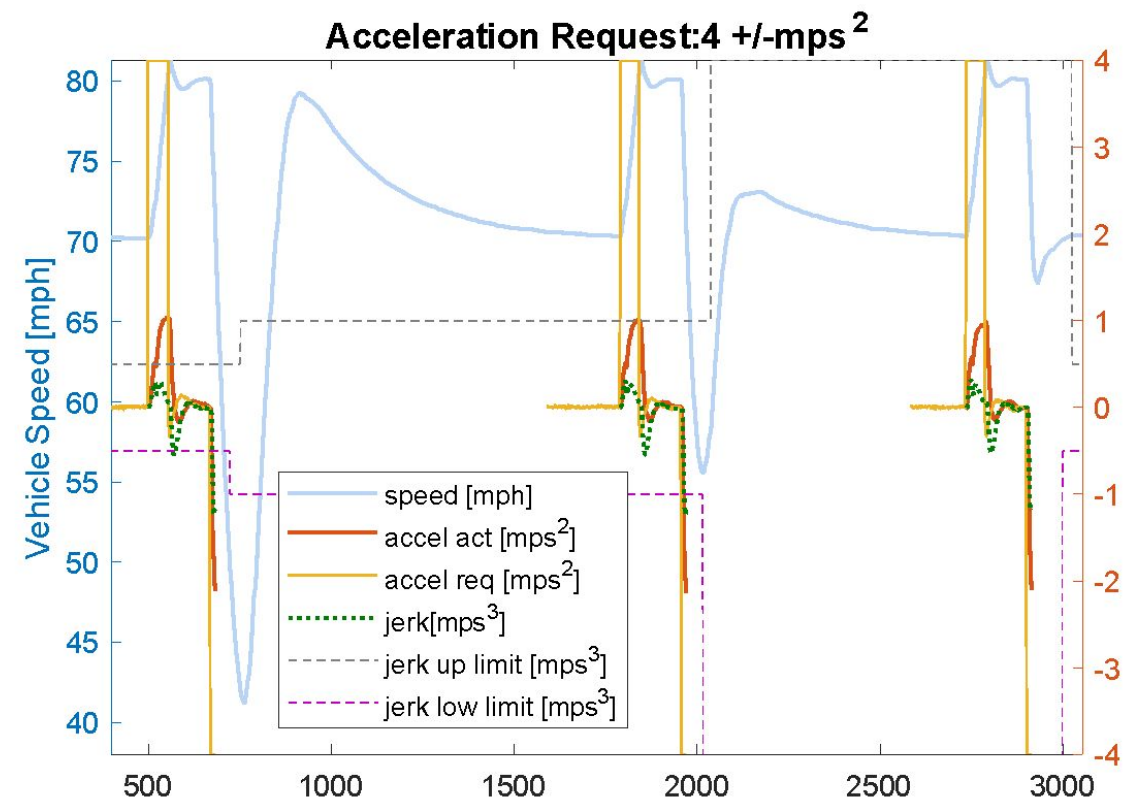


Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.03	-1.89	0.30	-0.98
$\pm 1 \text{ mps}^3$	1.01	-1.90	0.31	-1.01
$\pm 4 \text{ mps}^3$	1.01	-1.90	0.33	-0.97

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

70-80 mph

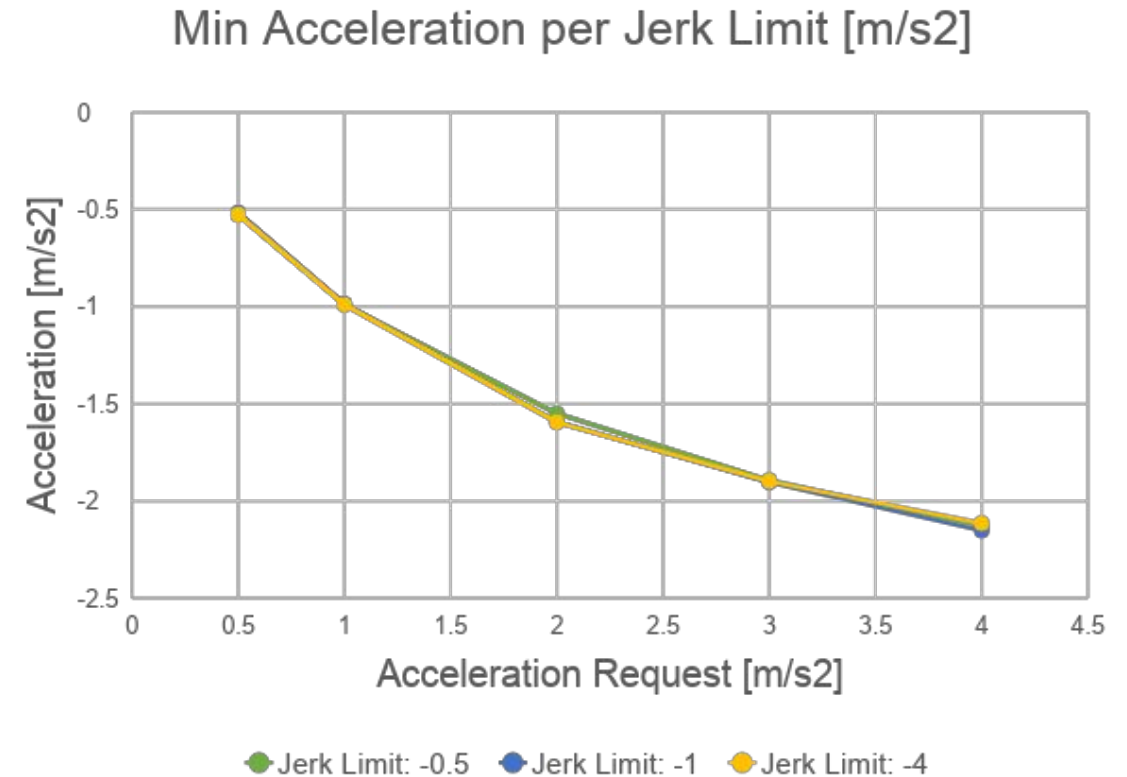
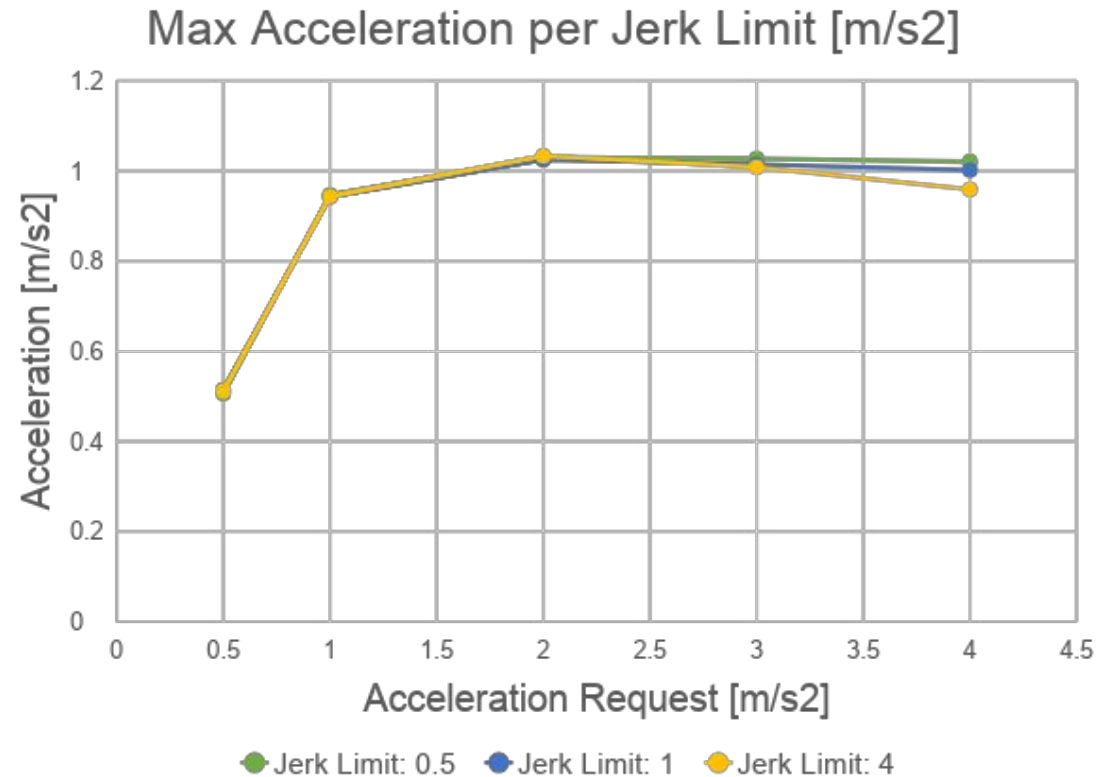


Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.02	-2.13	0.28	-1.23
$\pm 1 \text{ mps}^3$	1.00	-2.15	0.31	-1.24
$\pm 4 \text{ mps}^3$	0.96	-2.11	0.32	-1.18

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

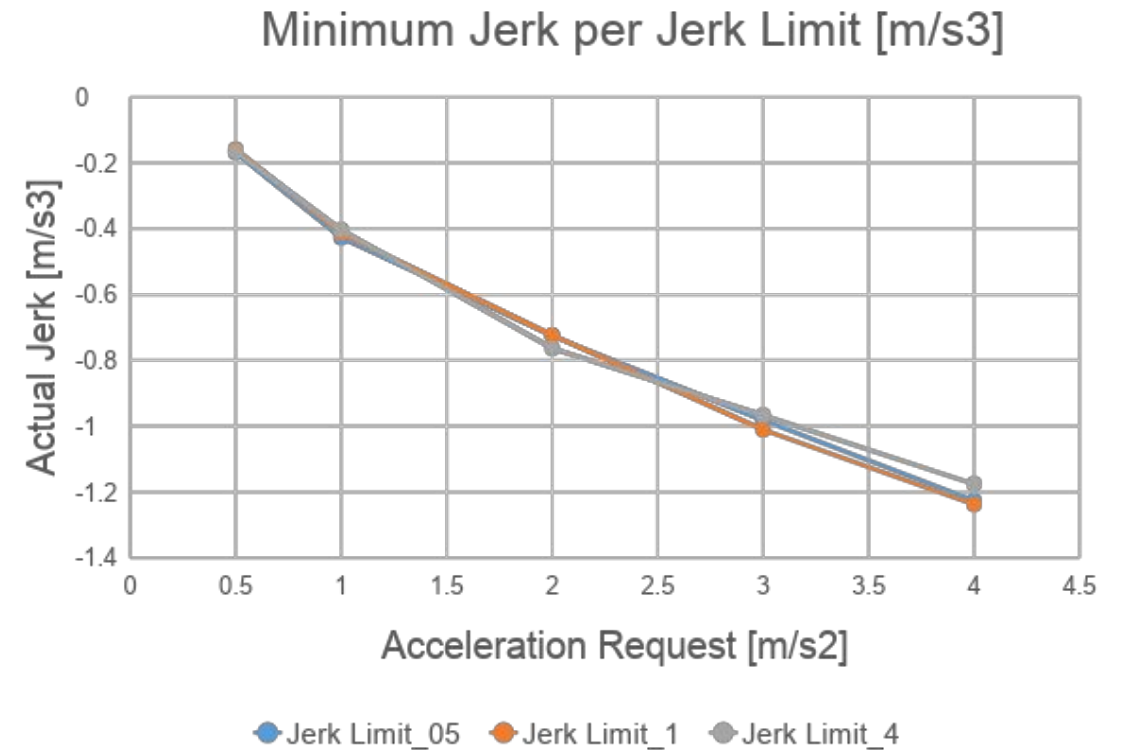
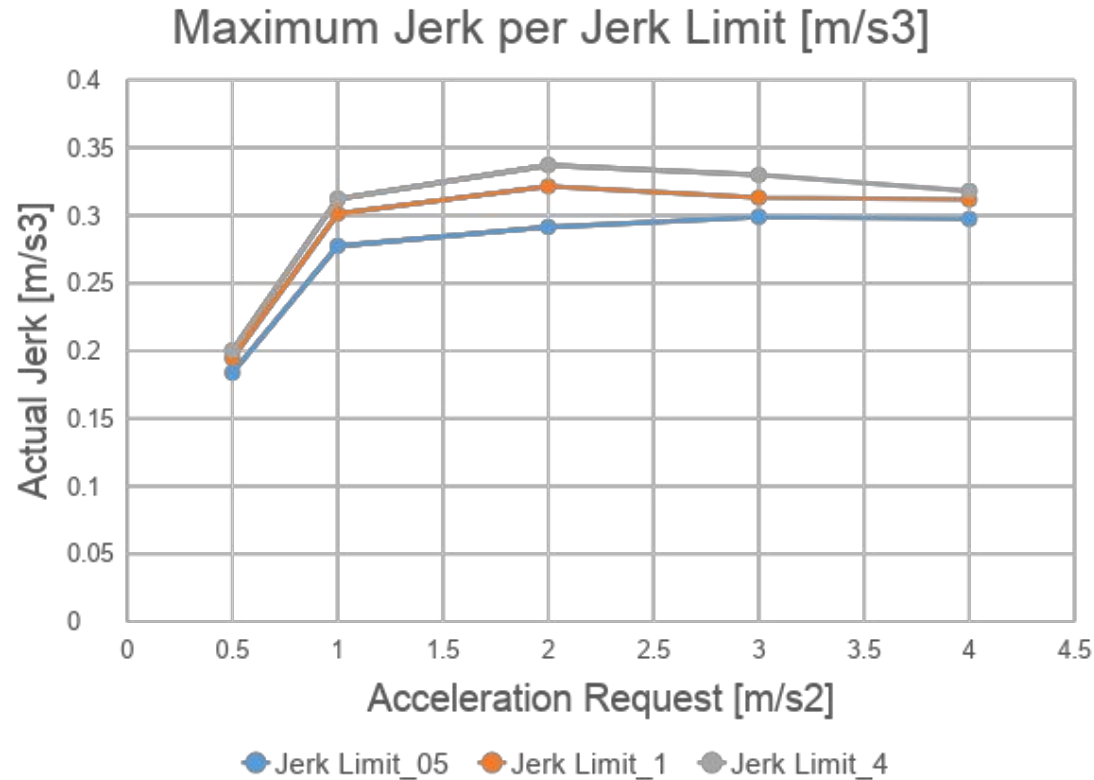
70-80 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

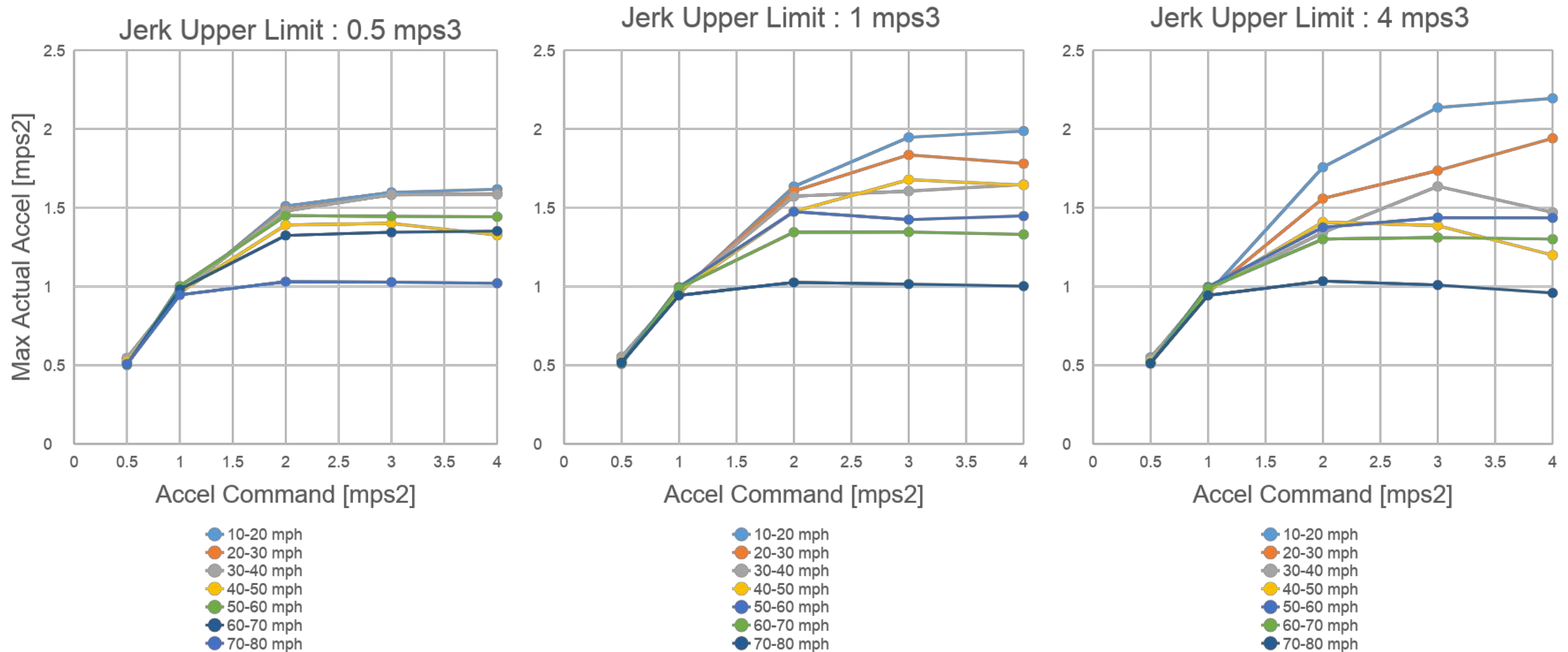
70-80 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

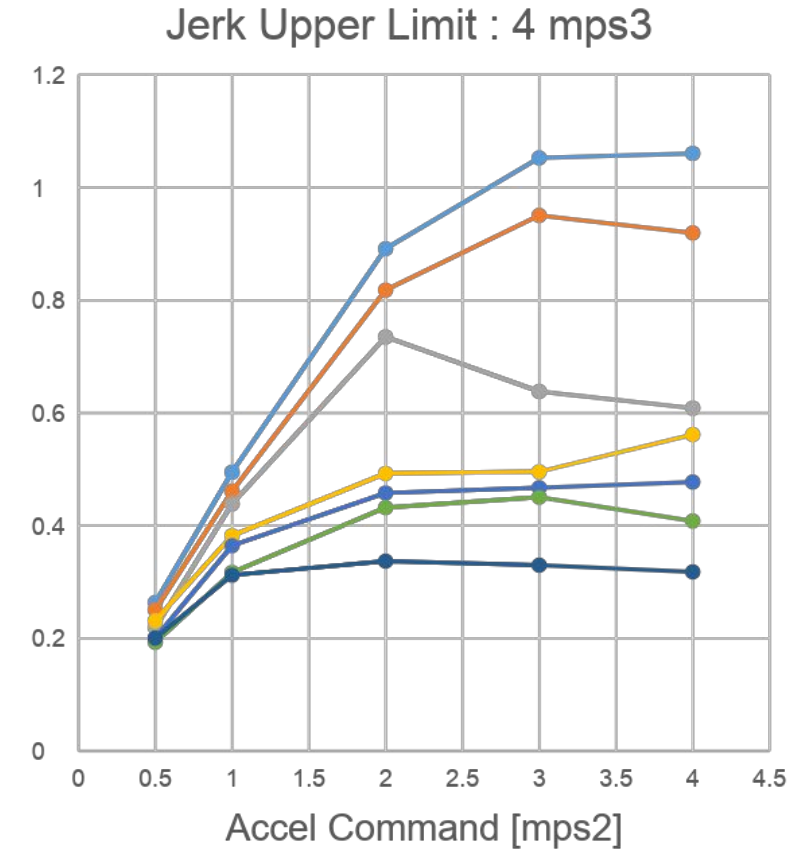
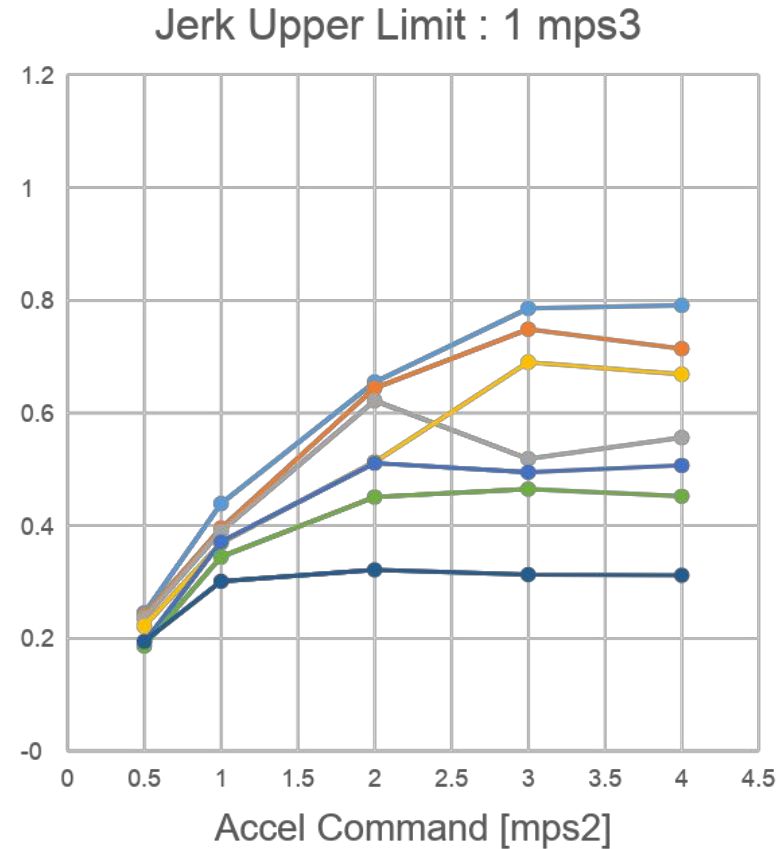
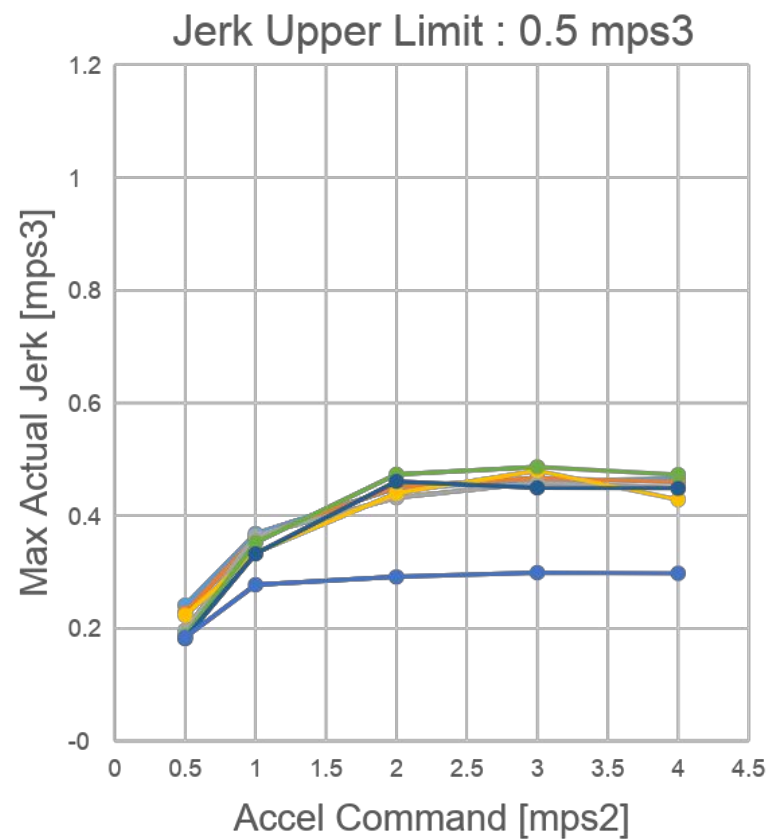
Max
Acceleration



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

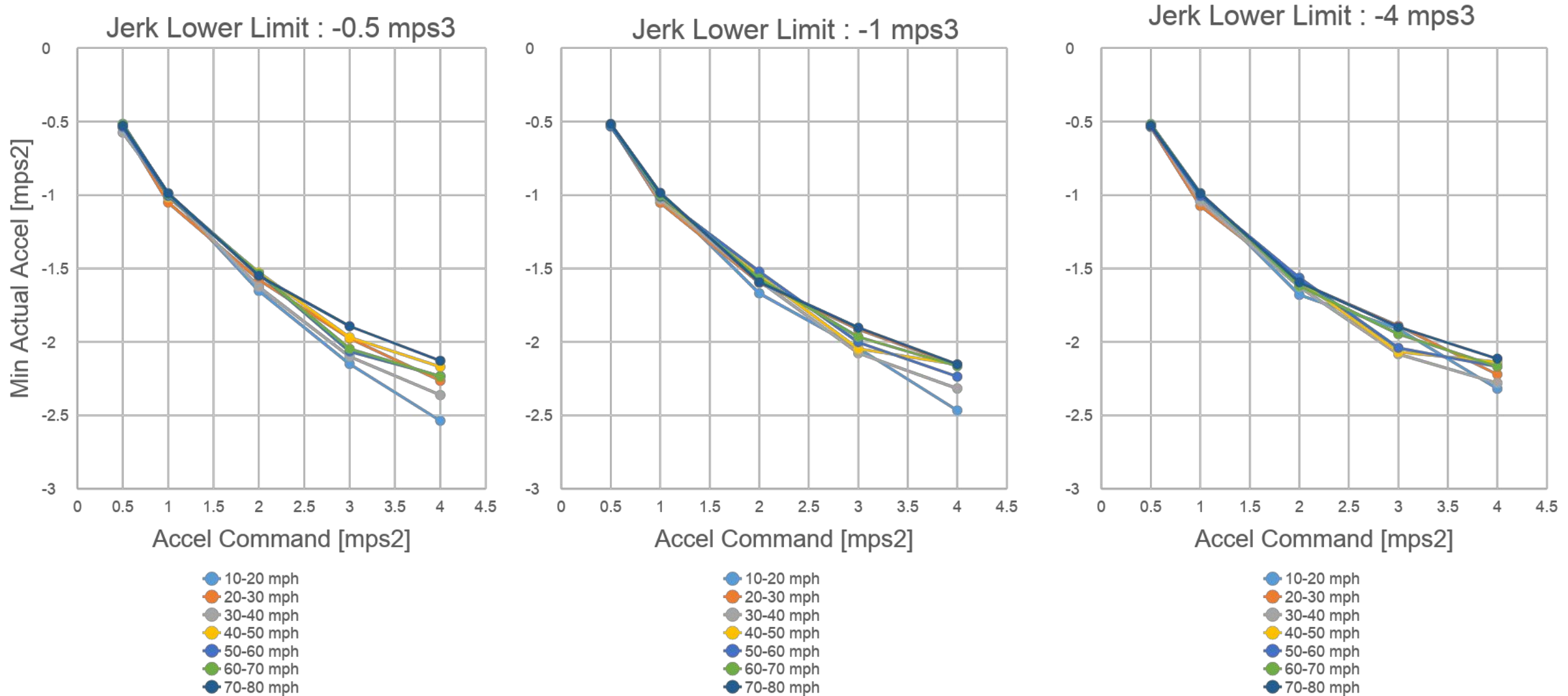
Max Jerk



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

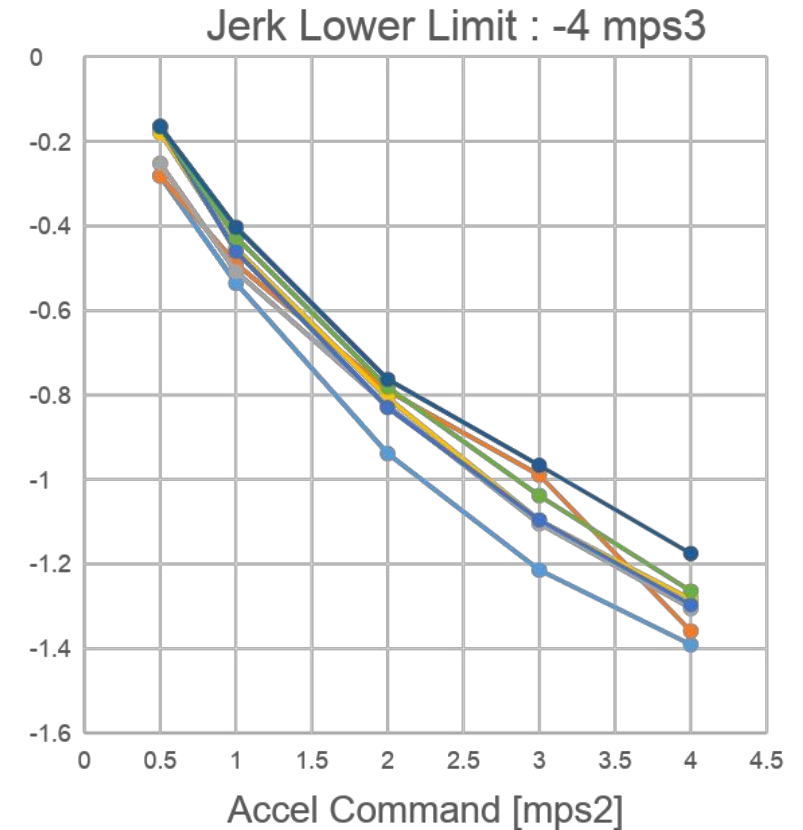
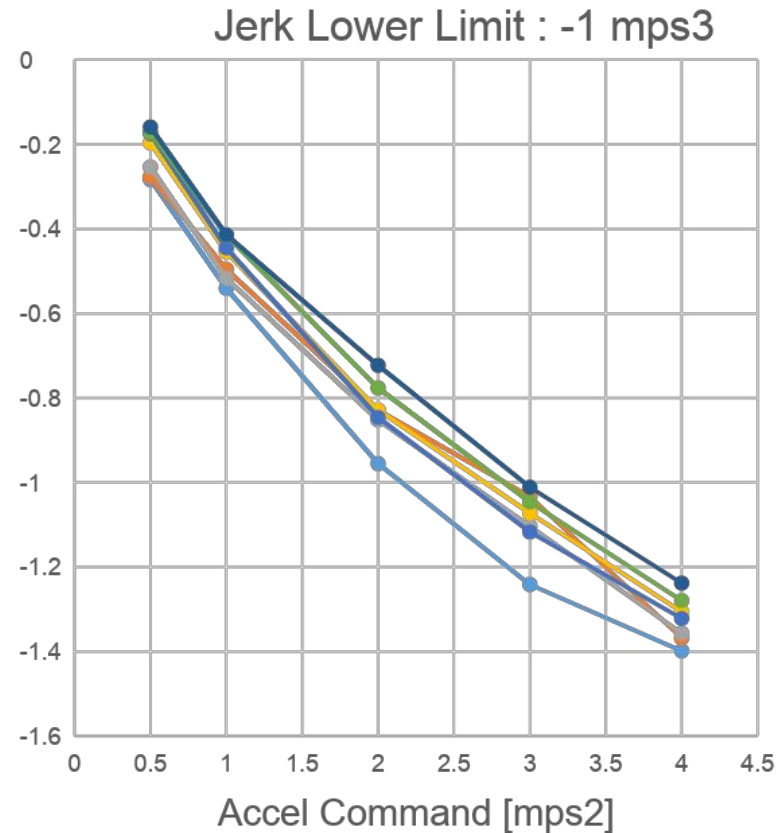
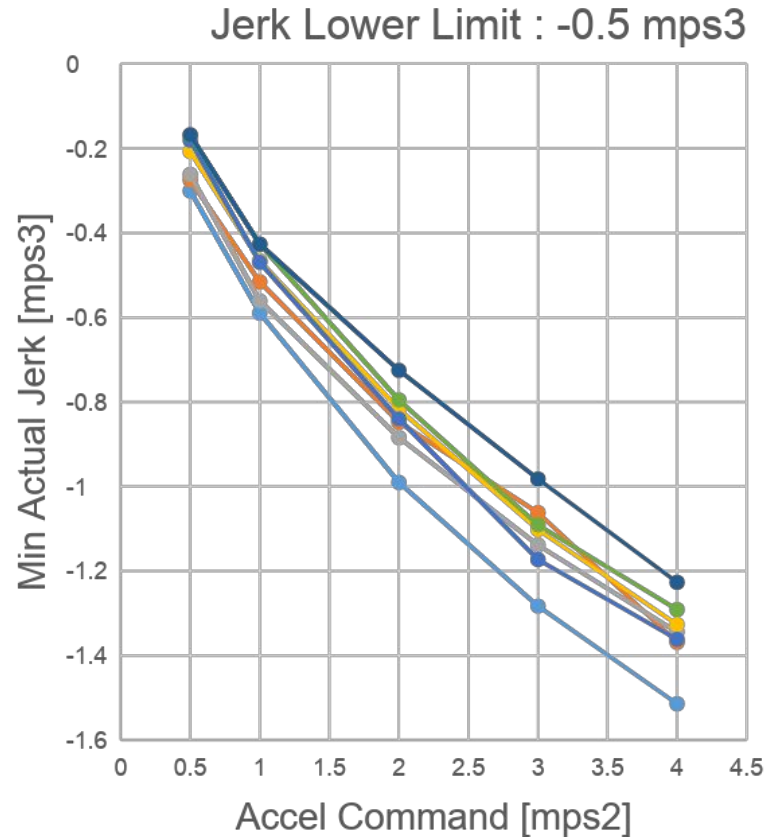
Min
Acceleration



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

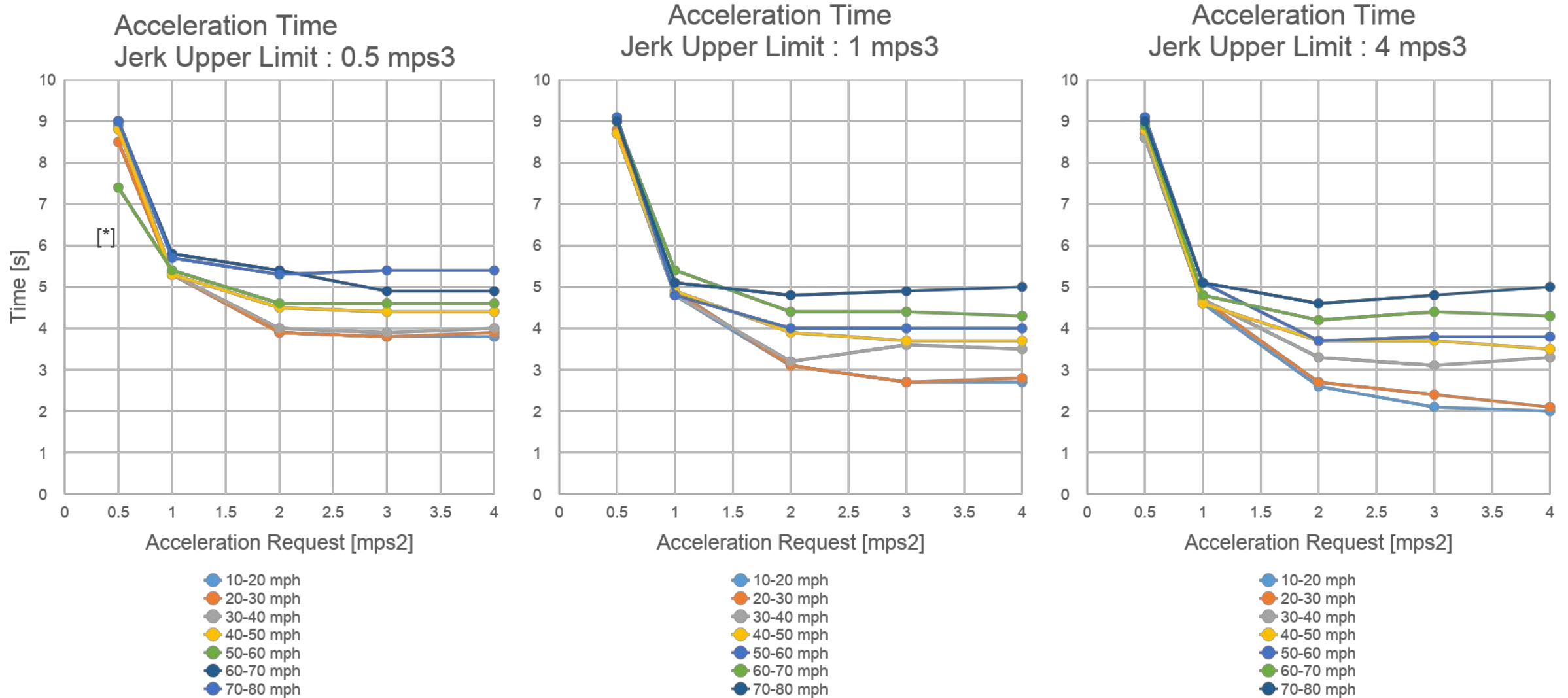
Min Jerk



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

Accel Time

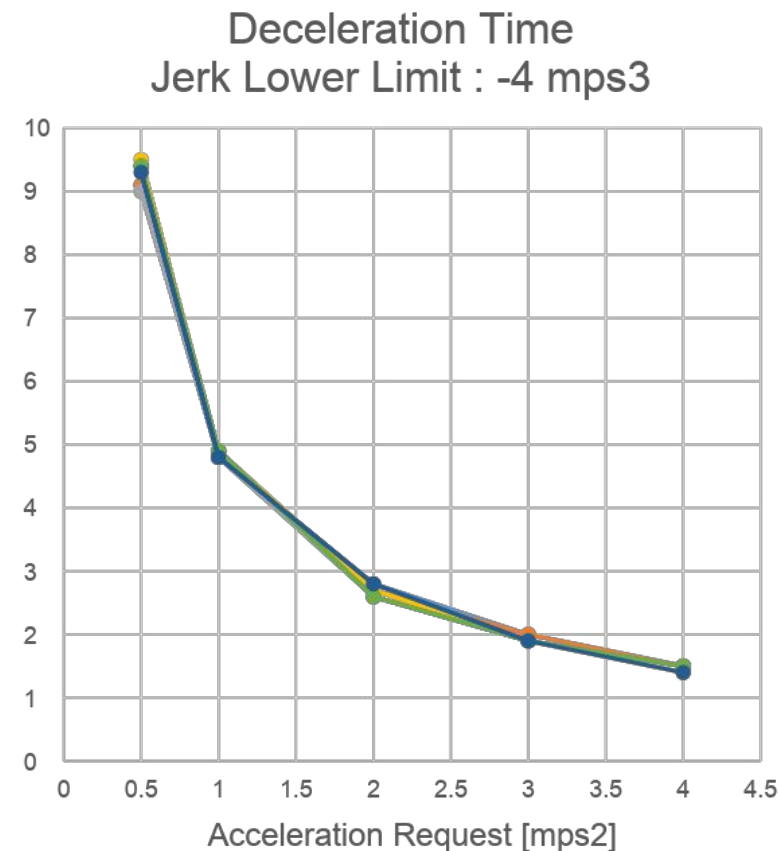
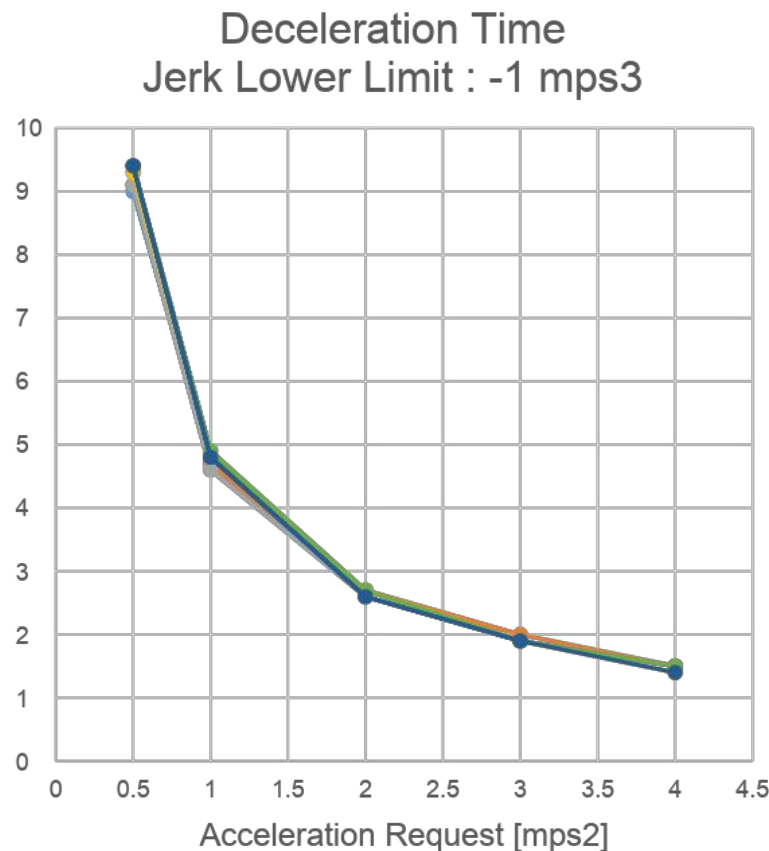
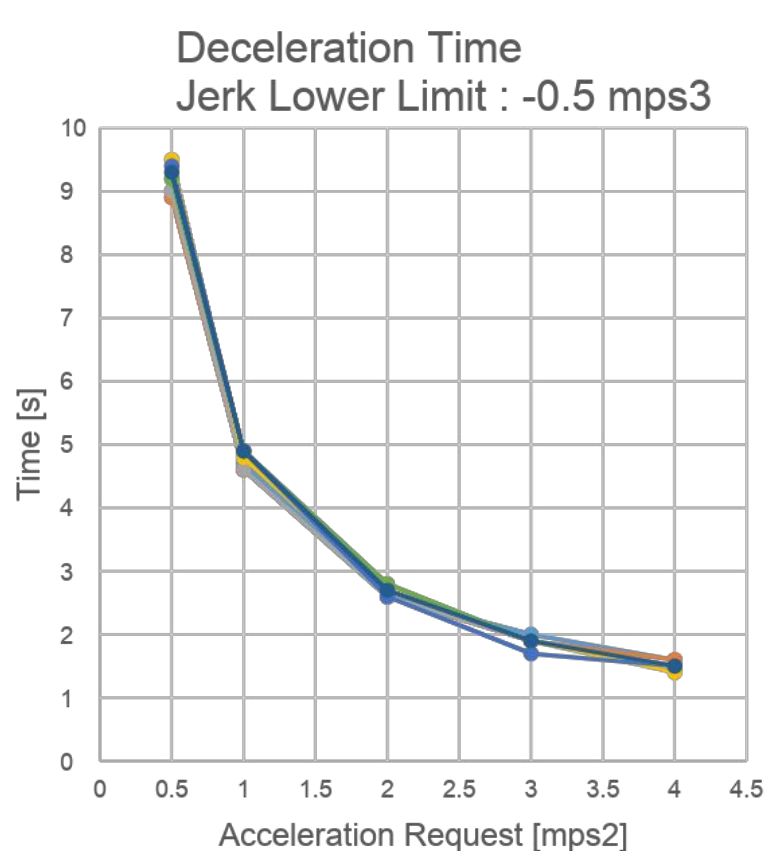


*this result is affected by the fact that the test has been started at 52 mph instead of 50 mph

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

Decel Time



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

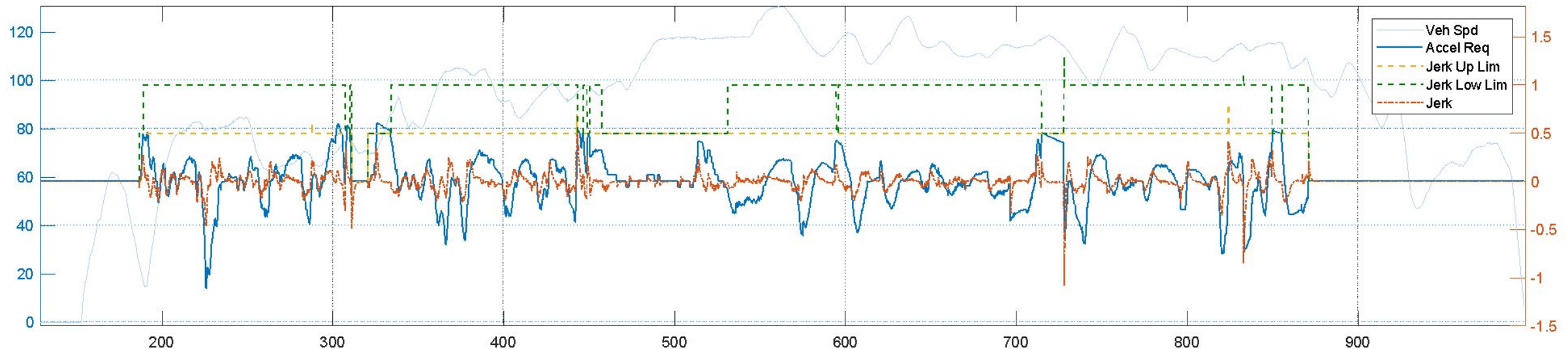
▪ Conclusions

- The higher the upper jerk limit, the higher is the actual max acceleration and the lower the acceleration time, for the same acceleration request
 - The lower the speed range the more visible the effect
 - The lower the acceleration request the less visible is the difference in max acceleration for the same jerk upper limit
- The higher the upper jerk limit, the higher is the actual max jerk for the same accel request
 - The lower the speed range the more visible the effect
 - The lower the acceleration request the less visible is the difference in max acceleration for the same jerk upper limit
- While these are general trends, there are cases where the transmission shifting affects the vehicle dynamic response
- These test did not show a correlation between the variations of the lower jerk limit and the vehicle dynamic behavior (min acceleration, deceleration time)
 - On-road data collected at a later time showed that both jerk limits vary in the positive range, therefore a negative jerk limit shows no impact on vehicle dynamic operations
 - This presentation has reported all the results for the sake of completeness

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – ACC On-Road Data Collection

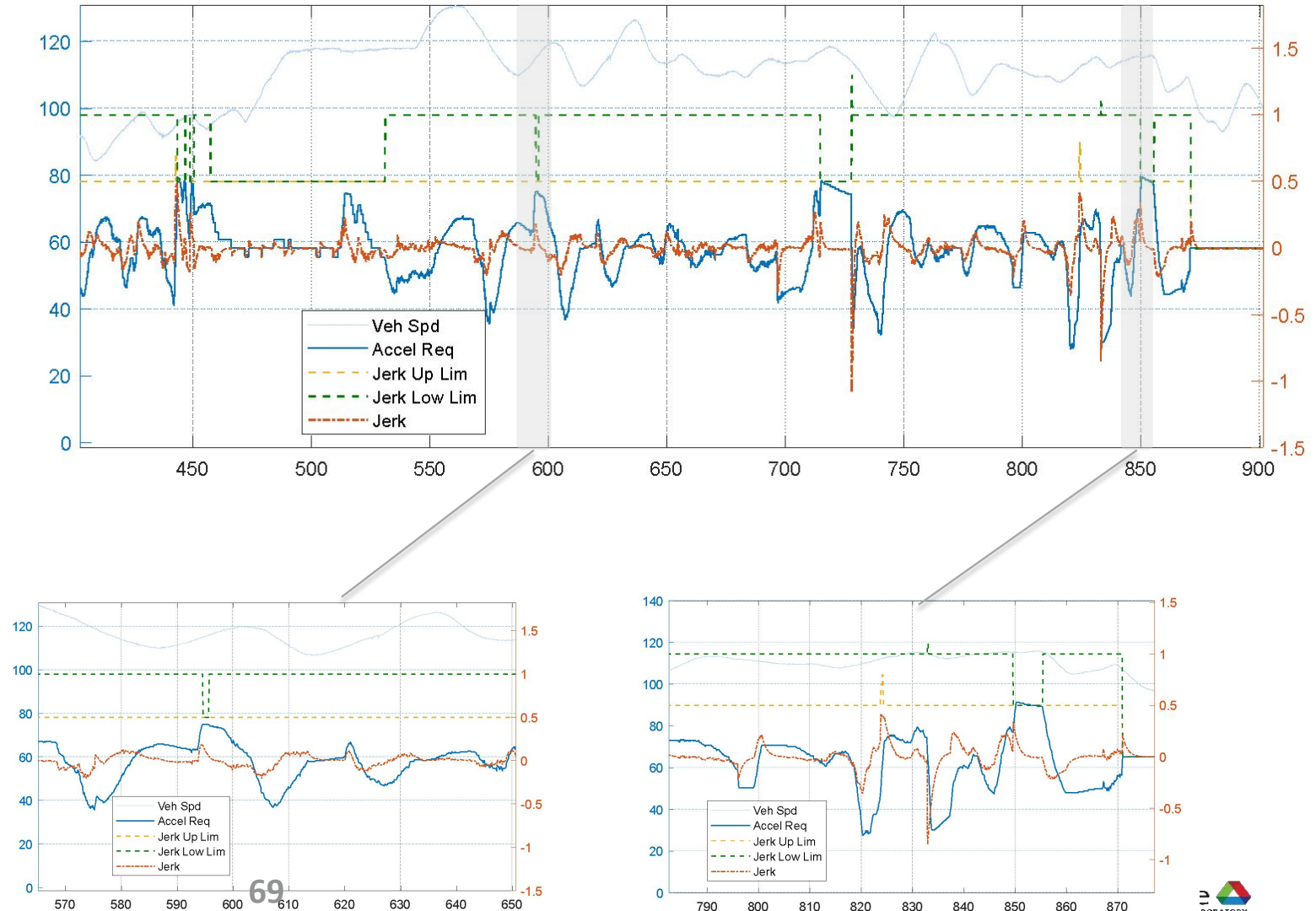
- Once the vehicle was taken off the chassis dynamometer, on-road highway driving data was collected



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – ACC On-Road Data Collection

- The graph shows the higher speed part of the driving
- The acceleration request signal originates in the smart cruise control module (same signal overridden on dyno). The jerk was calculated
- Data shows that the upper limit does not change except for two cases of higher positive jerk
- The lower limit changes more frequently becoming “stricter” for hard deceleration events (jerk changing rapidly)
- More data is needed to understand the correlation between the events and the signals

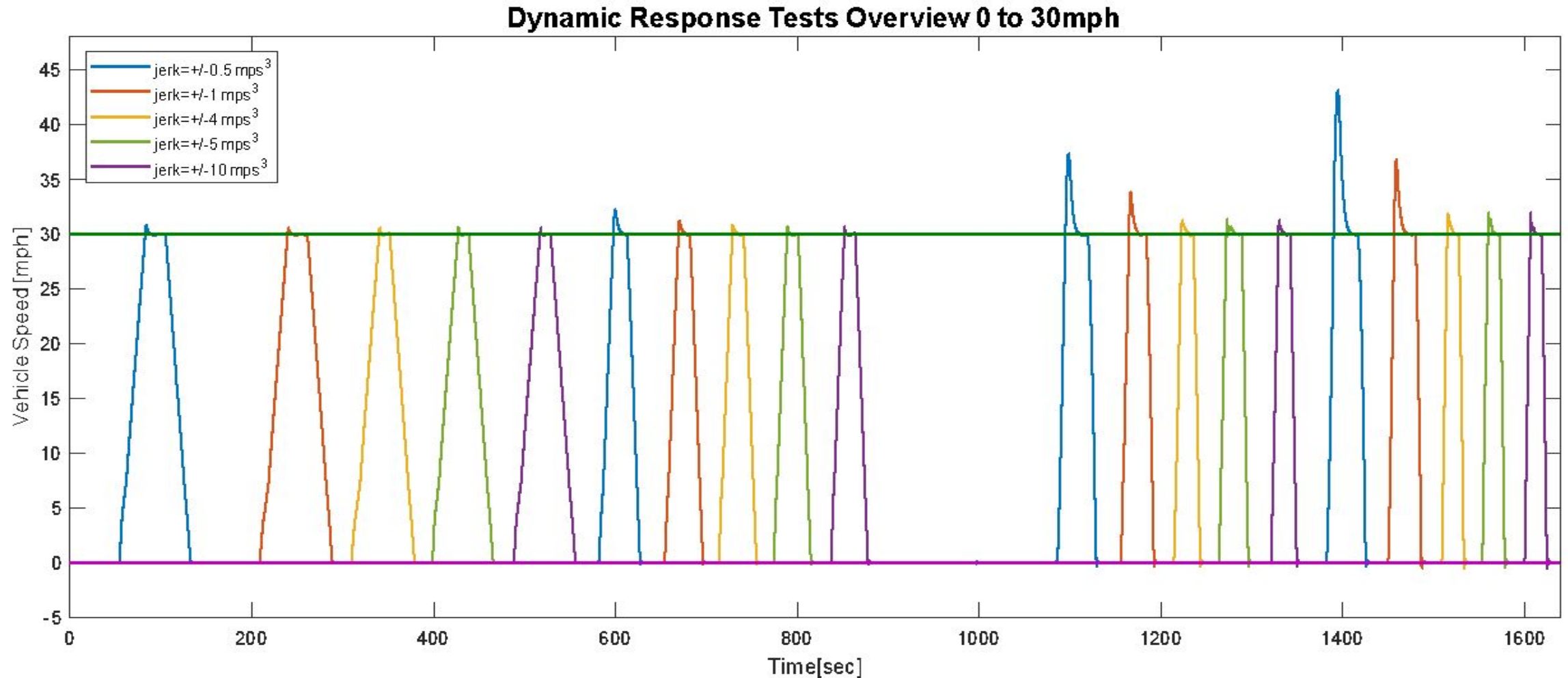


SECOND ROUND OF TESTING

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

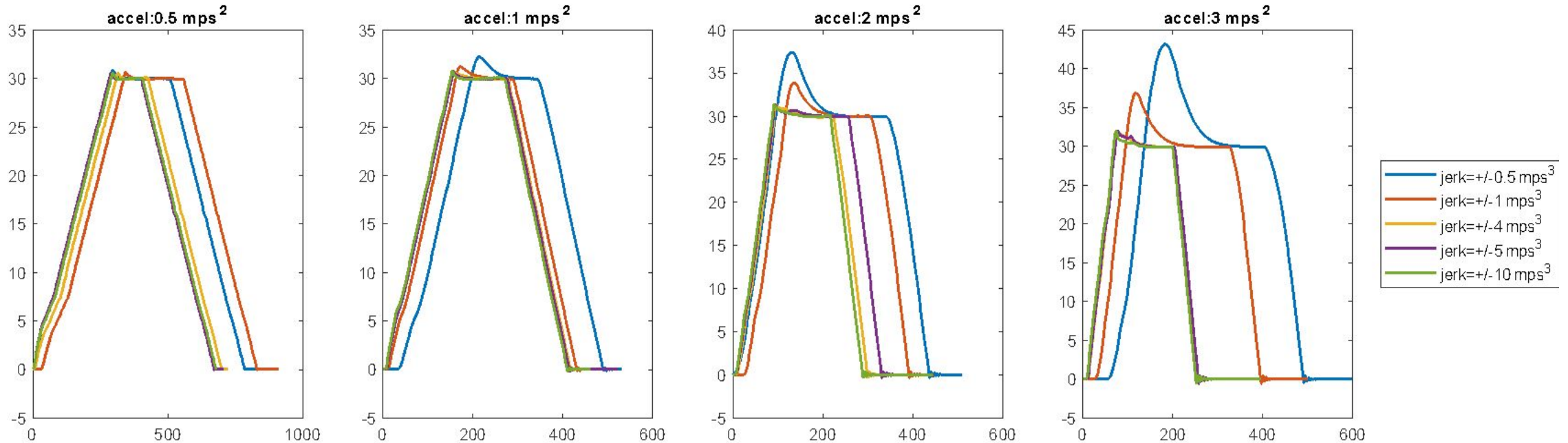
0-30 mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

0-30 mph



	accel req : 0.5 mps ²		accel req : 1 mps ²		accel req : 2 mps ²		accel req : 3 mps ²	
Jerk Limits	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time
±0.5 mps ³	27.90	27.90	15.80	14.90	9.30	9.60	7.80	8.80
±1 mps ³	30.10	27.50	15.00	14.40	9.00	8.80	6.60	7.10
±4 mps ³	30.00	27.40	14.50	14.00	8.30	7.70	5.80	5.50
±5 mps ³	27.90	27.20	14.50	14.00	8.20	7.40	6.20	5.50
±10 mps ³	28.50	27.30	14.30	14.00	8.10	7.20	6.00	5.30

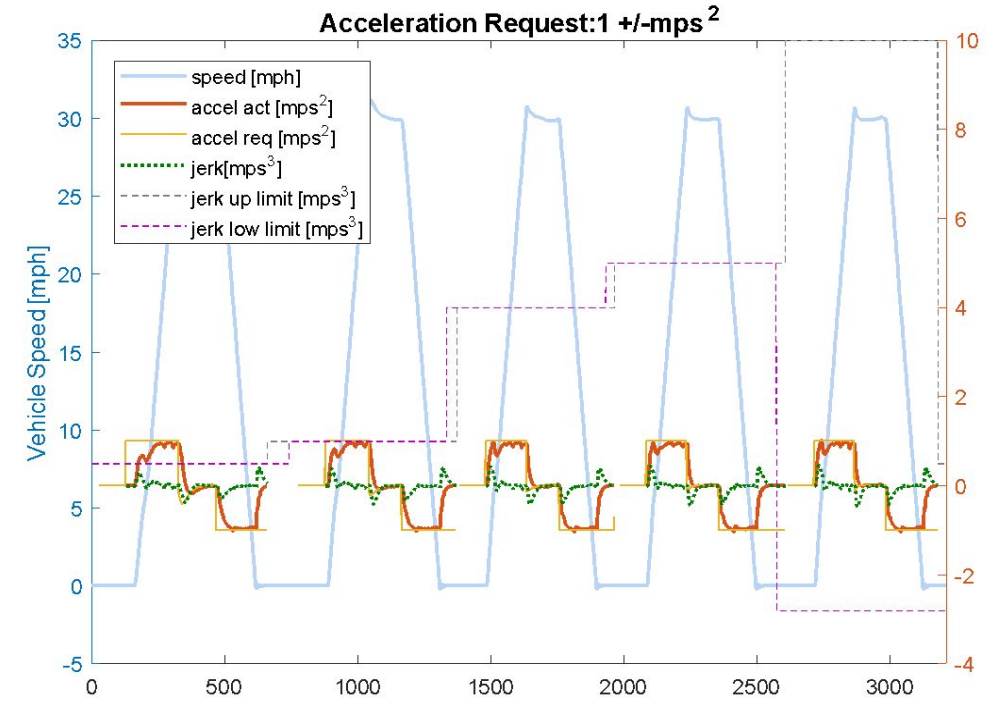
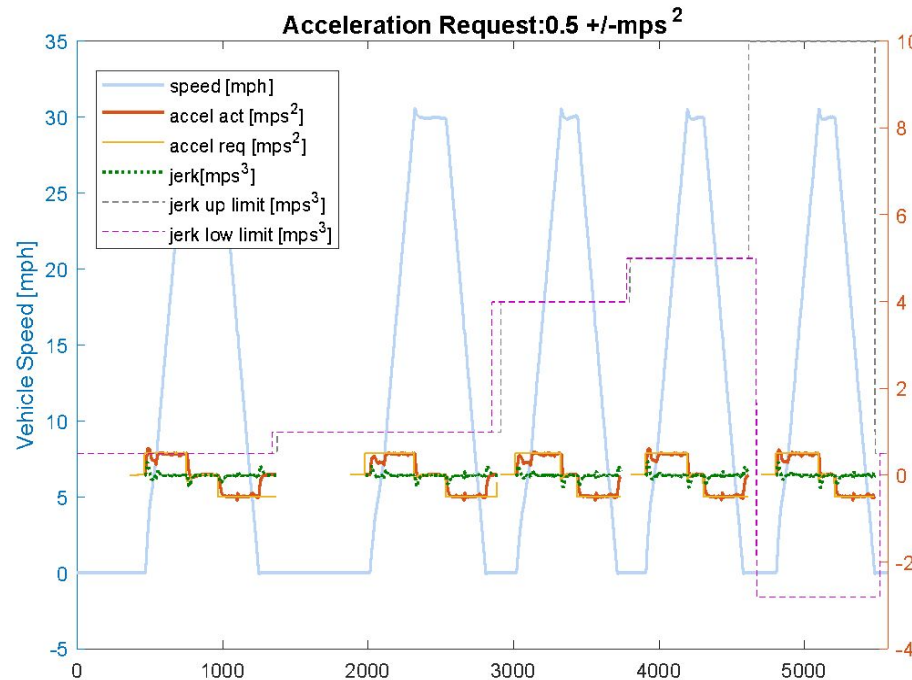
— jerk= $\pm 0.5 \text{ mps}^3$ — jerk= $\pm 1 \text{ mps}^3$ — jerk= $\pm 4 \text{ mps}^3$ — jerk= $\pm 5 \text{ mps}^3$ — jerk= $\pm 10 \text{ mps}^3$

— jerk= $\pm 0.5 \text{ mps}^3$ — jerk= $\pm 1 \text{ mps}^3$ — jerk= $\pm 4 \text{ mps}^3$ — jerk= $\pm 10 \text{ mps}^3$

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

0-30 mph



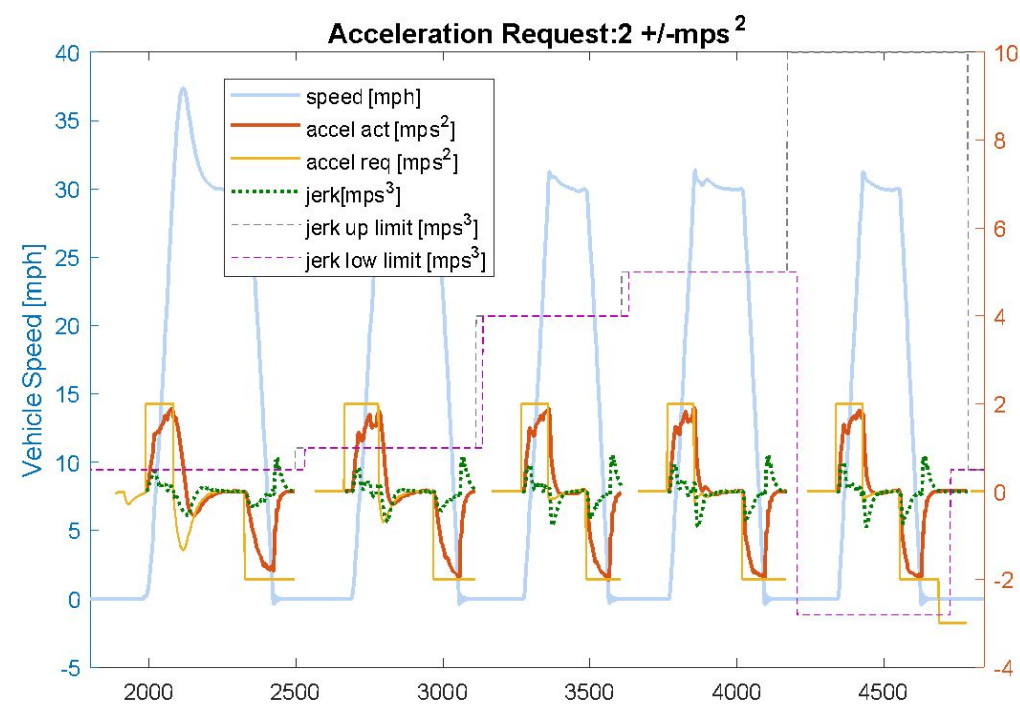
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.61	-0.58	0.31	-0.17
$\pm 1 \text{ mps}^3$	0.51	-0.58	0.22	-0.19
$\pm 4 \text{ mps}^3$	0.55	-0.59	0.22	-0.24
$\pm 5 \text{ mps}^3$	0.61	-0.58	0.32	-0.23
$\pm 10 \text{ mps}^3$	0.58	-0.58	0.31	-0.23

Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.00	-1.03	0.31	-0.27
$\pm 1 \text{ mps}^3$	0.98	-1.03	0.40	-0.34
$\pm 4 \text{ mps}^3$	1.00	-1.04	0.44	-0.43
$\pm 5 \text{ mps}^3$	0.98	-1.05	0.45	-0.43
$\pm 10 \text{ mps}^3$	1.03	-1.05	0.47	-0.43

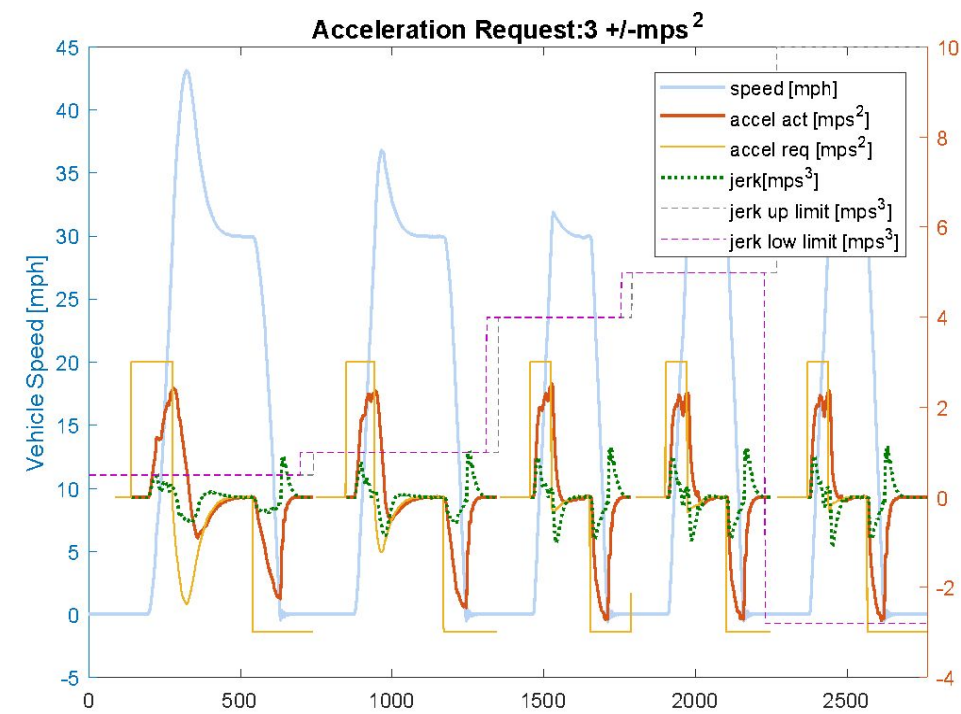
Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

0-30 mph



Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	-1.80	1.89	0.55	-0.33
$\pm 1 \text{ mps}^3$	-2.00	1.79	0.55	-0.43
$\pm 4 \text{ mps}^3$	-2.01	1.85	0.62	-0.61
$\pm 5 \text{ mps}^3$	-1.98	1.89	0.61	-0.69
$\pm 10 \text{ mps}^3$	-1.98	1.86	0.67	-0.73

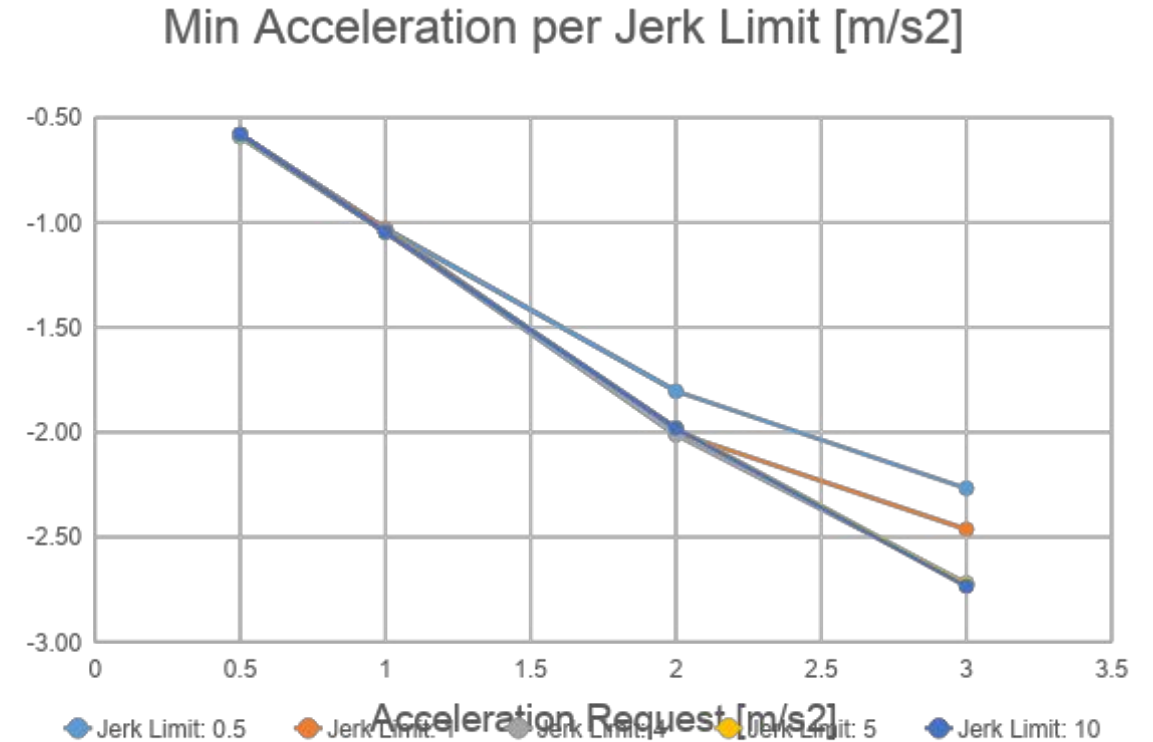
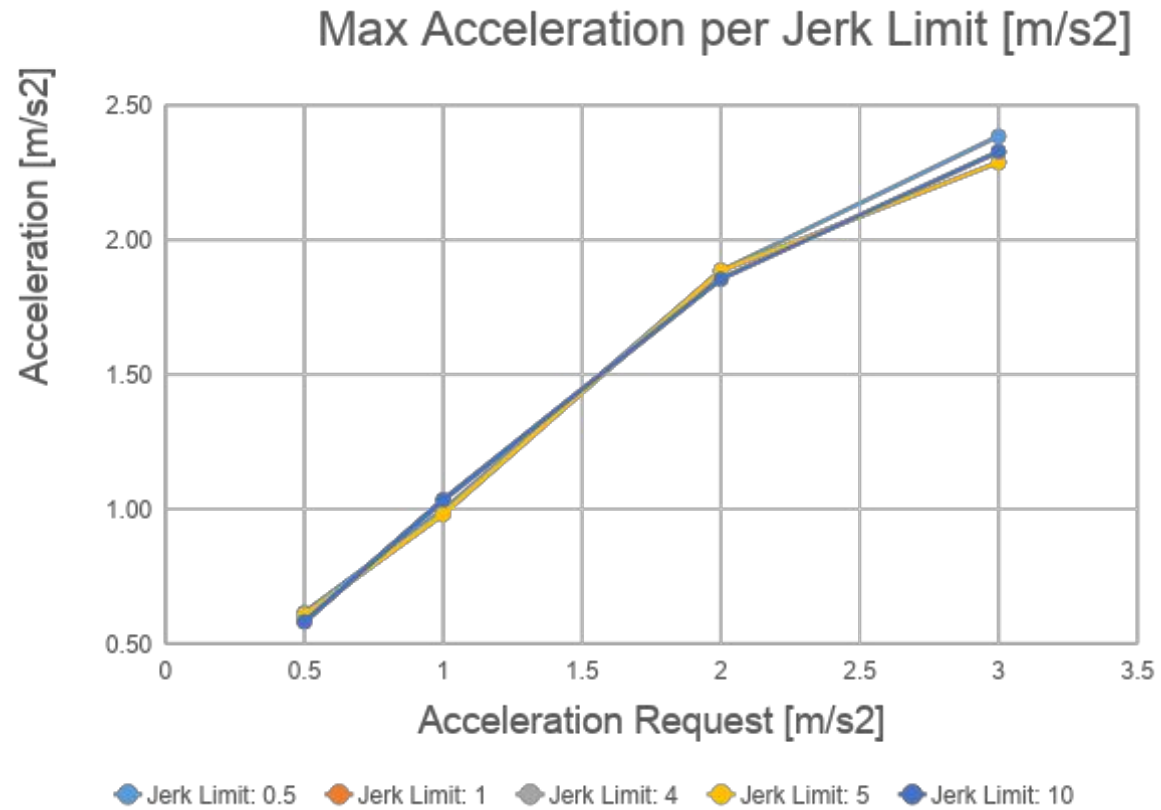


Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	2.38	-2.27	0.55	-0.36
$\pm 1 \text{ mps}^3$	2.32	-2.46	0.81	-0.56
$\pm 4 \text{ mps}^3$	2.46	-2.72	0.90	-0.92
$\pm 5 \text{ mps}^3$	2.29	-2.73	0.91	-0.90
$\pm 10 \text{ mps}^3$	2.33	-2.73	0.93	-0.92

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

0-30 mph

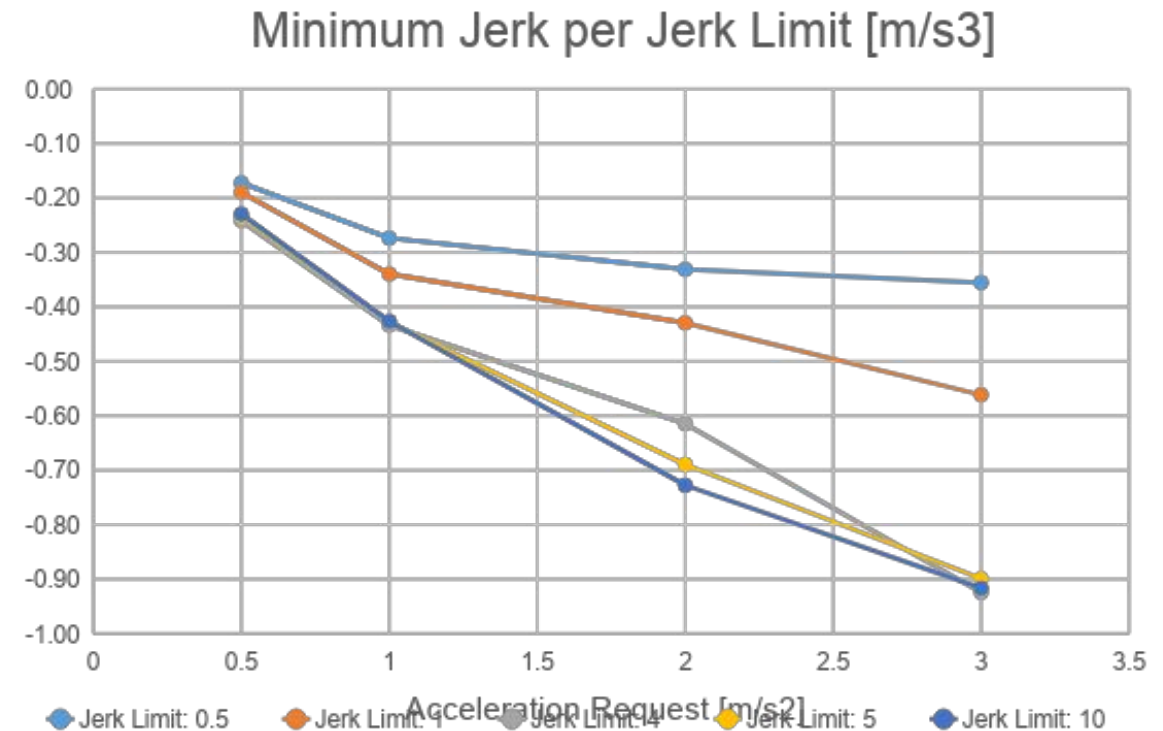
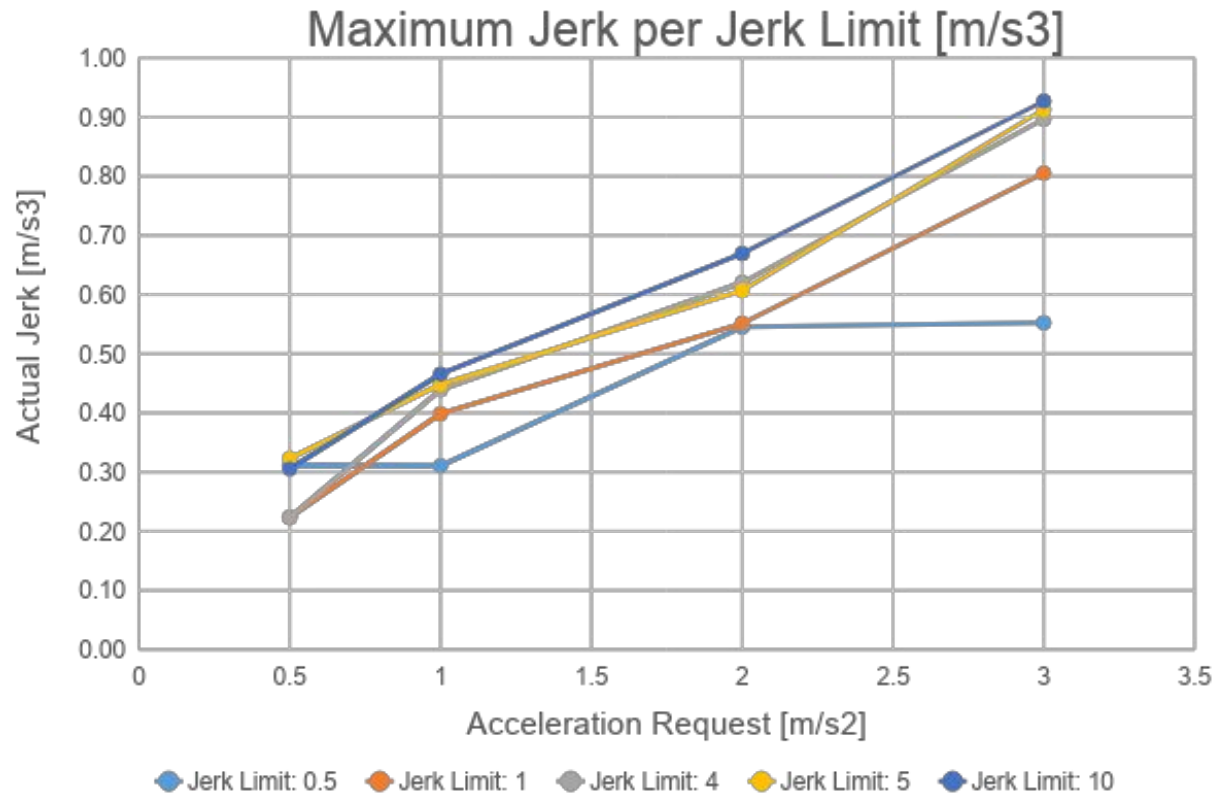


- The higher the jerk upper limit (the more “relaxed”), the higher is the maximum acceleration reached by the vehicle for higher acceleration requests (effect is not visible on lower acceleration requests)

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

0-30 mph



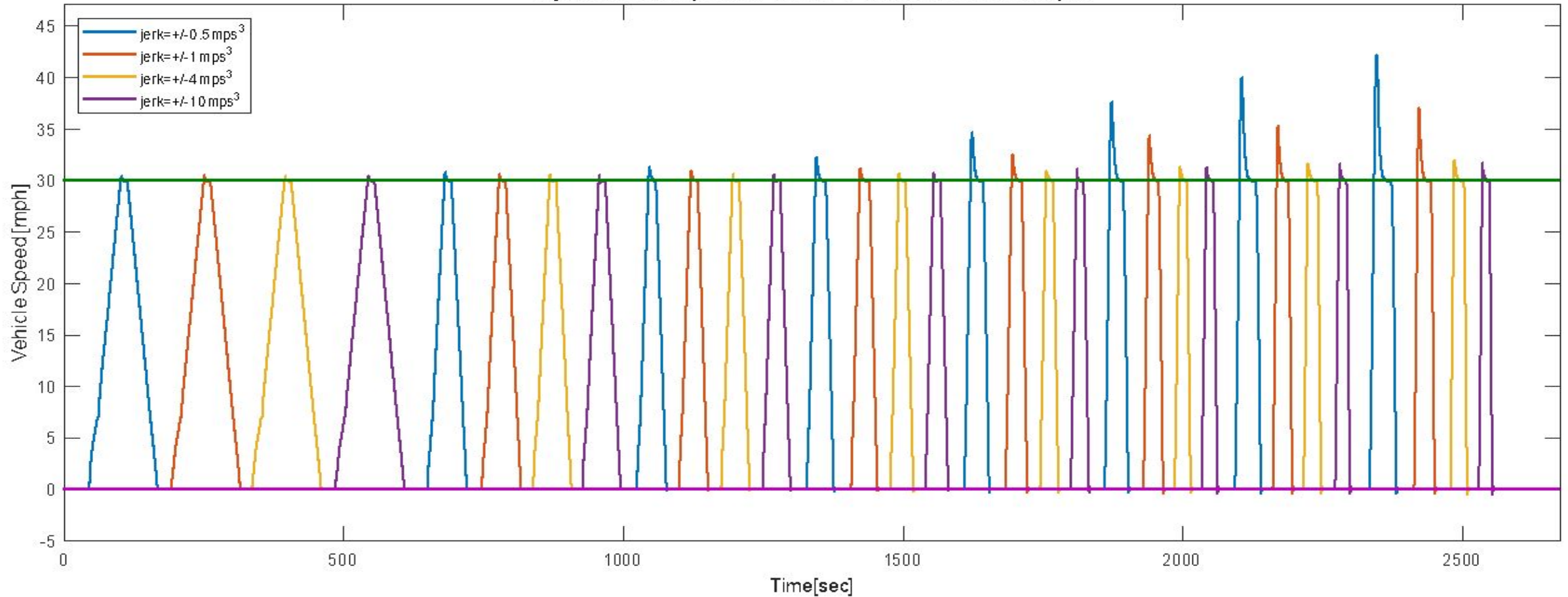
- On the maximum jerk the effect of the jerk limit is visible : the higher the jerk upper limit and the higher the acceleration request, the higher the actual jerk
- The effect of the jerk limit is not as visible on the minimum jerk

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

0-30 mph

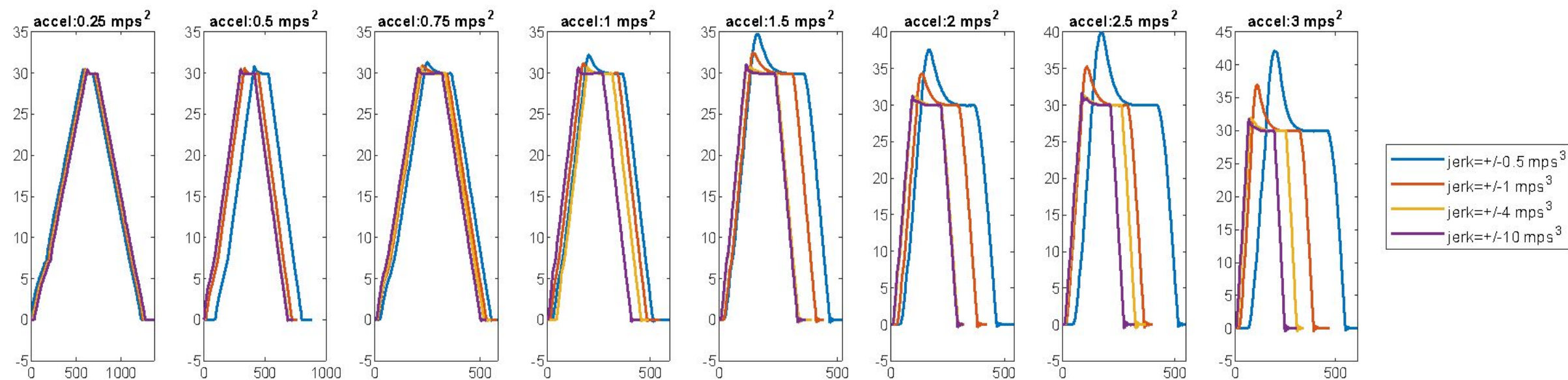
Dynamic Response Tests Overview 0 to 30mph



Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

0-30 mph

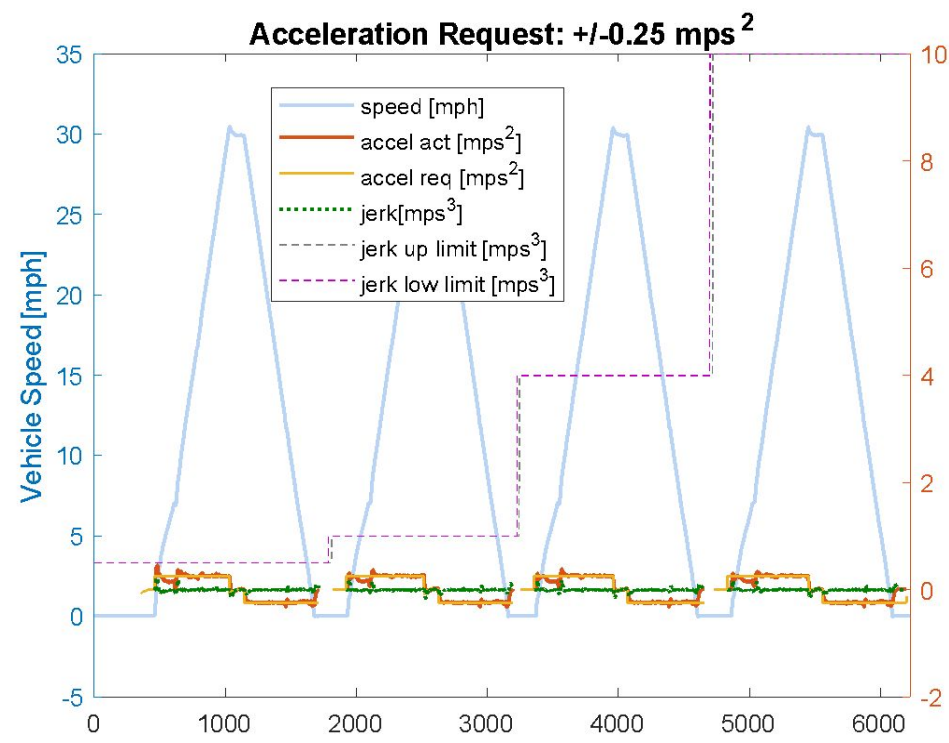


	accel req : 0.25 mps ²		accel req : 0.5 mps ²		accel req : 0.75 mps ²		accel req : 1 mps ²		accel req : 1.5 mps ²		accel req : 2 mps ²		accel req : 2.5 mps ²		accel req : 3 mps ²	
Jerk Limits mps ³	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time	Accel Time	Decel Time
±0.5																
±1																
±4																
±10																

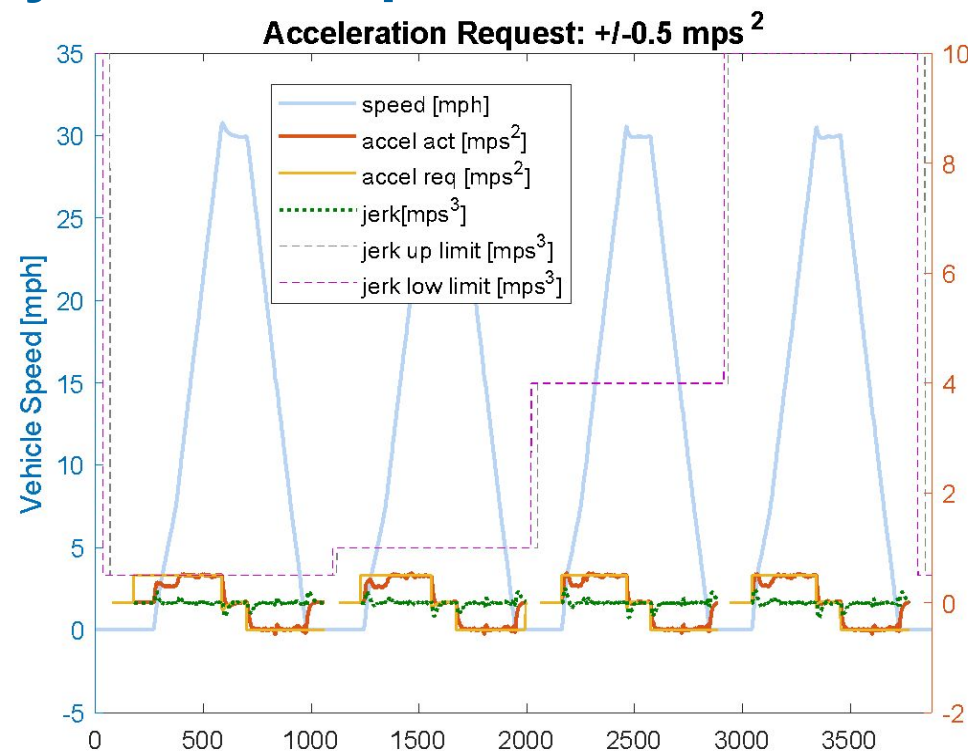
Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

0-30 mph



Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.46	-0.33	0.22	-0.09
$\pm 1 \text{ mps}^3$	0.36	-0.34	0.19	-0.11
$\pm 4 \text{ mps}^3$	0.38	-0.33	0.18	-0.10
$\pm 10 \text{ mps}^3$	0.33	-0.34	0.18	-0.10

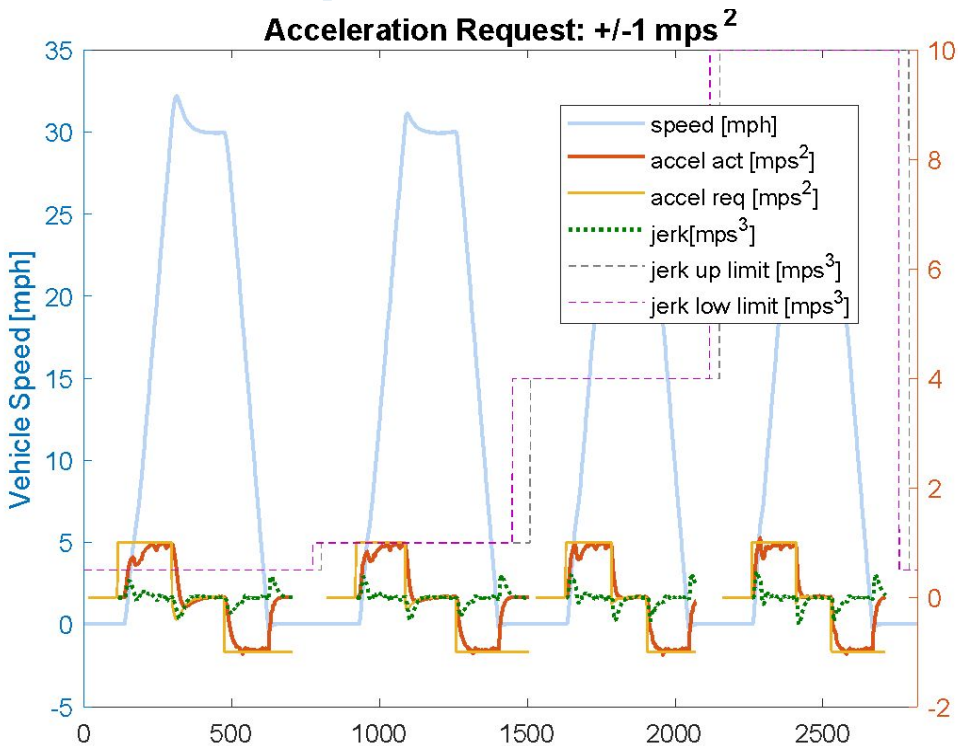
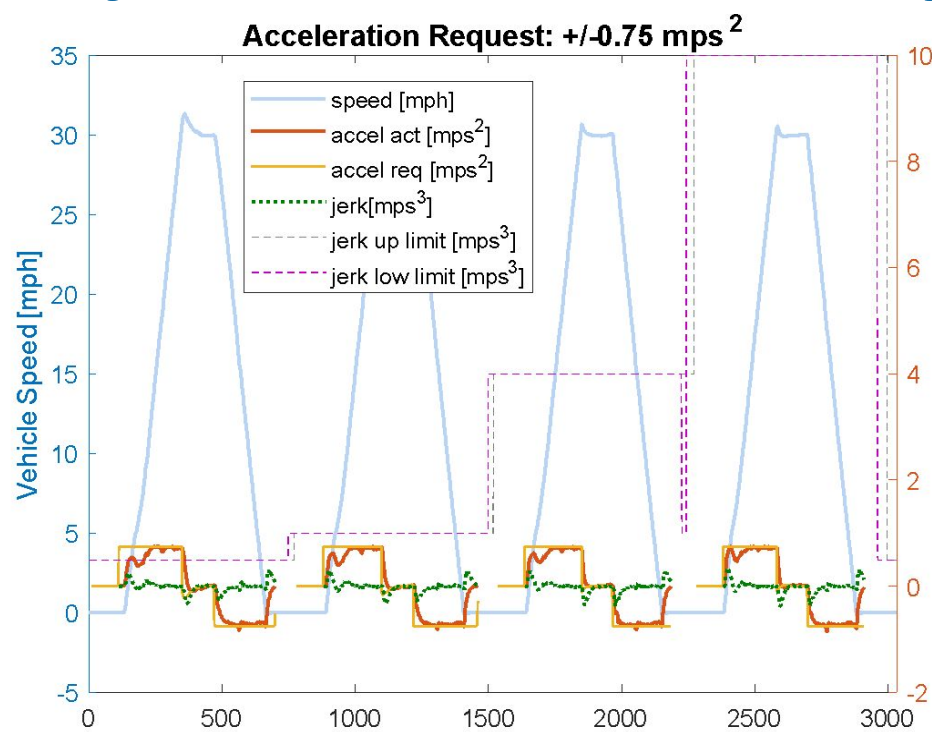


Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.51	-0.58	0.20	-0.18
$\pm 1 \text{ mps}^3$	0.53	-0.58	0.20	-0.21
$\pm 4 \text{ mps}^3$	0.54	-0.57	0.27	-0.25
$\pm 10 \text{ mps}^3$	0.52	-0.58	0.25	-0.25

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

0-30 mph



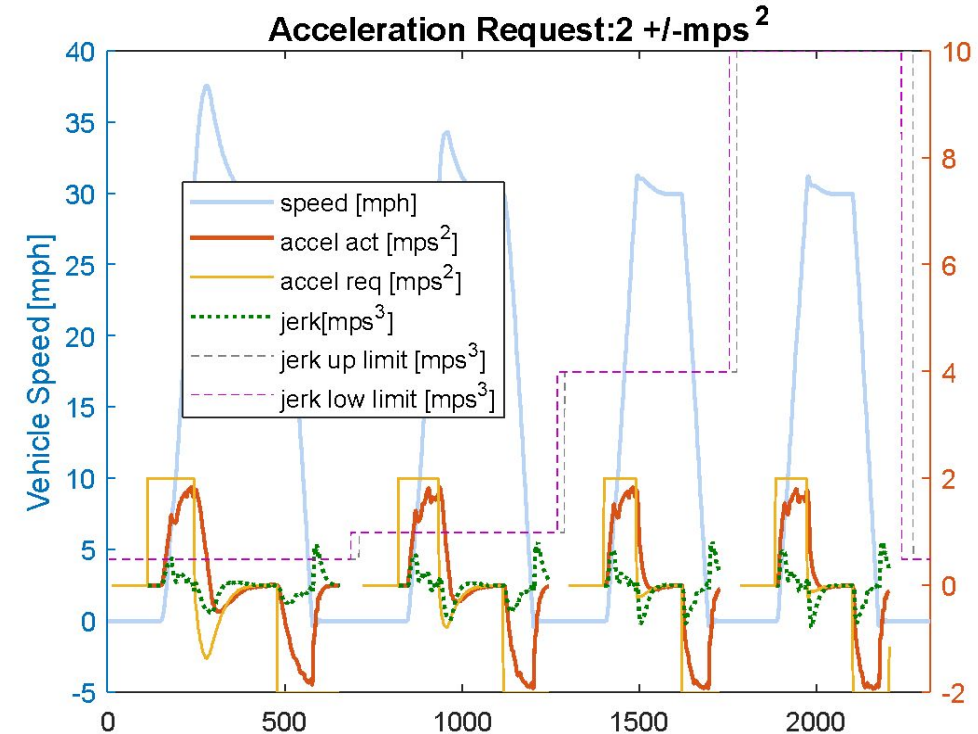
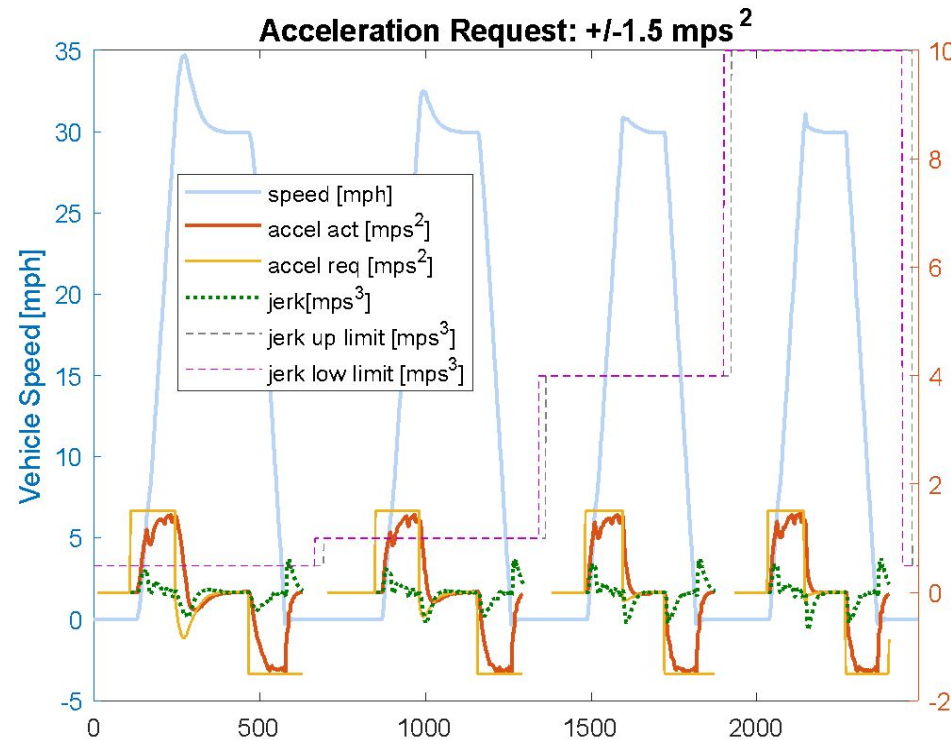
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.74	-0.83	0.25	-0.24
$\pm 1 \text{ mps}^3$	0.75	-0.82	0.31	-0.27
$\pm 4 \text{ mps}^3$	0.74	-0.82	0.29	-0.37
$\pm 10 \text{ mps}^3$	0.75	-0.85	0.36	-0.36

Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	0.98	-1.04	0.32	-0.29
$\pm 1 \text{ mps}^3$	0.97	-1.03	0.38	-0.35
$\pm 4 \text{ mps}^3$	0.98	-1.06	0.42	-0.43
$\pm 10 \text{ mps}^3$	1.08	-1.05	0.48	-0.43

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

0-30 mph



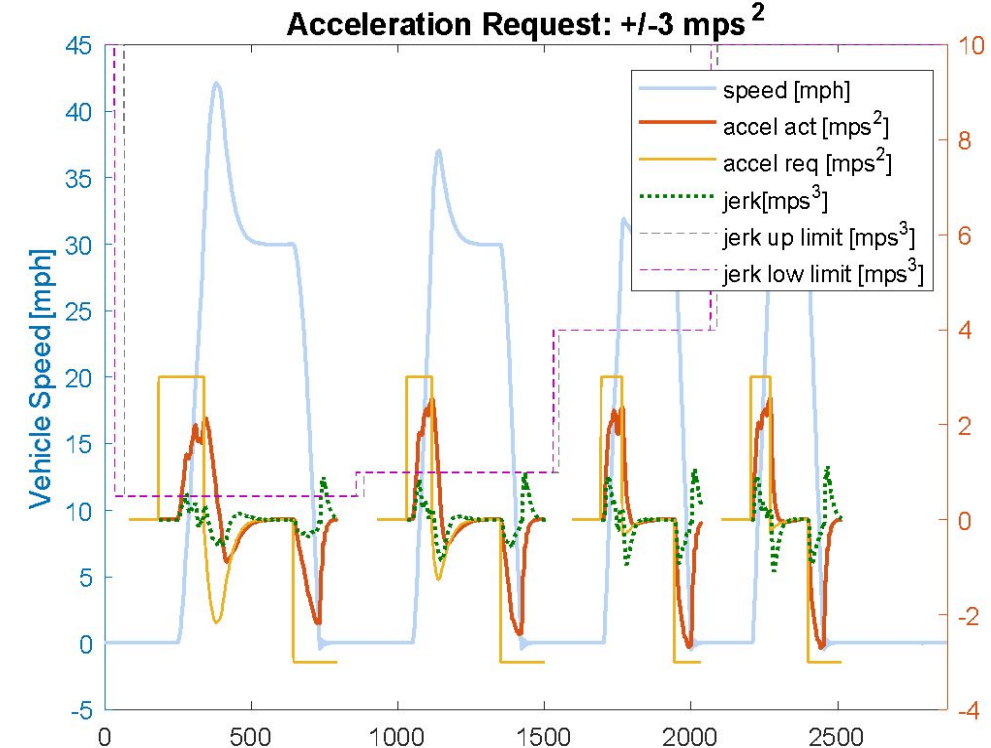
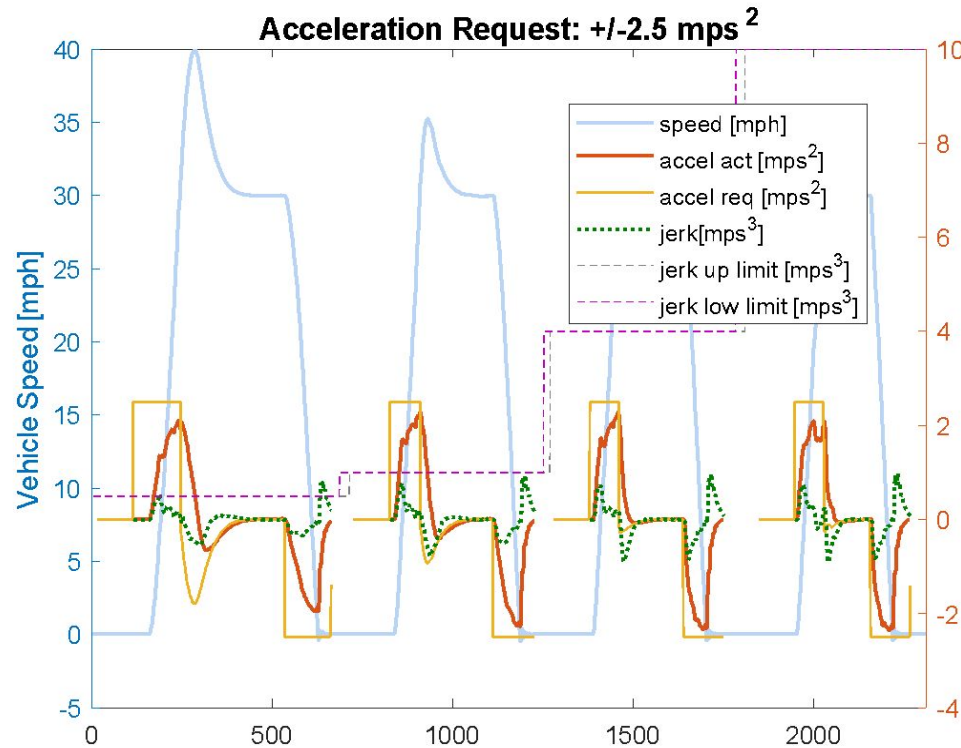
Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.43	-1.50	0.41	-0.33
$\pm 1 \text{ mps}^3$	1.44	-1.51	0.45	-0.45
$\pm 4 \text{ mps}^3$	1.43	-1.50	0.52	-0.57
$\pm 10 \text{ mps}^3$	1.46	-1.51	0.53	-0.57

Jerk Limits	Max Accel [mps^2]	Min Accel [mps^2]	Max Jerk [mps^3]	Min Jerk [mps^3]
$\pm 0.5 \text{ mps}^3$	1.84	-1.87	0.53	-0.32
$\pm 1 \text{ mps}^3$	1.81	-1.94	0.58	-0.50
$\pm 4 \text{ mps}^3$	1.83	-1.94	0.66	-0.71
$\pm 10 \text{ mps}^3$	1.80	-1.93	0.66	-0.71

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

0-30 mph



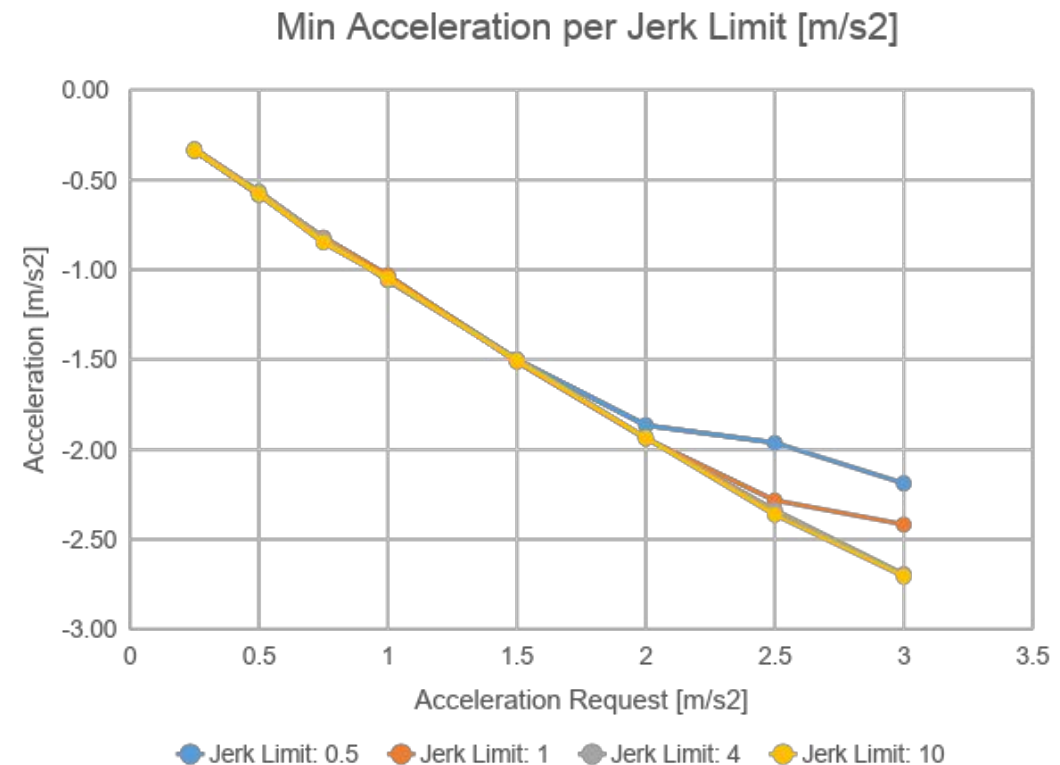
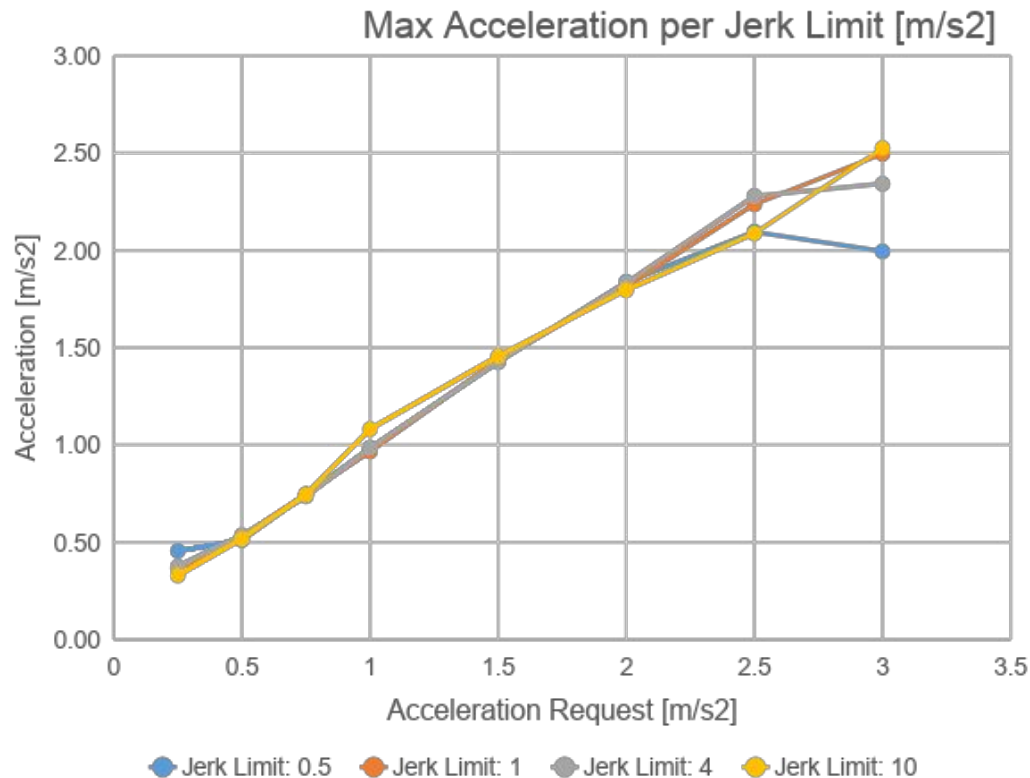
Jerk Limits	Max Accel [mps ²]	Min Accel [mps ²]	Max Jerk [mps ³]	Min Jerk [mps ³]
$\pm 0.5 \text{ mps}^3$	2.10	-1.96	0.55	-0.34
$\pm 1 \text{ mps}^3$	2.24	-2.28	0.78	-0.50
$\pm 4 \text{ mps}^3$	2.28	-2.34	0.72	-0.85
$\pm 10 \text{ mps}^3$	2.09	-2.36	0.74	-0.82

Jerk Limits	Max Accel [mps ²]	Min Accel [mps ²]	Max Jerk [mps ³]	Min Jerk [mps ³]
$\pm 0.5 \text{ mps}^3$	2.00	-2.19	0.55	-0.35
$\pm 1 \text{ mps}^3$	2.50	-2.42	0.85	-0.58
$\pm 4 \text{ mps}^3$	2.34	-2.69	0.92	-0.91
$\pm 10 \text{ mps}^3$	2.53	-2.71	0.92	-0.92

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

0-30 mph

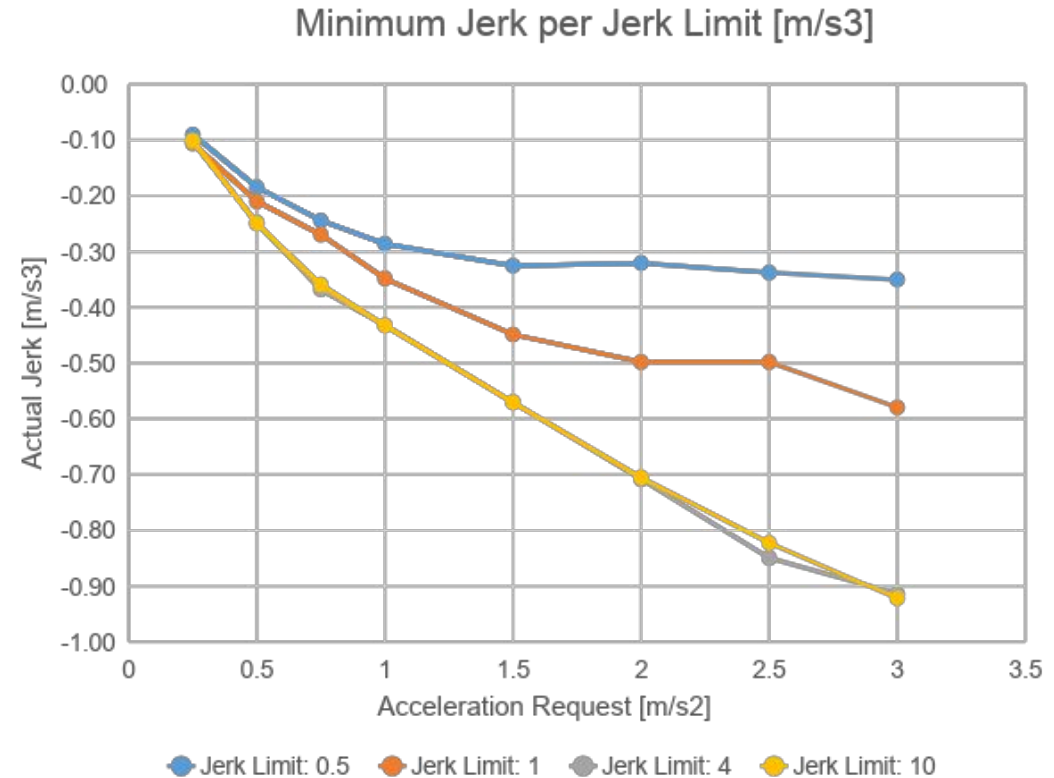
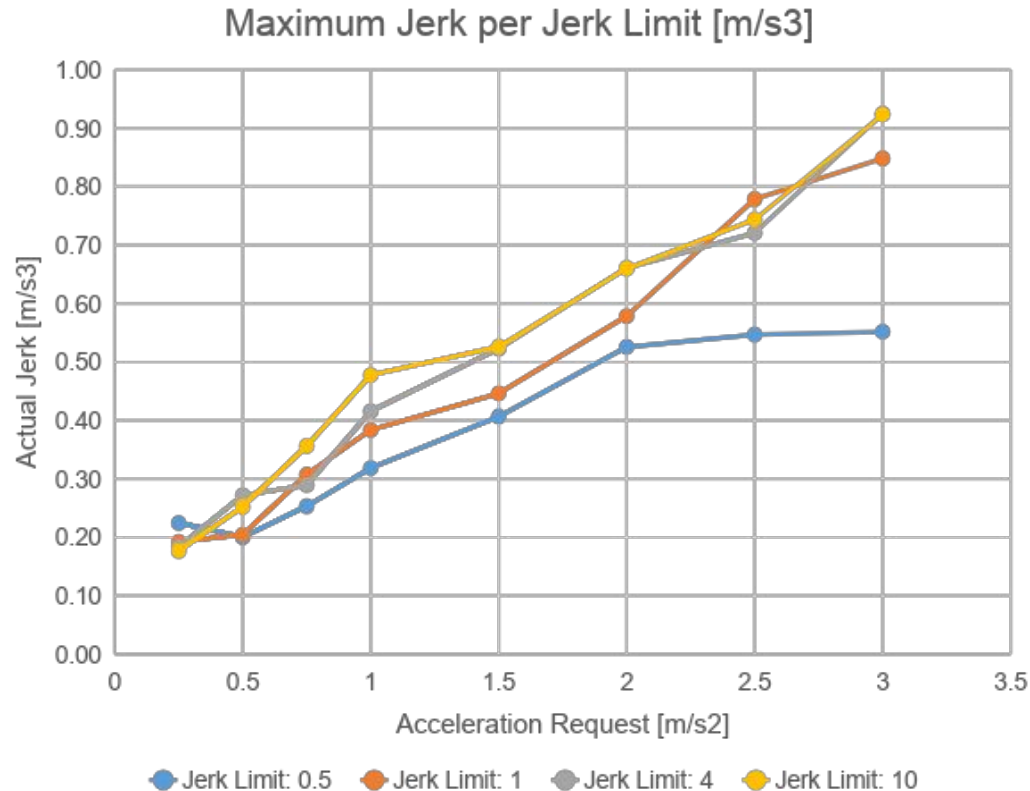


- The higher the jerk upper limit (the more “relaxed”), the higher is the maximum acceleration reached by the vehicle for higher acceleration requests (effect is not visible on lower acceleration requests)

Task 1- Expanding the XIL Research Fleet

2021 Hyundai Sonata – Chassis Dyno Dynamic Response Tests

0-30 mph



- On the maximum jerk the effect of the jerk limit is visible : the higher the jerk upper limit and the higher the acceleration request, the higher the actual jerk
- The effect of the jerk limit is not as visible on the minimum jerk